

Dimensioning

Basic Topics

Advanced Topics

Exercises

Dimensioning: Basic Topics

Summary

2-1) Detailed Drawings

2-2) Learning to Dimension

2-3) Dimension Appearance and Techniques.

2-4) Dimensioning and Locating Simple Features.

2-6) Dimension Choice.

Dimensioning: Advanced Topics

2-5) Dimensioning and Locating Advanced Features.

Dimensioning: Exercises

Exercise 2-1: Dimension type

Exercise 2-2: Spacing and readability 1

Exercise 2-3: Spacing and readability 2

Exercise 2-4: Duplicate dimensions

Exercise 2-5: Dimension placement

Exercise 2-6: Circular and rectangular views

Exercise 2-7: Dimensioning and locating features

Exercise 2-8: Advanced features

Exercise 2-9: Dimension accuracy

Dimensioning

Summary

Dimensioning Summary

What will we learn?

- *How to dimension an orthographic projection using proper dimensioning techniques.*

Key points

- *Dimensions affect how a part is manufactured.*

Dimensioning

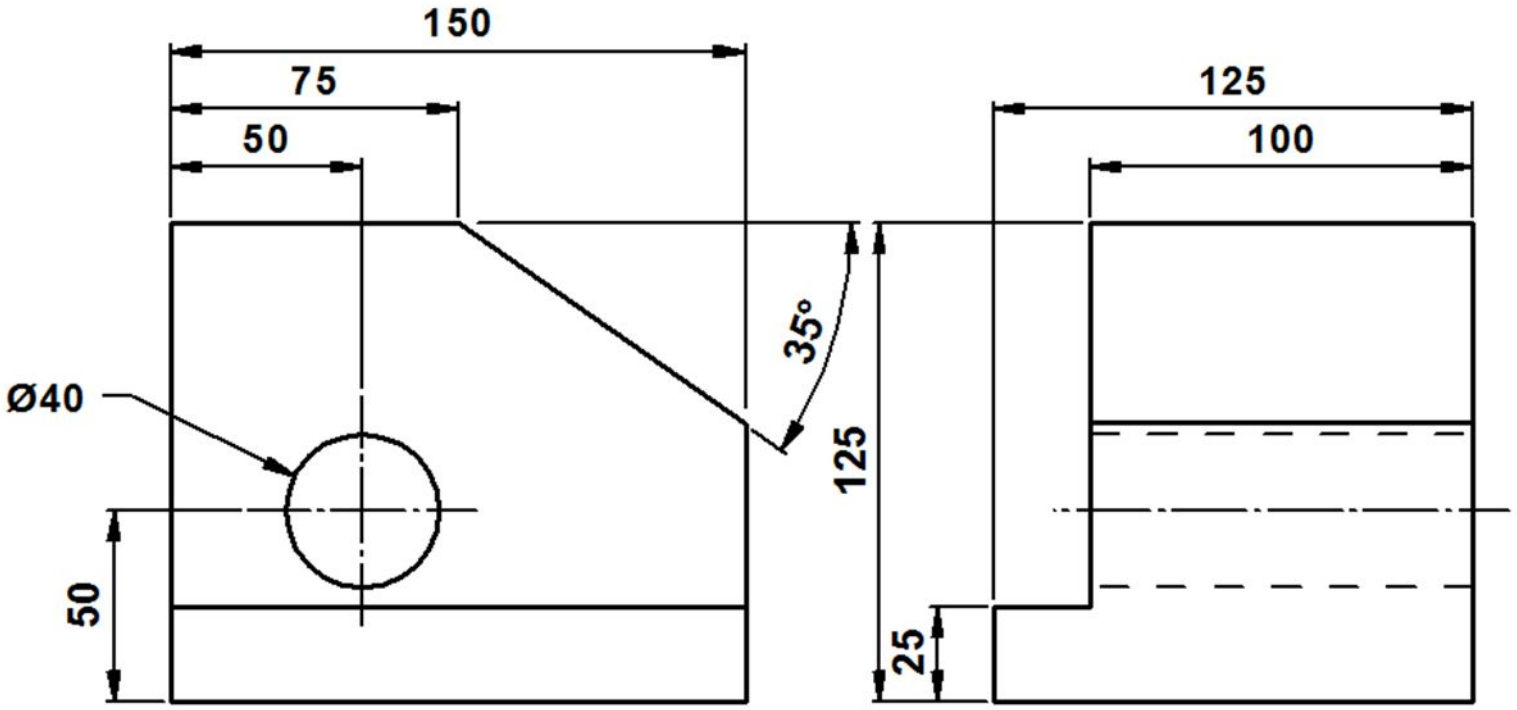
2-1) Detailed Drawings

Detailed Drawings

- Orthographic Projection: A shape description of an object (front, top, right-side views).

- Detailed Drawing: An orthographic projection, complete with all the dimensions and specifications needed to manufacture the object.

Detailed Drawing Example



Detailed Drawings

- *Can we dimension an object using the dimensions that we used to draw the object?*

No

These are not necessarily the same dimensions required to manufacture it.

We need to follow accepted standards.

(British Standard BS 8888 : 2011)

Dimensioning

2-2) Learning To Dimension

Learning to Dimension

What is our goal when dimensioning a part?

- *Basically, dimensions should be given in a clear and concise manner and should include everything needed to produce and inspect the part exactly as intended by the designer.*

Learning to Dimension

Proper dimensioning techniques require the knowledge of the following three areas.

- *Dimension Appearance/Technique*
- *Dimensioning and Locating Features*
- *Dimension Choice*

Learning to Dimension

1. <u>Dimension Appearance/Technique:</u>
   Dimensions use special lines, arrows, symbols and text.

   a) The lines used in dimensioning
   b) Types of dimensions
   c) Dimension symbols
   d) Dimension spacing and readability
   e) Dimension placement

Learning to Dimension

2. <u>Dimensioning and Locating Features:</u>
   Different types of features require unique methods of dimensioning.

Learning to Dimension

3. <u>Dimension Choice:</u> Your choice of dimensions will directly influence the method used to manufacture the part.

   a) *Units and decimal places.*
   b) *Locating feature using datums.*
   c) *Dimension accuracy and error build up.*

Dimensioning

2-3) Dimension Appearance and Techniques

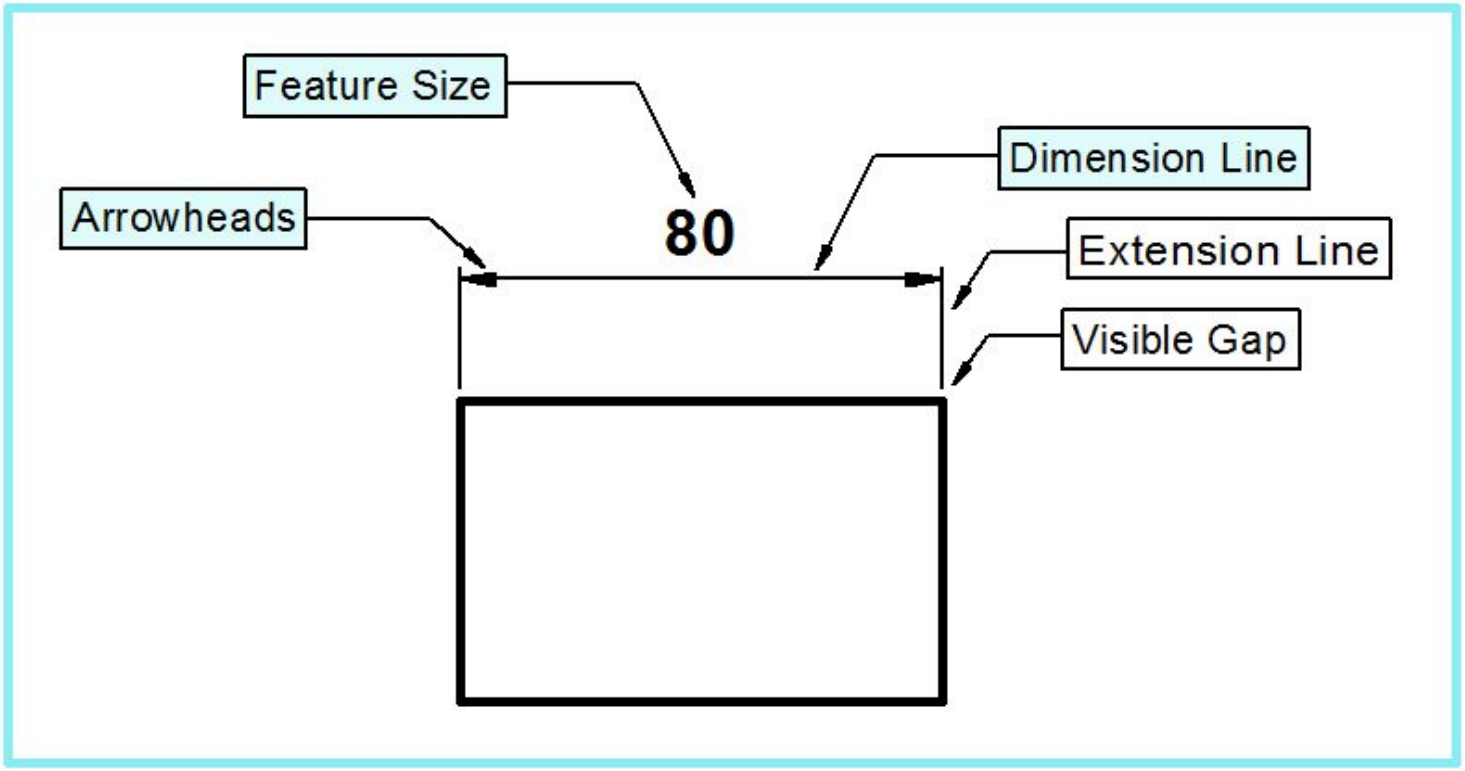
Lines used in Dimensioning

- Dimensioning requires the use of
  - ***Dimension lines***
  - ***Extension lines***
  - ***Leader lines***

- All three-line types are drawn thin so that they will not be confused with visible lines.

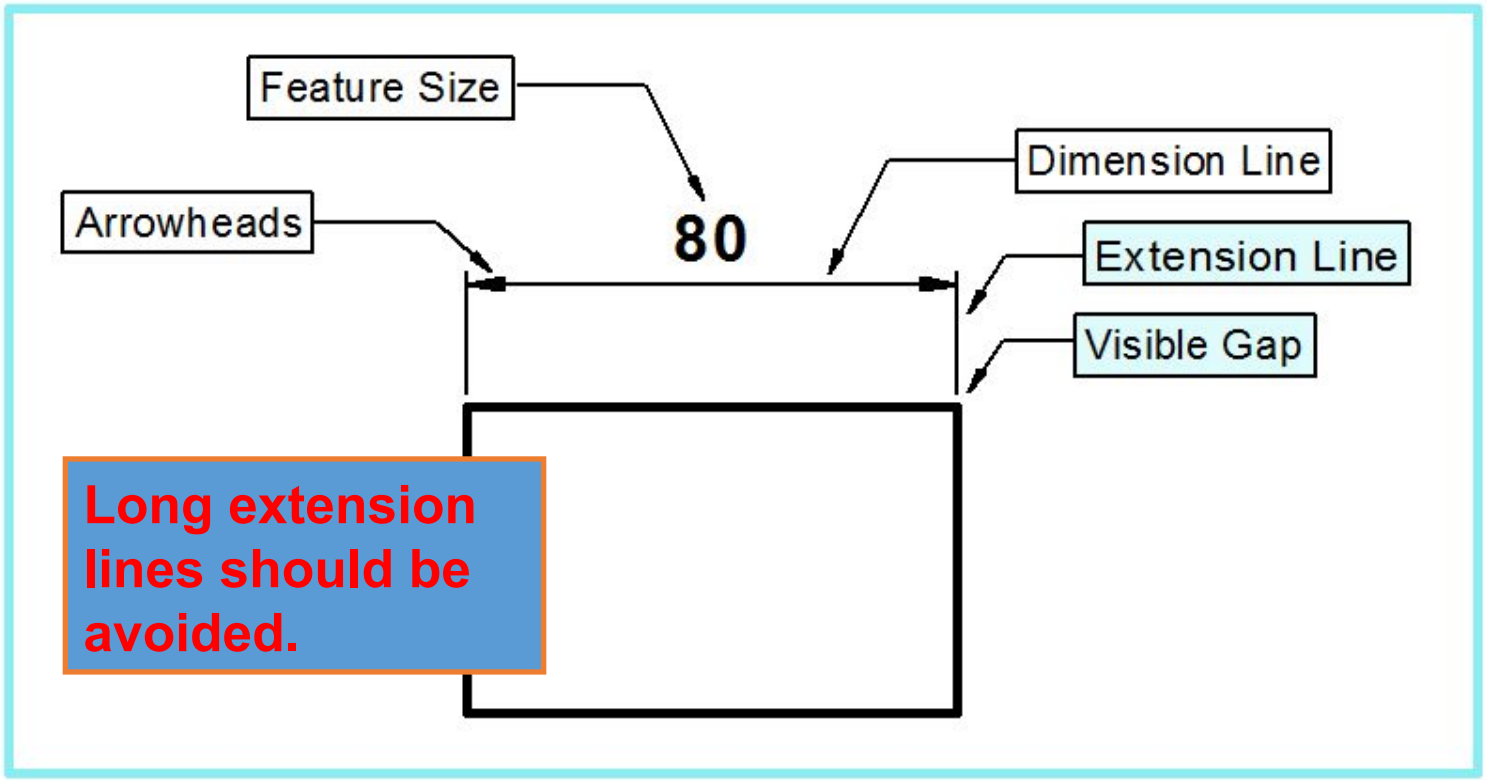
Dimension Line

- Dimension line: A line terminated by arrowheads, which indicates the direction and extent of a dimension.



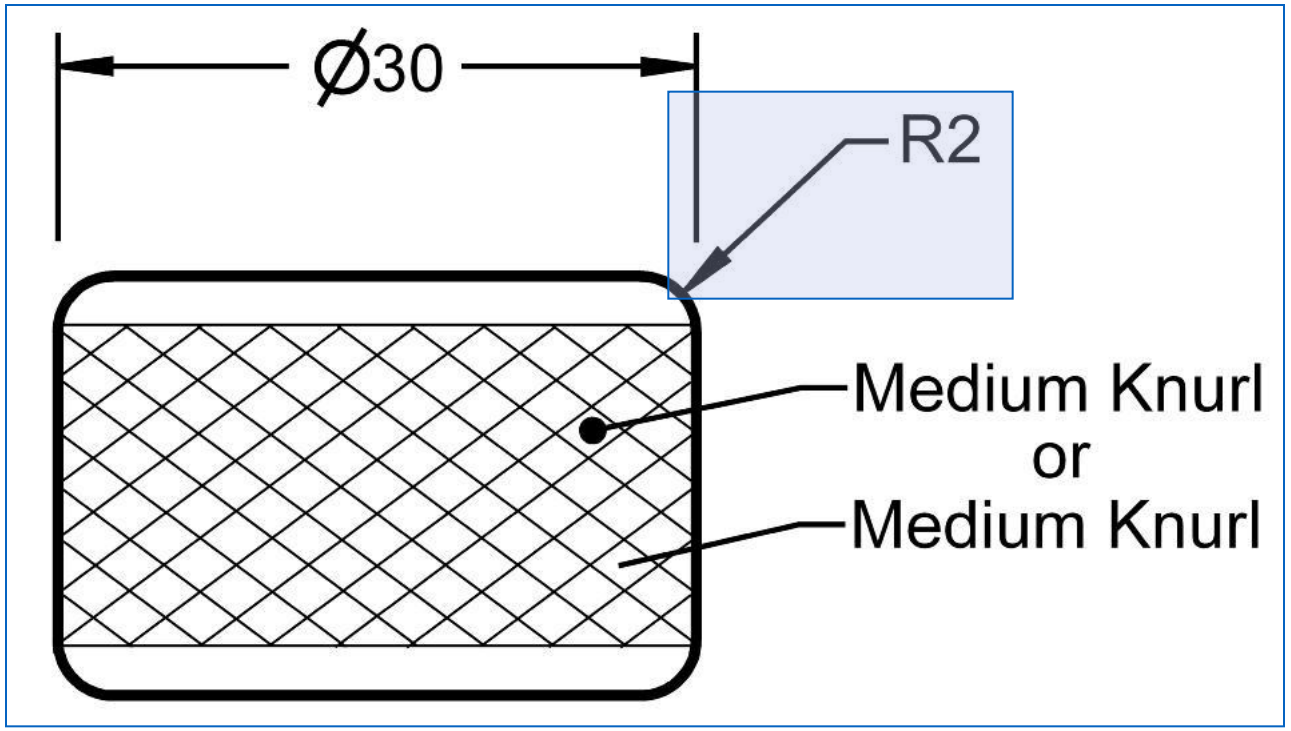
Extension Line

- Extension line: An extension line is a thin solid line that extends from a point on the drawing to which the dimension refers.



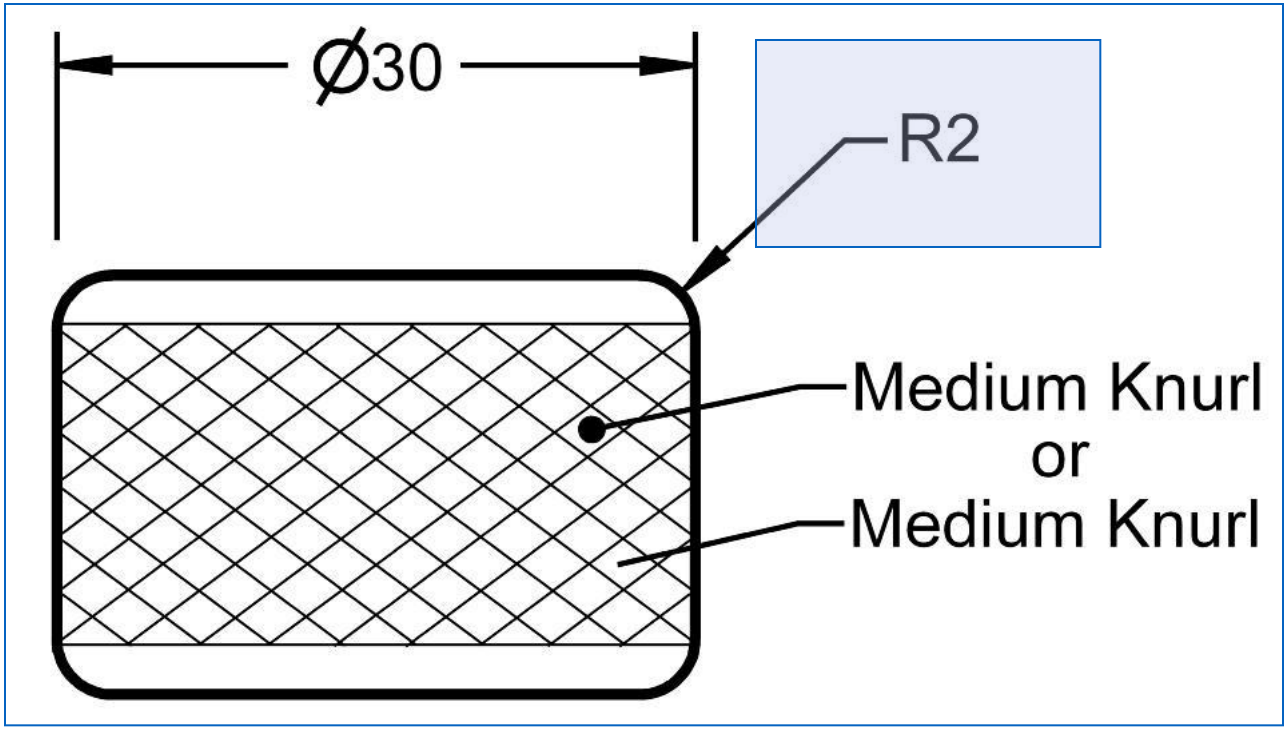
Leader Line

- Leader Line: A straight inclined thin solid line that is usually terminated by an arrowhead.



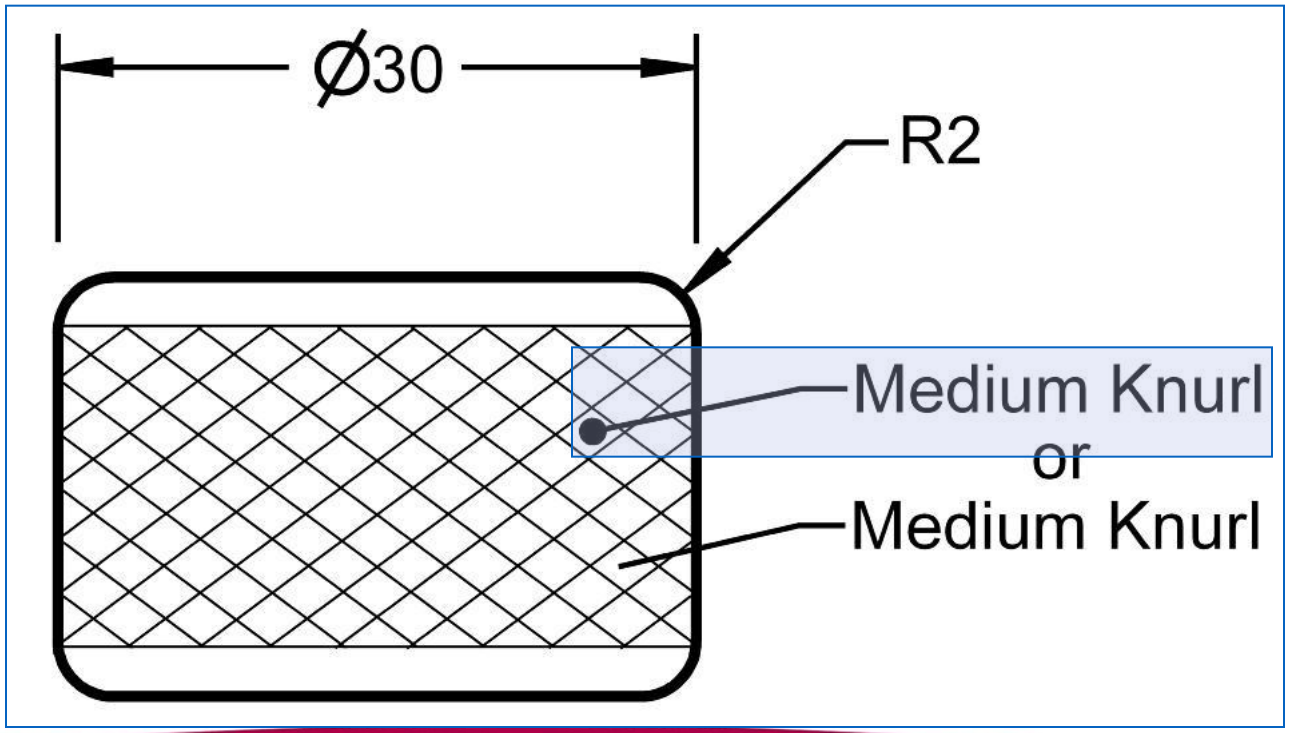
Leader Line

- Leaders may be terminated:
  - *with an arrow, if it ends on the outline of an object.*



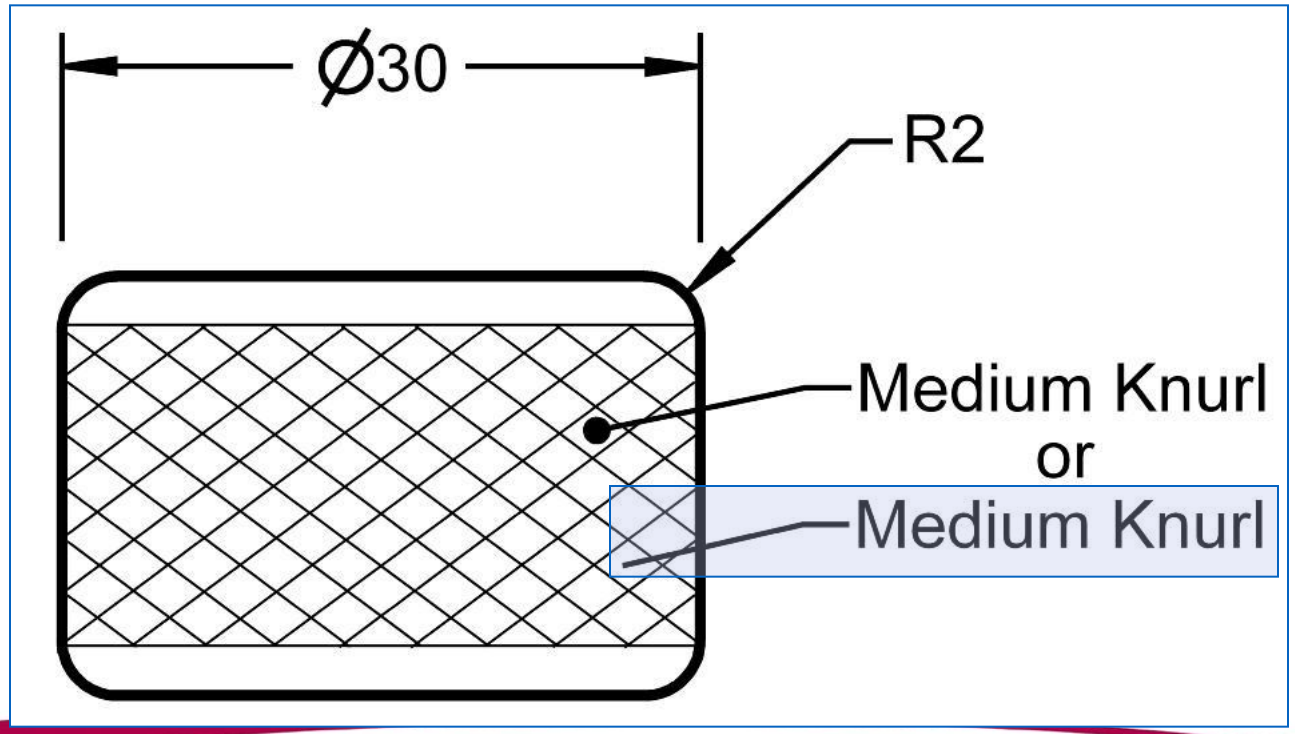
Leader Line

- Leaders may be terminated:
  - *with a dot if it ends within the outline of an object.*



Leader Line

- Leaders may be terminated:
  - *without an arrowhead or dot, if it ends within the outline of an object.*



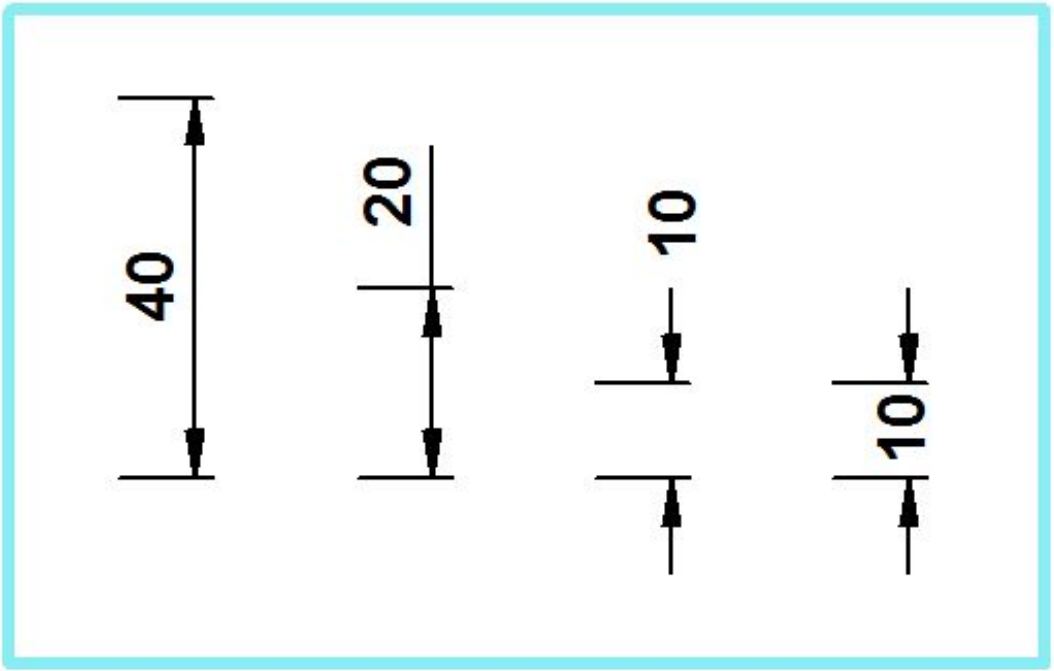
Leader Line

Avoid!

- *Crossing leaders.*
- *Long leaders.*
- *Leaders that are parallel to adjacent dimension, extension or section lines.*
- *Small angles between the leader and the terminating surface.*

Arrowheads

- Arrowheads are drawn between the extension lines if possible. If space is limited, they may be drawn on the outside.



Types of Dimensions

- Dimensions are given in the form of *linear distances*, *angles*, and *notes*.
  - *<u>Linear distances:</u> They are usually arranged horizontally or vertically but may also be aligned with a particular feature of the part.*

Types of Dimensions

- Dimensions are given in the form of *linear distances*, *angles*, and *notes*.
  - *<u>Angles:</u> Used to give the angle between two surfaces or features of a part.*

Types of Dimensions

- Dimensions are given in the form of *linear distances*, *angles*, and *notes*.
  - *<u>Notes:</u> Used to dimension diameters, radii, chamfers, threads, and other features that can not be dimensioned by the other two methods.*

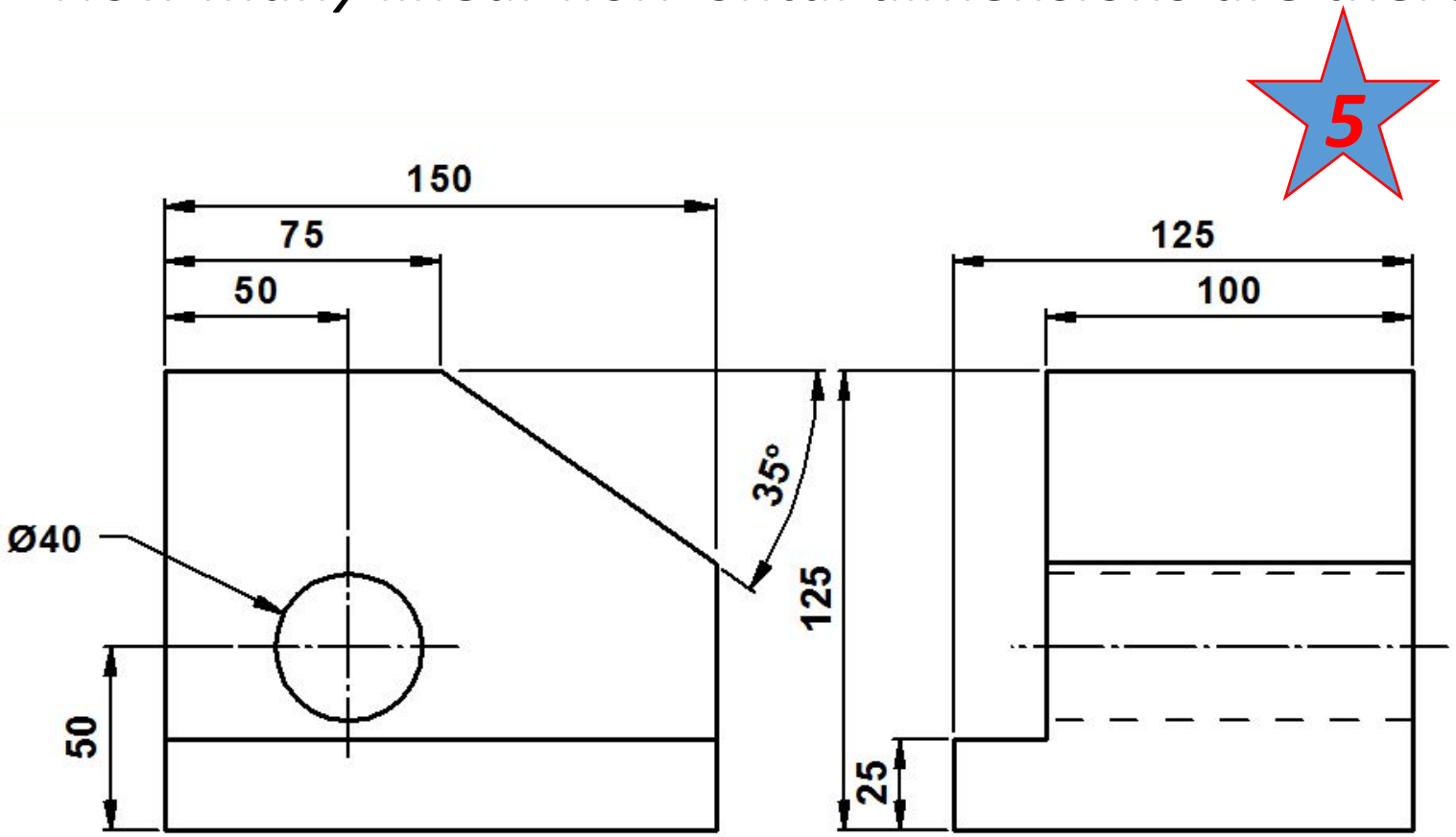


Exercise 2-1

Dimension types

Exercise 2-1

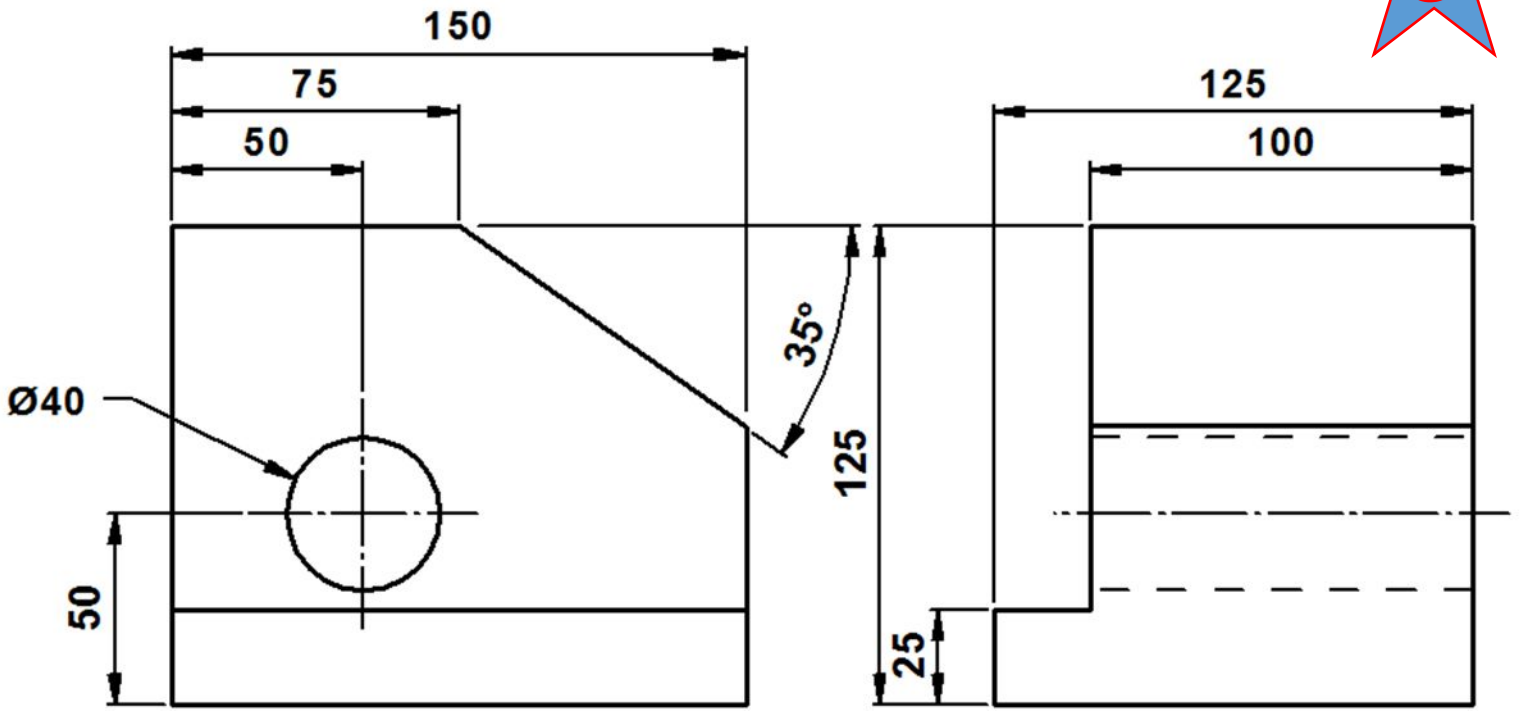
- Count the different types of dimensions.
  - *How many linear horizontal dimensions are there?*



Exercise 2-1

- Count the different types of dimensions.
  - *How many linear vertical dimensions are there?*

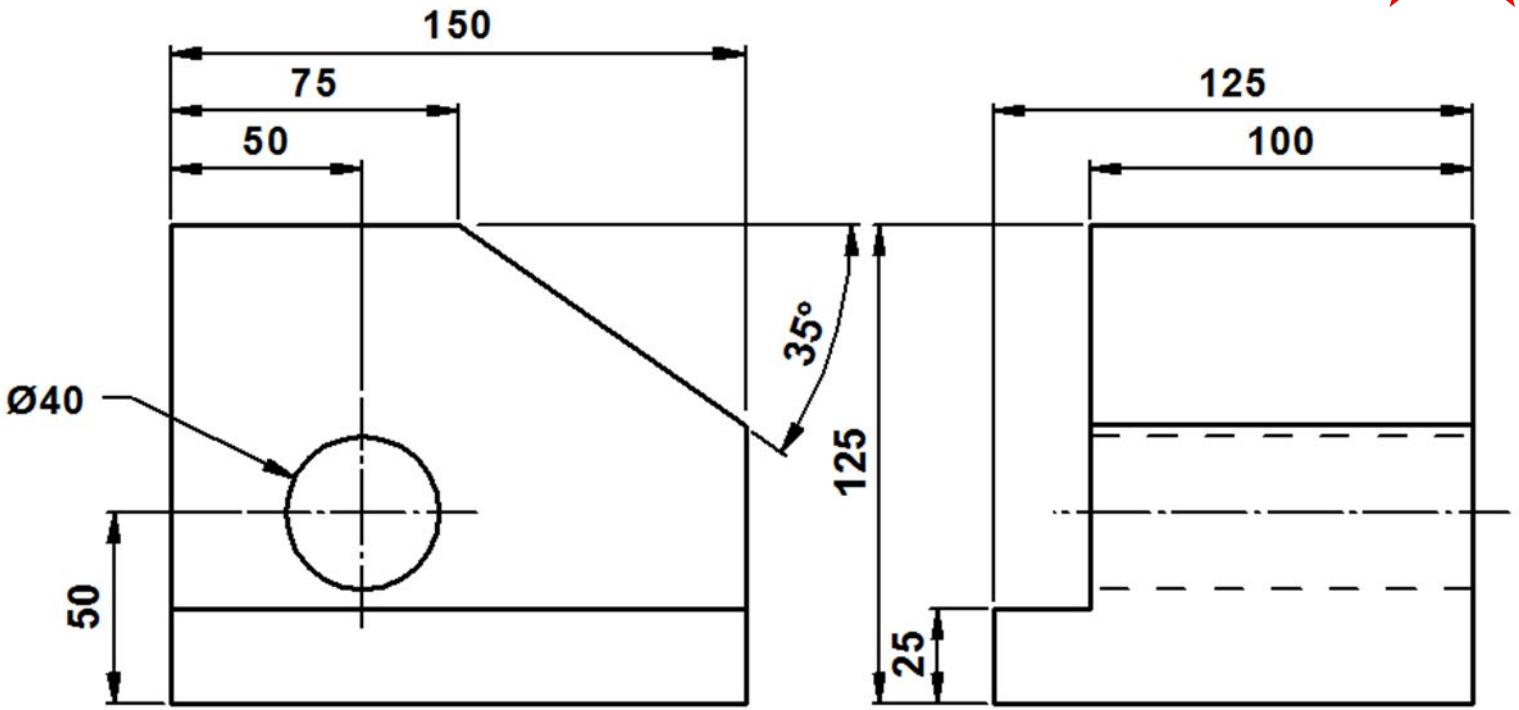
3



Exercise 2-1

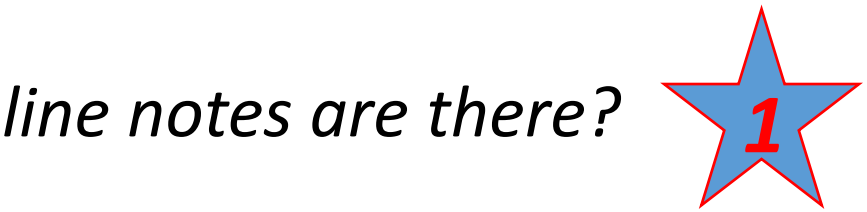
- Count the different types of dimensions.
  - *How many angular dimensions are there?*

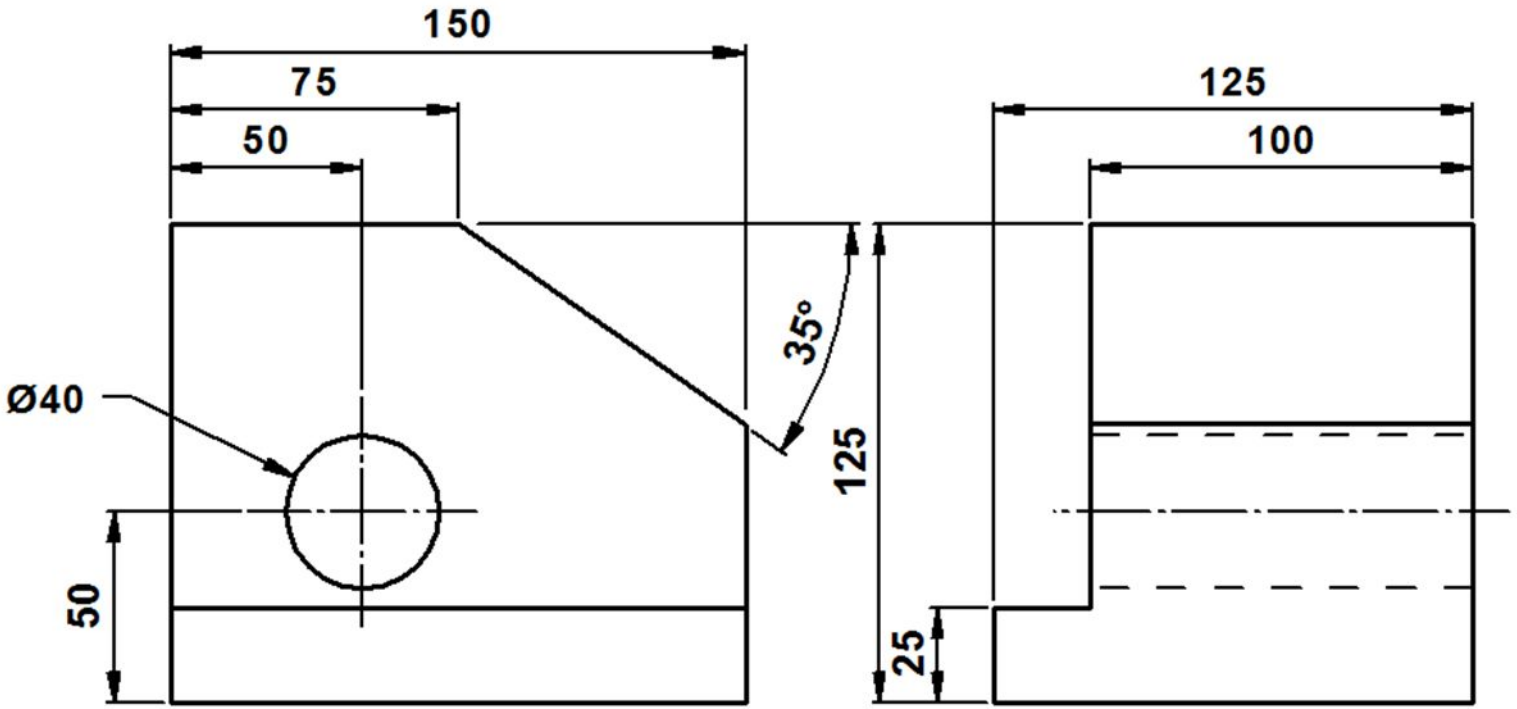




Exercise 2-1

- Count the different types of dimensions.
  - *How many leader line notes are there?*





Lettering

- Lettering should be legible, easy to read, and uniform throughout the drawing.
  - *Upper case letters should be used for all lettering unless a lower case is required.*
  - *The minimum lettering height is **3mm**.*

Dimensioning Symbols

- Dimensioning symbols replace text.
  - *The goal of using dimensioning symbols is to eliminate the need for language translation.*
- Why is it important to use symbols.
  - *How many products are designed in the UK?*
  - *How many products are manufactured or assembled in the UK?*

Dimensioning Symbols

Symbols and Abbreviations

AF	Across flats
ASSY	Assembly
CRS	Centres
CL	Centre line

CBORE	Counter bore
CYL	Cylinder or cylindrical
20°	Degree symbol
DIA	Diameter (in a note)
Ø	Diameter (preceding a dim)
DRG	Drawing
EQUI SP	Equally spaced
EXT	External
FIG.	Figure

CHAM	Chamfer
CH HD	Cheese head
CSK	Countersunk
CSK HD	Countersunk head

NO.	Number
NTS	Not to scale
PCD	Pitch circle diameter
R	Radius (preceding a dim)
RAD	Radius (in a note)
RH	Right hand
RD HD	Round head
SCR	Screw or screwed
SPHERE	Spherical

HEX	Hexagon
HEX HD	Hexagon head
I/D	Inside diameter
O/D	Outside diameter
INT	Internal
LH	Left hand
LG	Long
√	Machining symbol
M	Metric screw thread
MATL	Material
MAX	Maximum
MIN	Minimum
mm	Millimetre

SØ	Spherical diameter (only preceding a dimension)
SR	Spherical radius (only preceding a dimension)
SFACE	Spotface
SQ	Square (in a note)
□	Square (preceding a dim)
STD	Standard
THD	Thread
THK	Thick
TYP	Typical or typically
UCT	Undercut
VOL	Volume
WT	Weight

Dimensioning Rules

Spacing and Readability

1. Dimensions should be easy to read and minimise the possibility for conflicting interpretations.
2. The spacing between dimension lines should be uniform throughout the drawing.
3. Do not dimension inside an object or have the dimension line touch the object unless clearness is gained.
4. Dimension text should be place above the dimension line for horizontal dimensions. Vertical dimension text is placed vertically above the dimension line on the LHS. This is so they can be read from the bottom of the drawing.
5. Dimension text should not cross dimension, extension or visible lines.

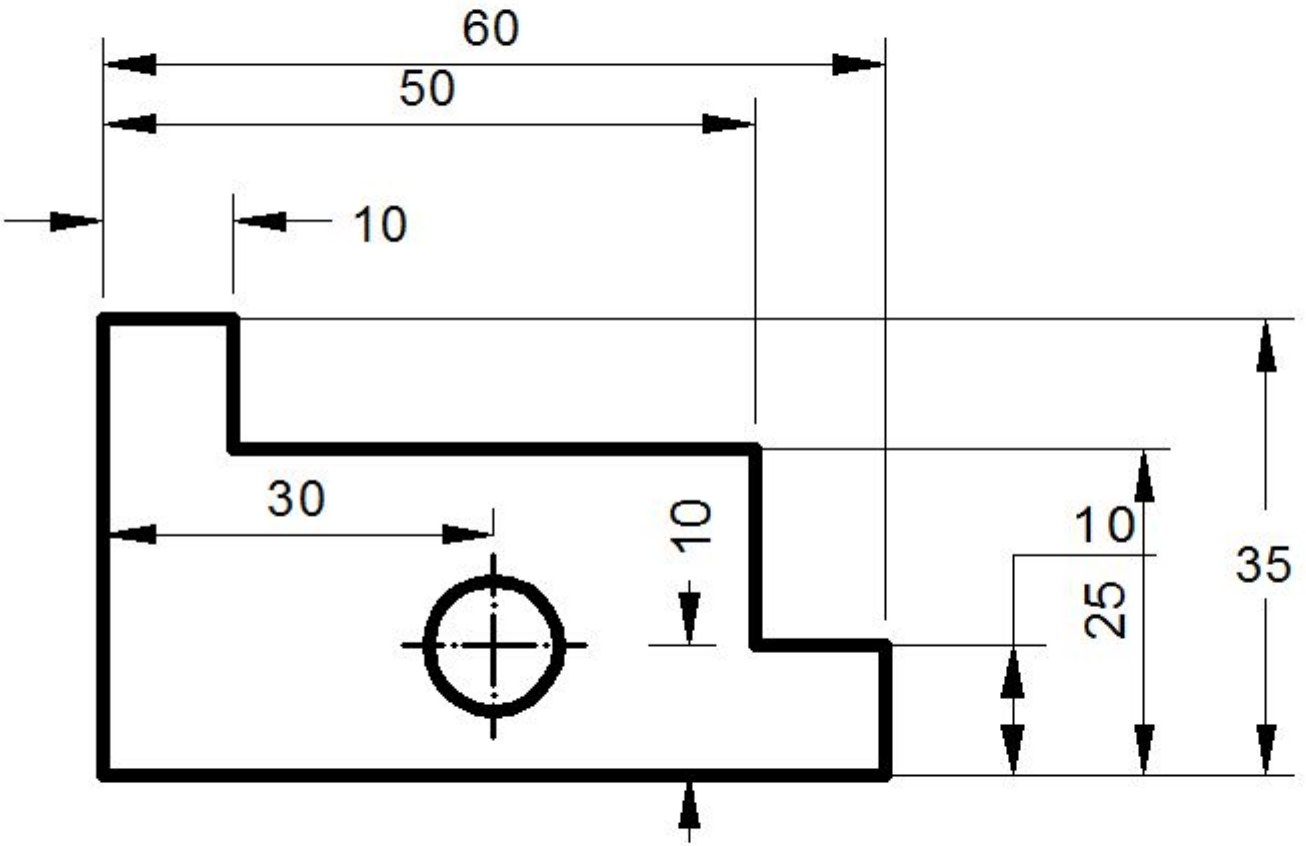


Exercise 2-2

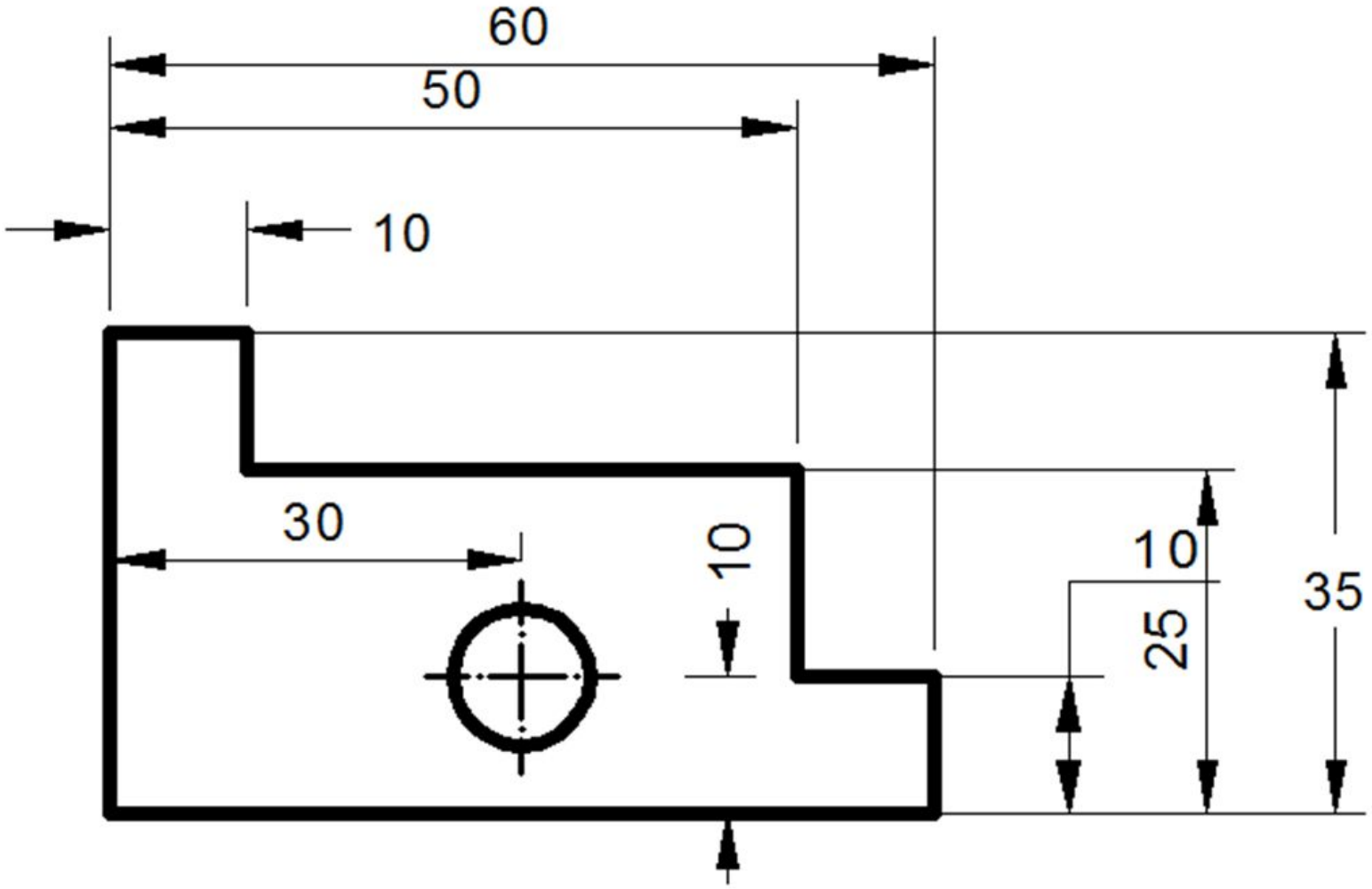
Spacing and Readability 1

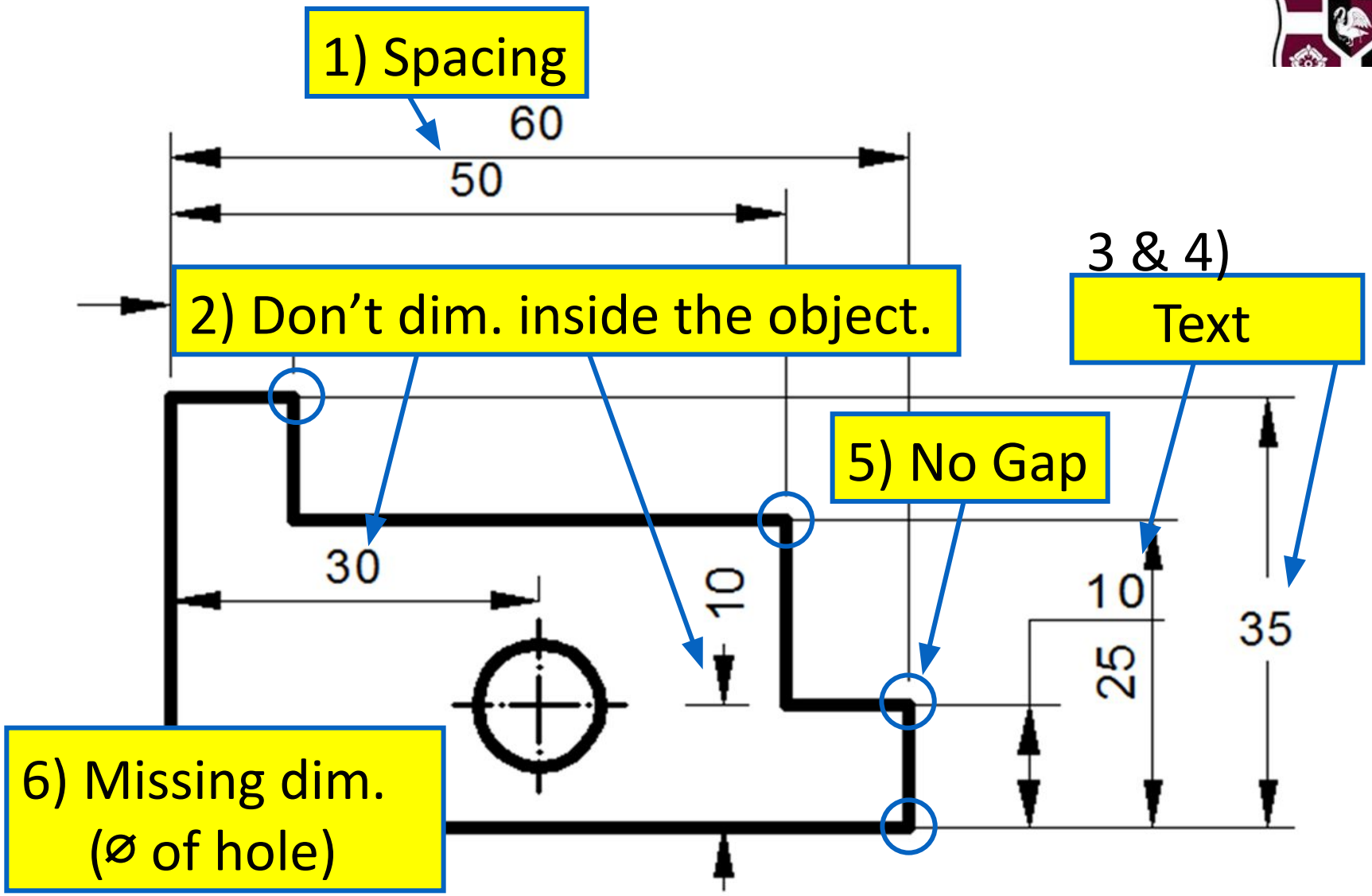
Exercise 2-2

- List the dimensioning mistakes and then dimension the object correctly.

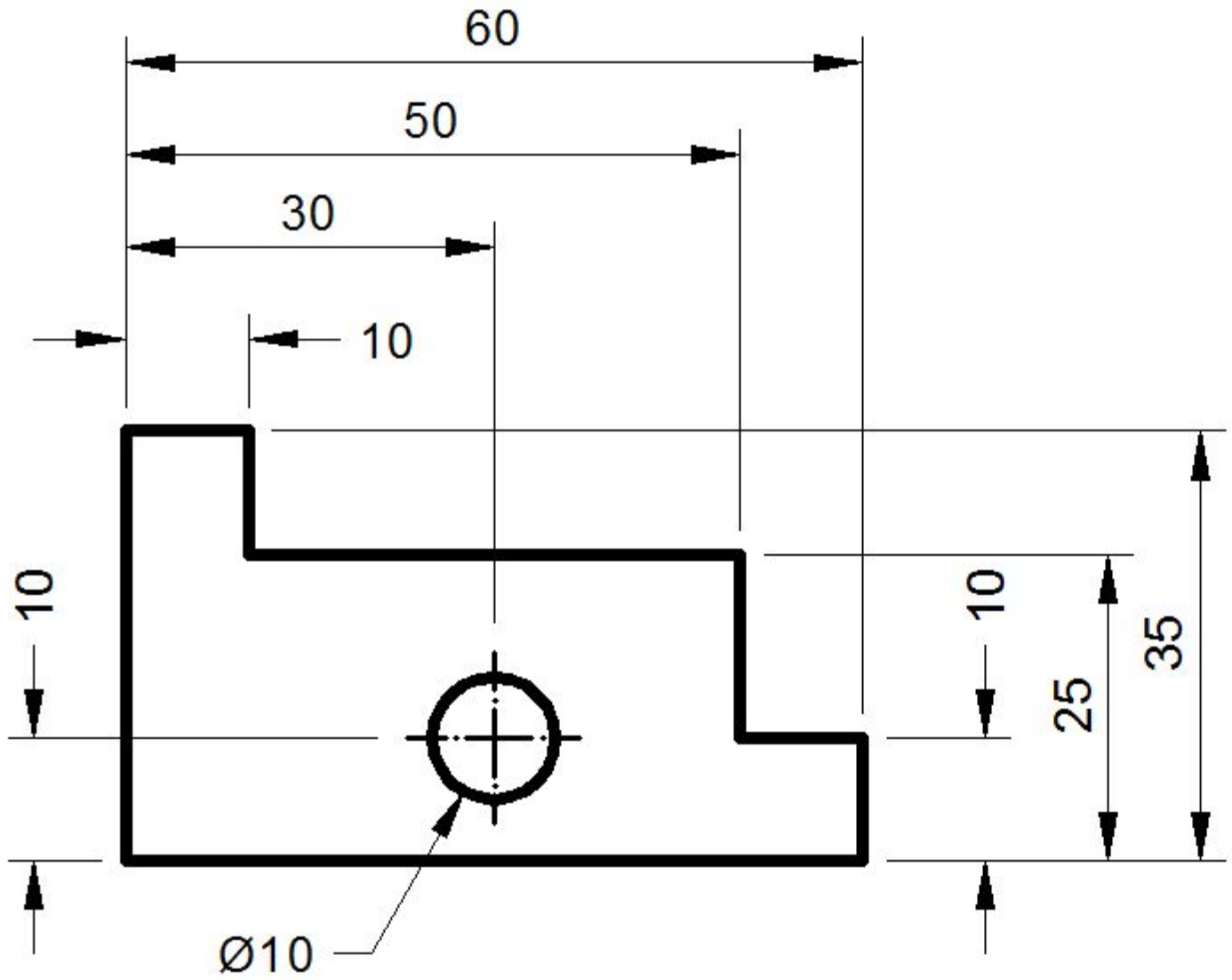


What are the 6 dimensioning mistakes?




1) Spacing
60
50
2) Don’t dim. inside the object.
3 & 4)
Text
5) No Gap
30
10
10
25
35
6) Missing dim.
(⌀ of hole)

Correctly Dimensioned



Dimensioning Rules

Spacing and Readability (Cont…)

6. Dimension lines should not cross extension lines or other dimension lines.
   - *Extension lines can cross other extension lines or visible lines.*
7. Extension lines and centrelines should not connect between views.
8. Leader lines should be straight, not curved, and point to the centre of the arc or circle at an angle between 30° – 60°.

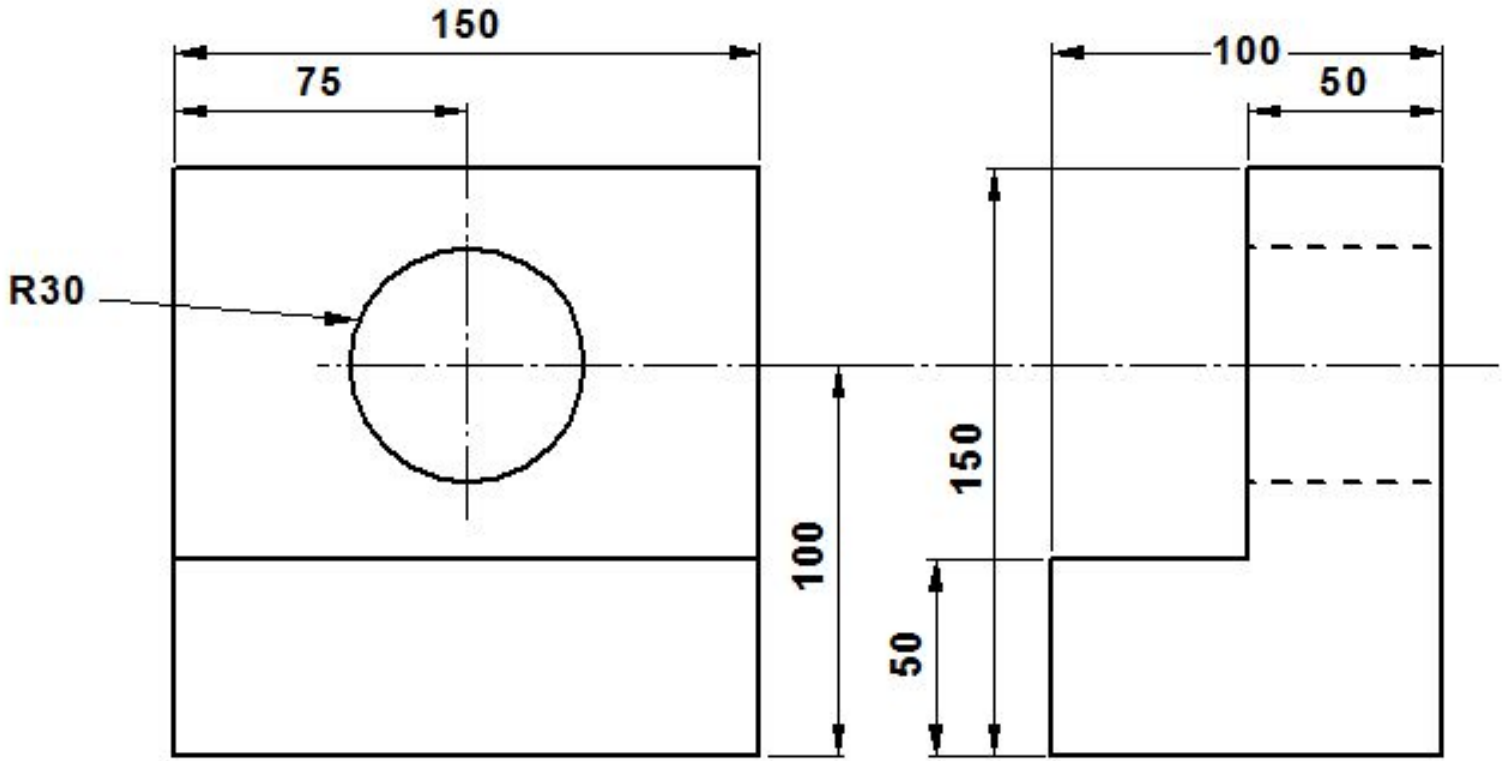


Exercise 2-3

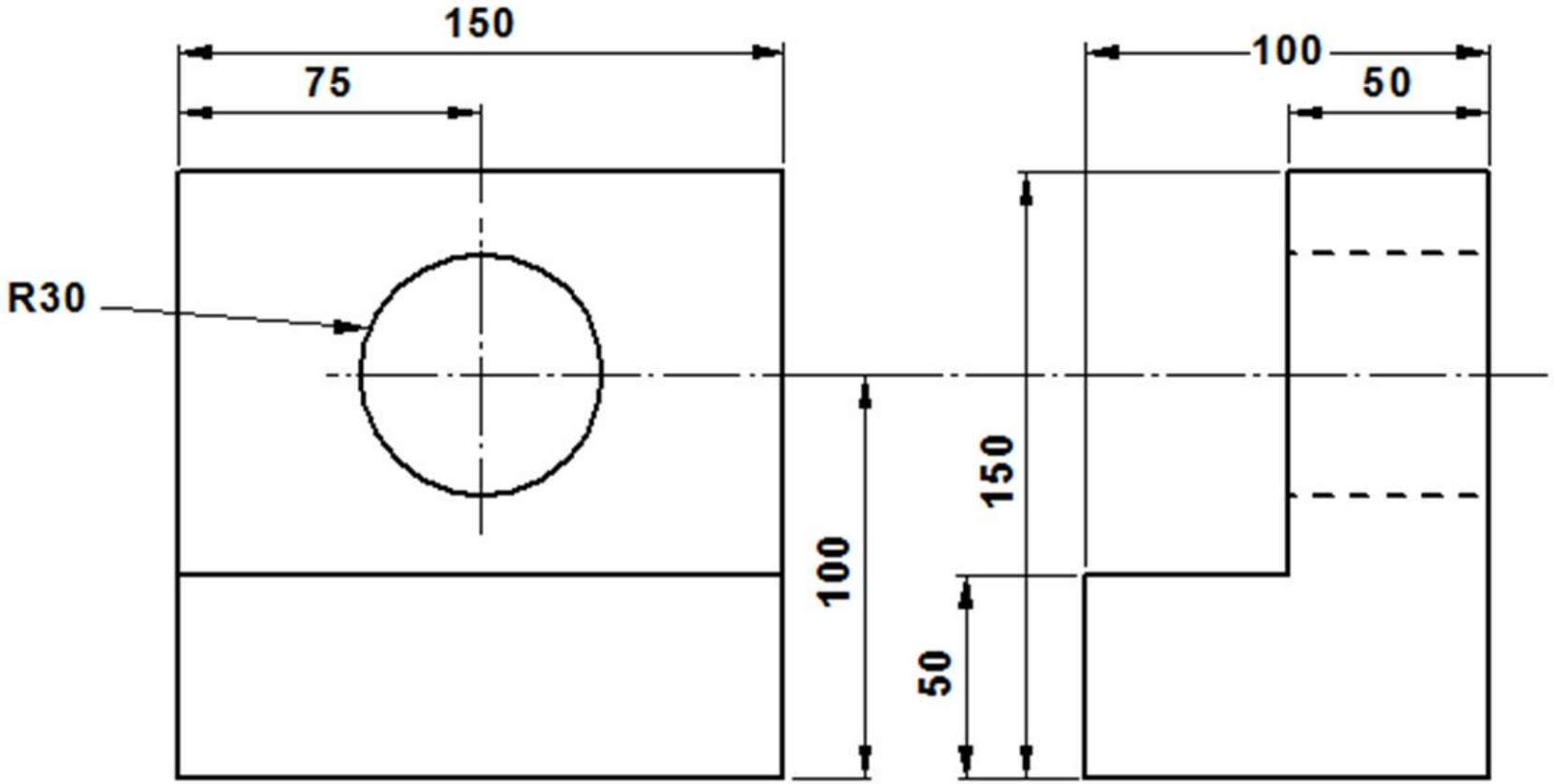
Spacing and Readability 2

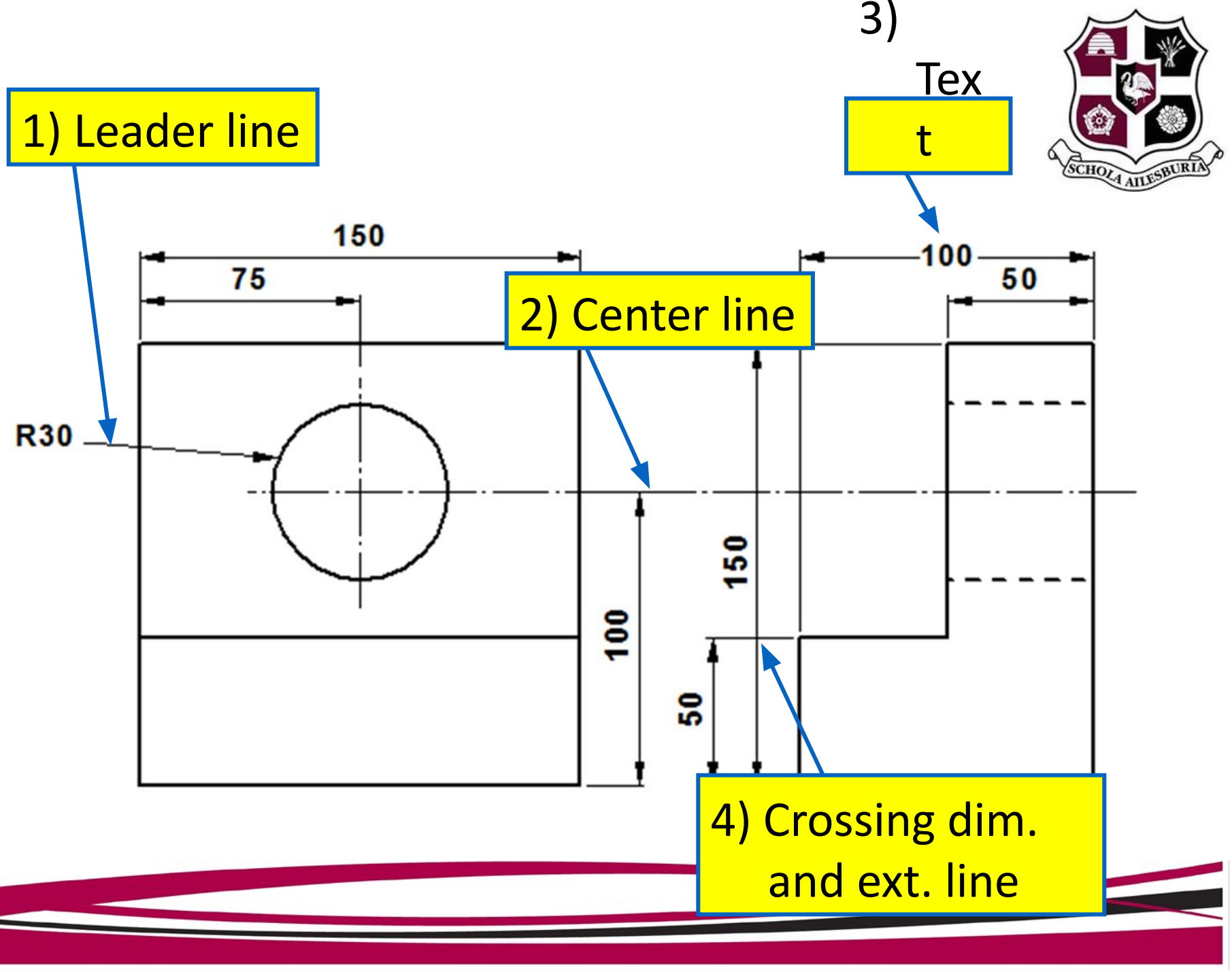
Exercise 2-3

- List the dimensioning mistakes and then dimension the object correctly.

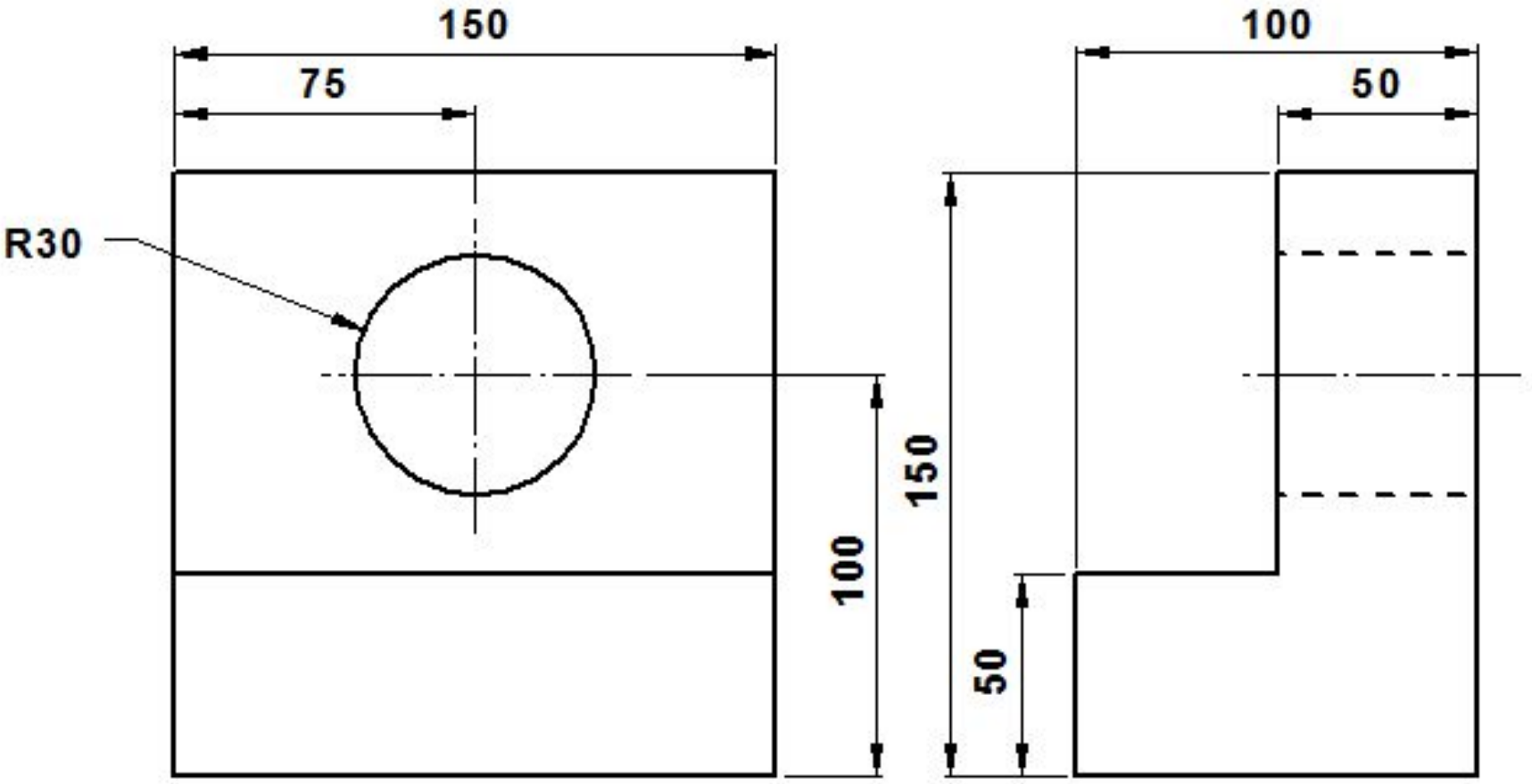


What are the 4 dimensioning mistakes?




3)
Tex
t
1) Leader line
150
75
100
50
2) Center line
R30
100
150
50
4) Crossing dim.
and ext. line

Correctly Dimensioned



Dimensioning Rules

Spacing and Readability (Cont…)

9. Dimensions should not be duplicated, or the same information given in two different ways.
   - *If a reference dimension is used, the size value is placed within parentheses (e.g. (10)).*

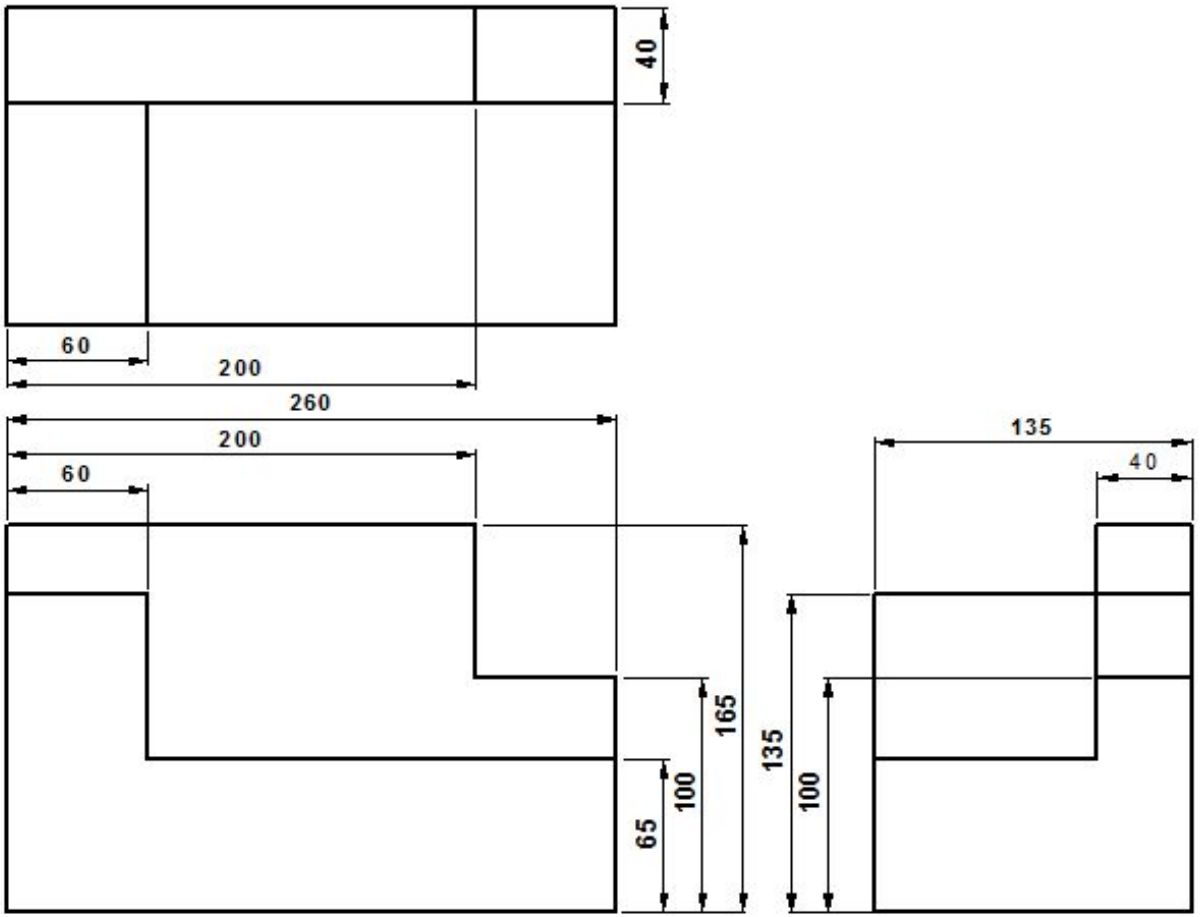


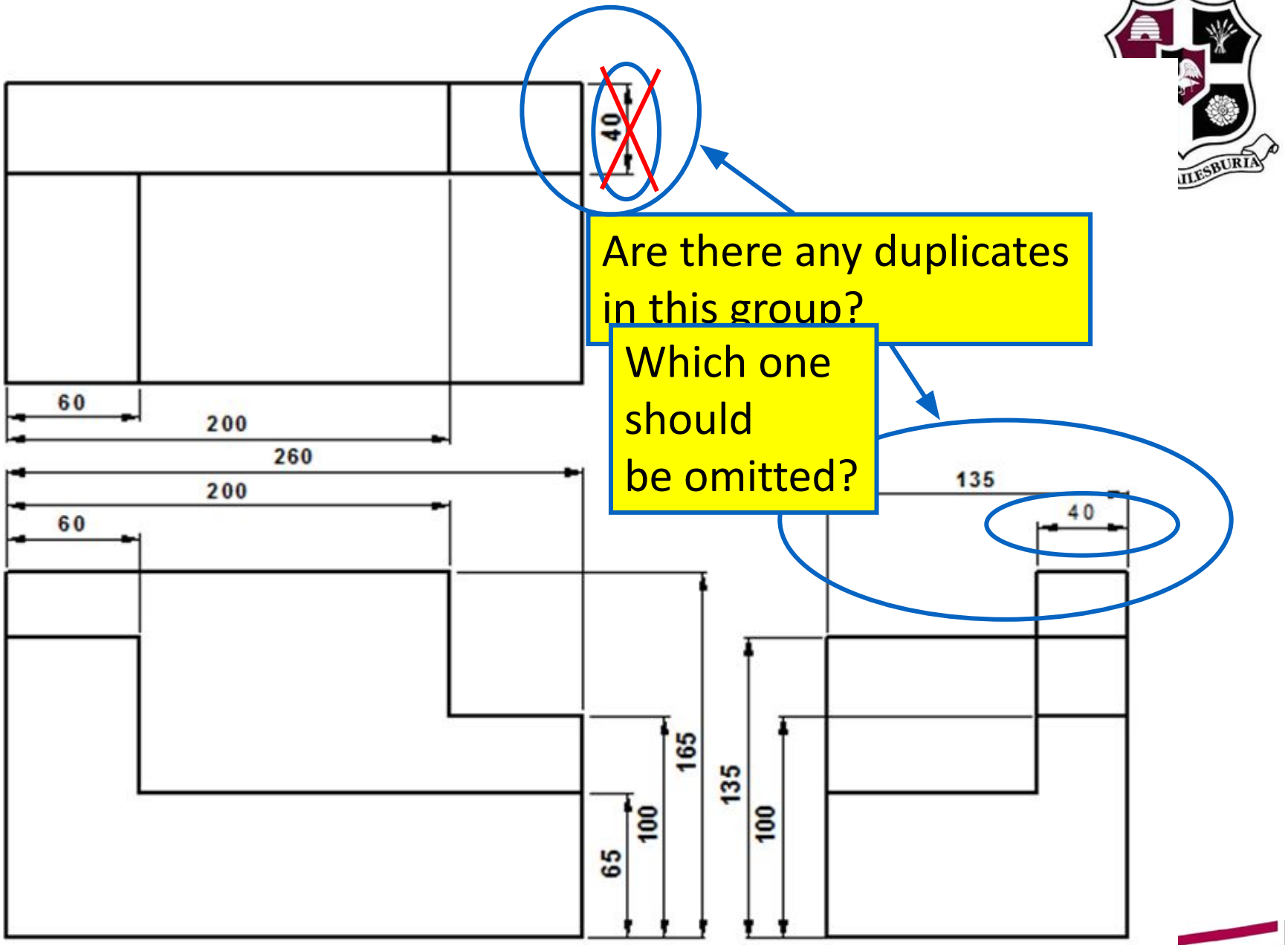
Exercise 2-4

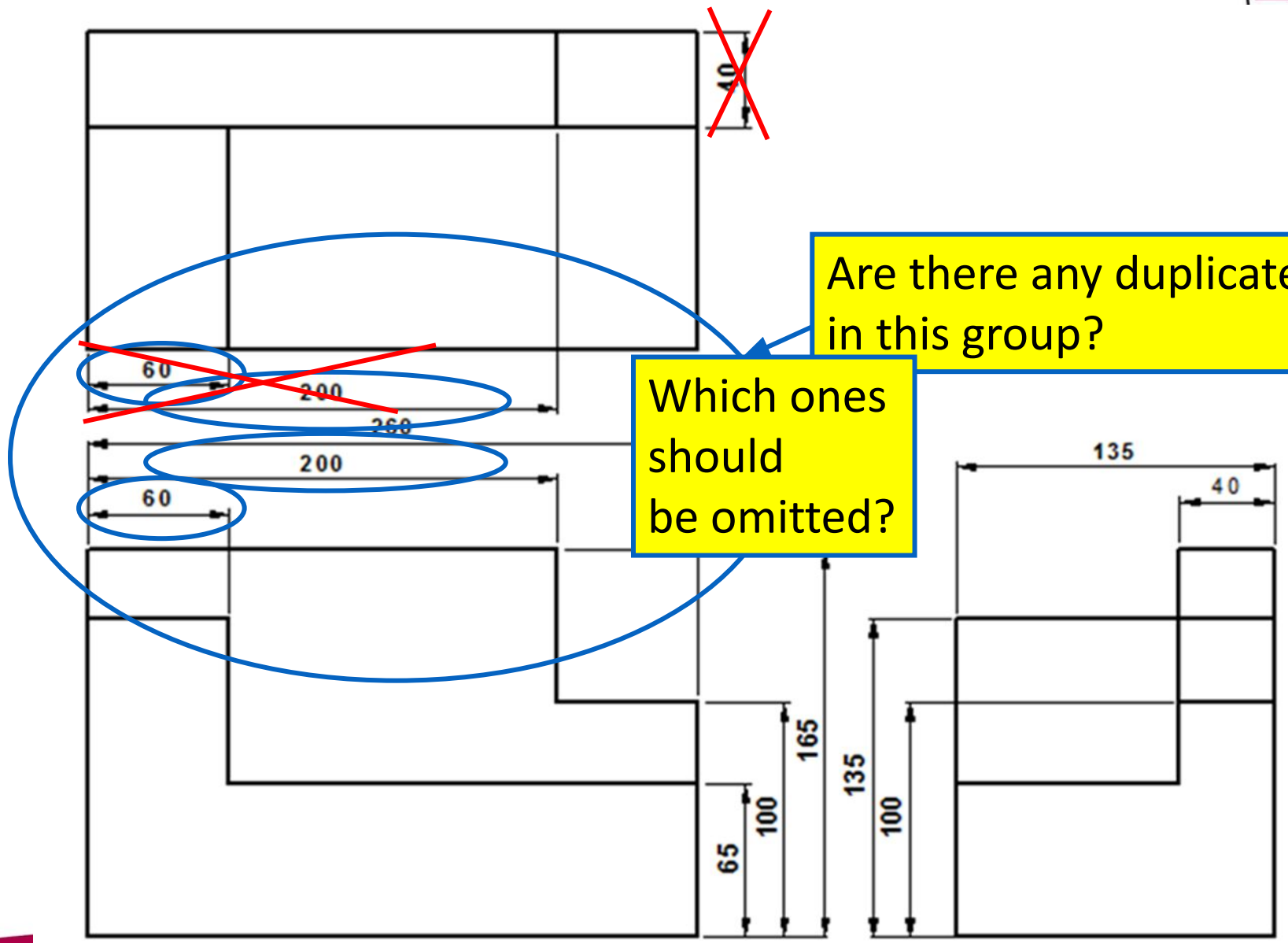
Duplicate dimensions

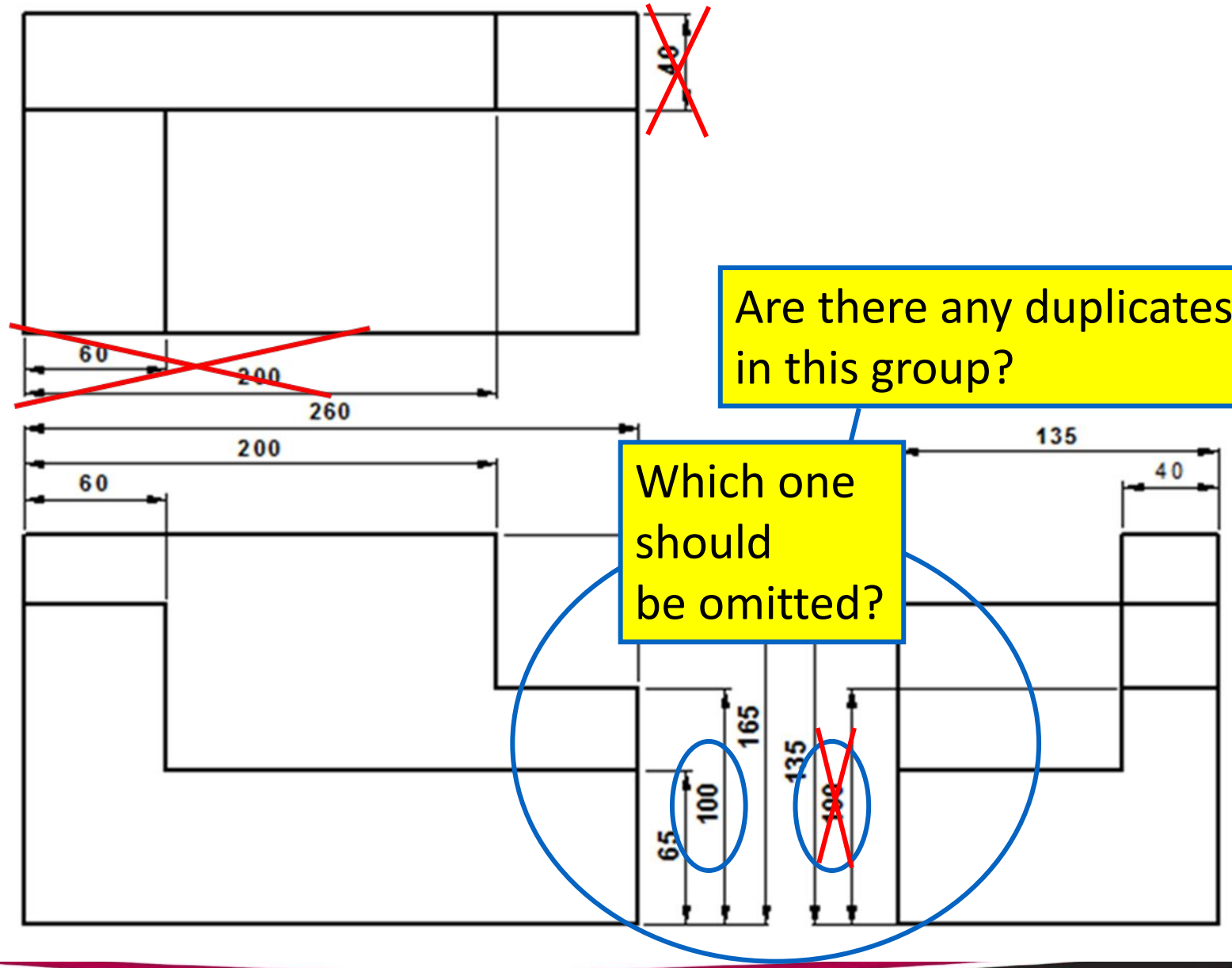
Exercise 2-4

- Find the duplicate dimensions and cross out the ones that you feel should be omitted.




40
Are there any duplicates in this group?
Which one should be omitted?
60
200
260
200
60
135
40
165
100
65
135
100


40
60
200
260
200
60
Are there any duplicates in this group?
Which ones should be omitted?
135
40
165
100
65
135
100


40
60
200
260
200
60
135
40
Are there any duplicates in this group?
Which one should be omitted?
165
100
65
135
100

Dimensioning Rules

Dimension Placement

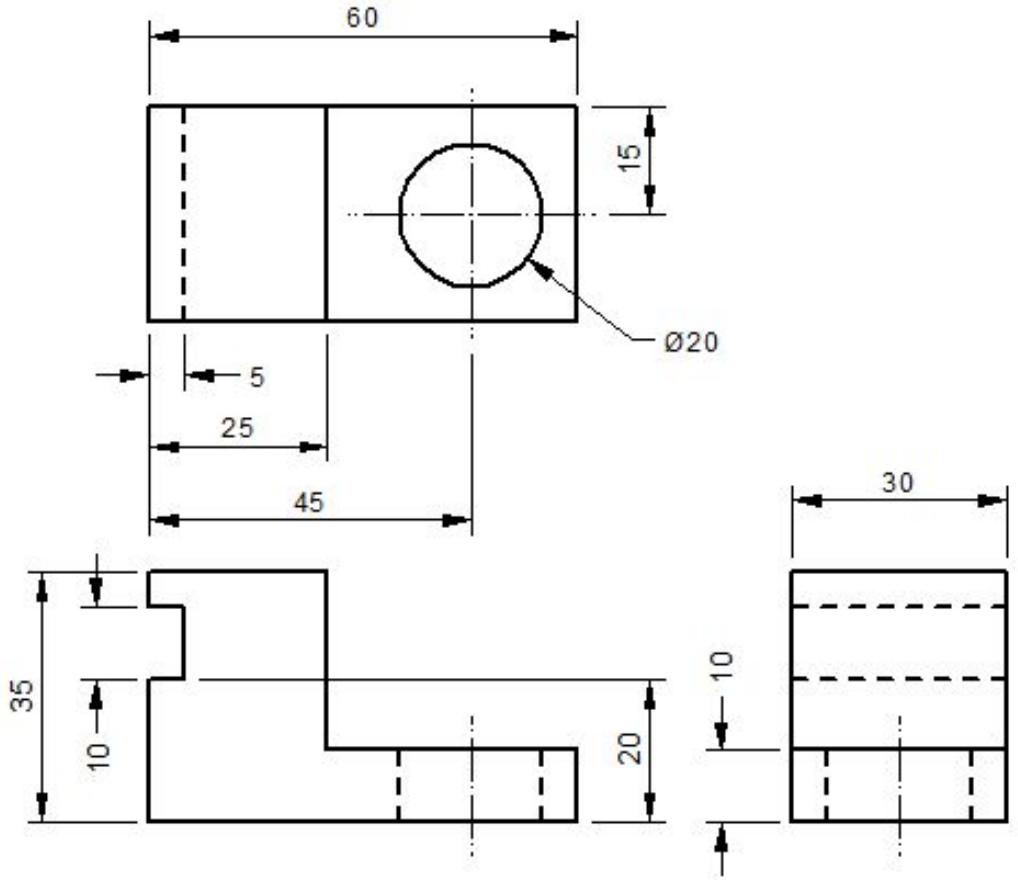
1. Dimensions should be placed in such a way as to enhance the communication of your design.
2. Dimensions should be grouped whenever possible.
3. Dimensions should be placed between views, unless clearness is promoted by placing some outside.
4. Dimensions should be attached to the view where the shape is shown best.
5. Do not dimension hidden lines.

Exercise 2-5

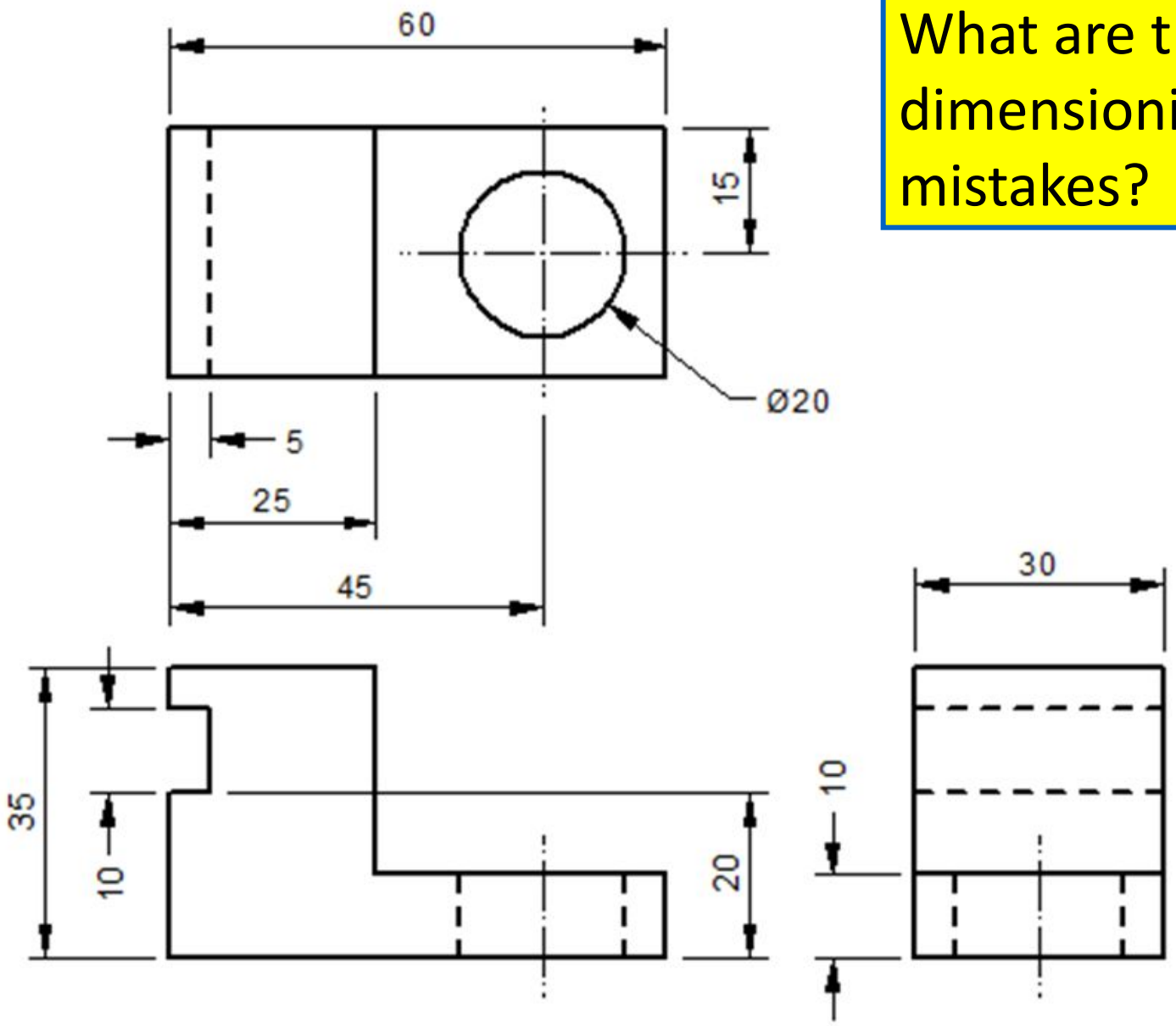
Dimension placement

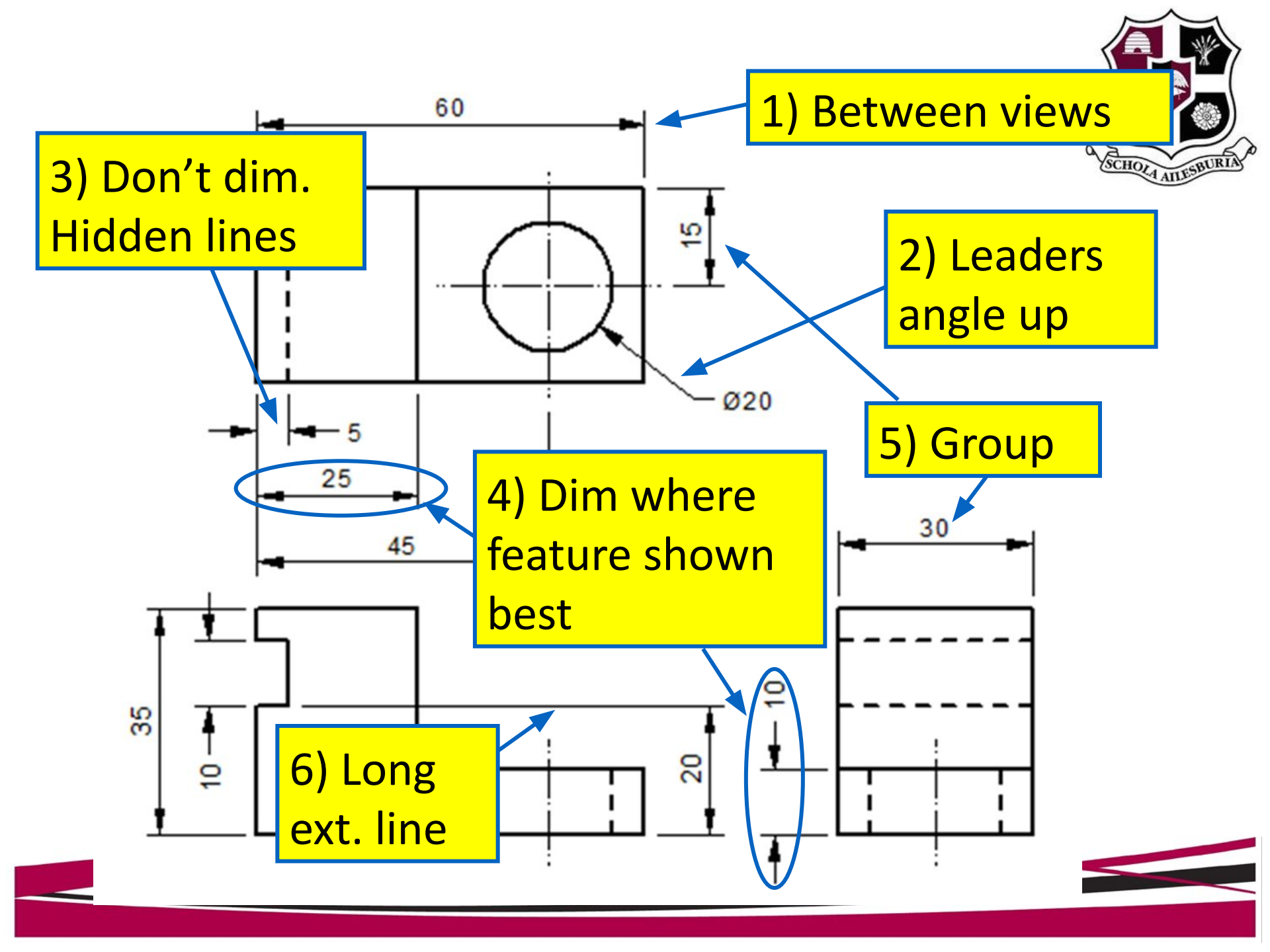
Exercise 2-5

- List the dimensioning mistakes and then dimension the object correctly.

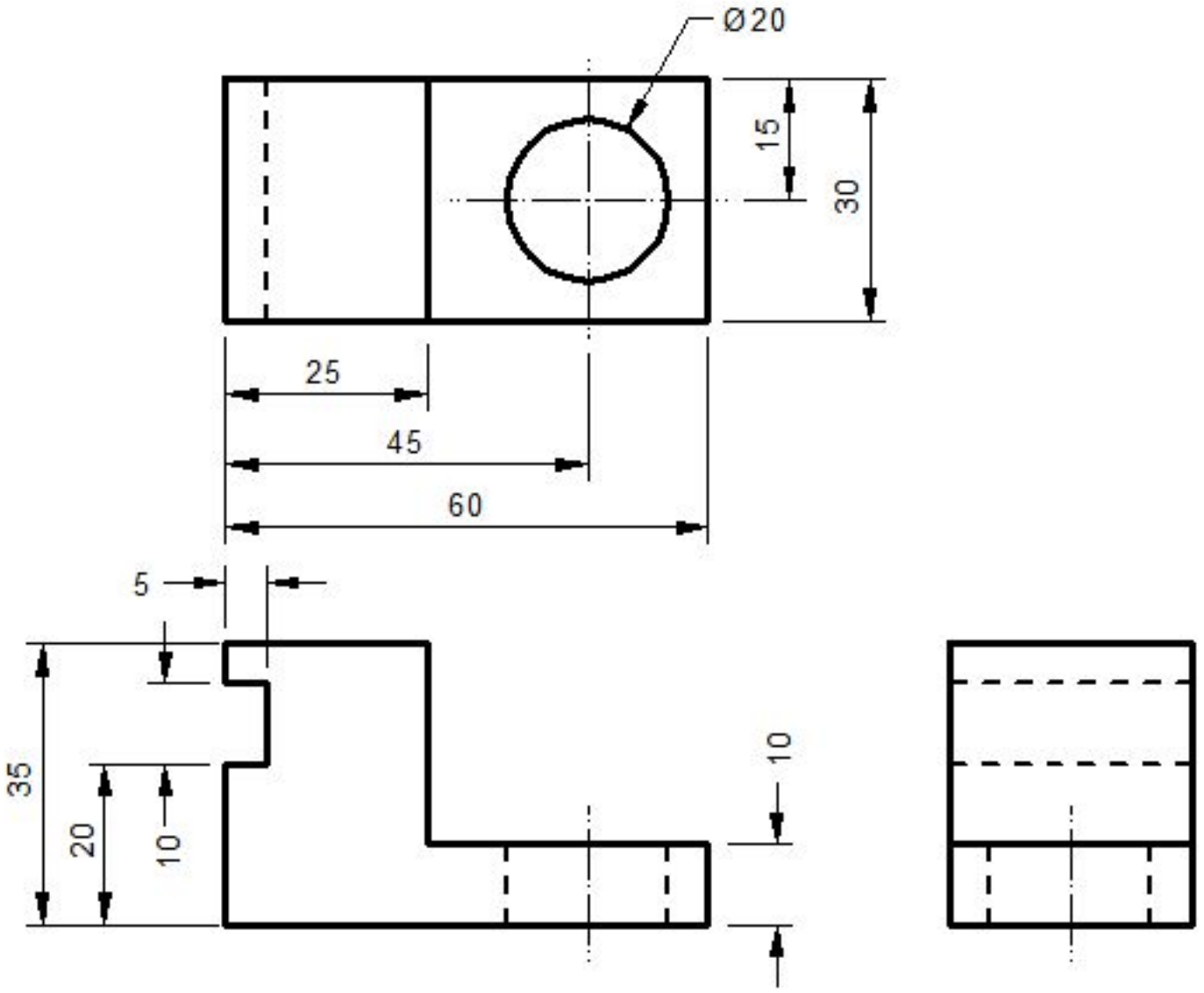


What are the 6 dimensioning mistakes?




1) Between views
3) Don't dim. Hidden lines
2) Leaders angle up
5) Group
4) Dim where feature shown best
6) Long ext. line
60
15
Ø20
5
25
45
30
35
10
20
10

Correctly Dimensioned

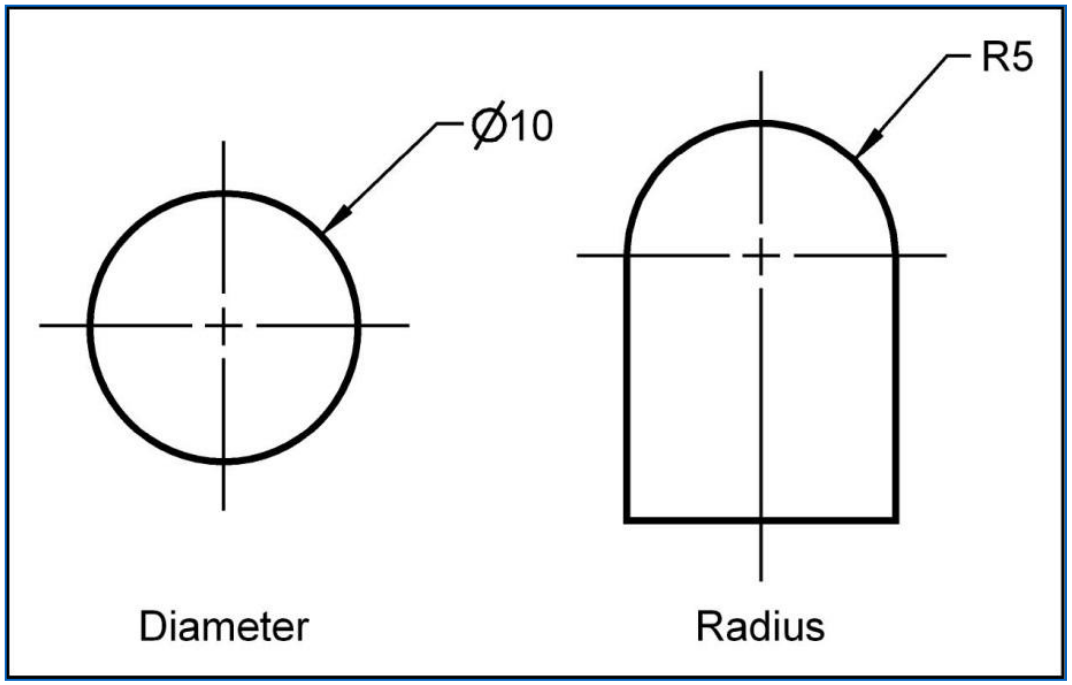


Dimensioning

2-4) Dimensioning and Locating Simple Features

Dimensioning Features

- A circle is dimensioned by its diameter and an arc by its radius using a leader line and a note.

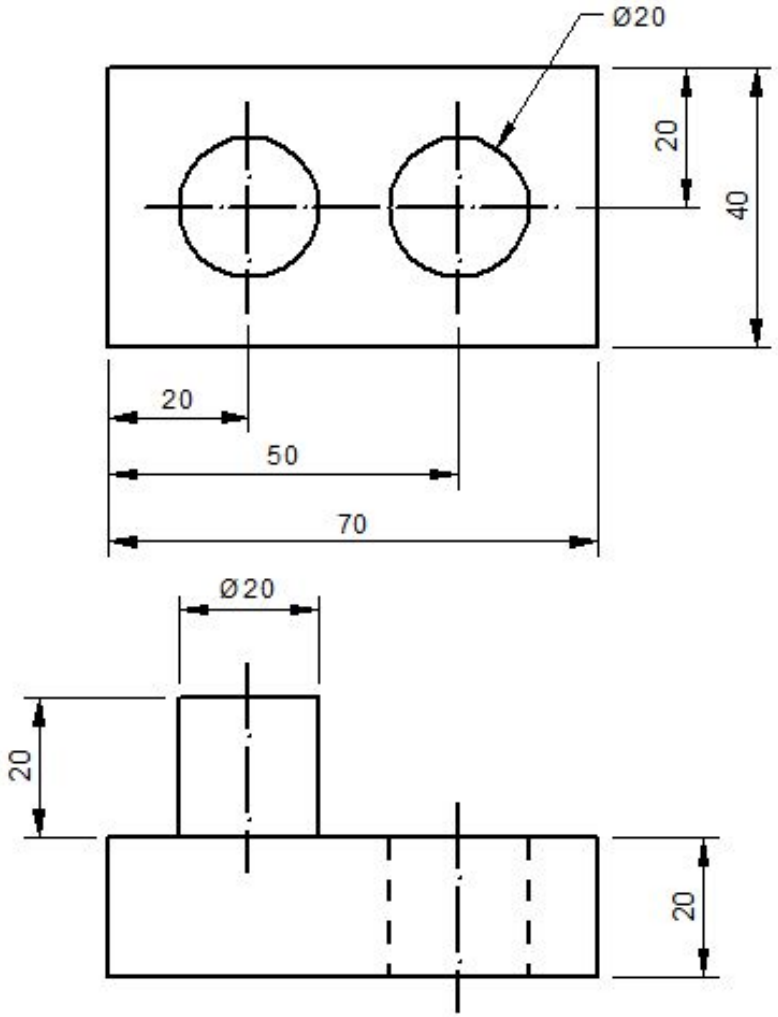


Exercise 2-6

Circular and rectangular views

Exercise 2-6

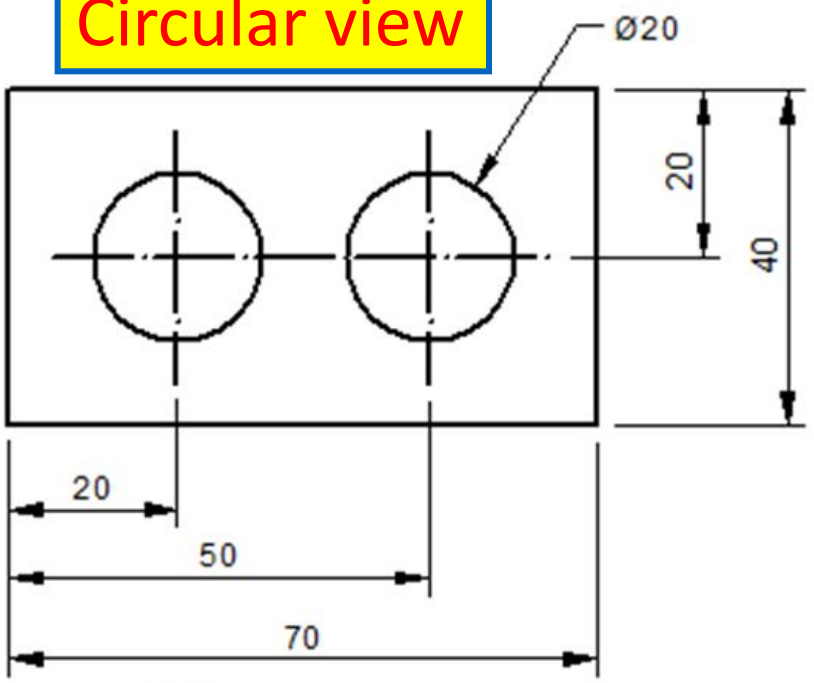
a. Answer questions about the cylindrical and hole features of the part shown.

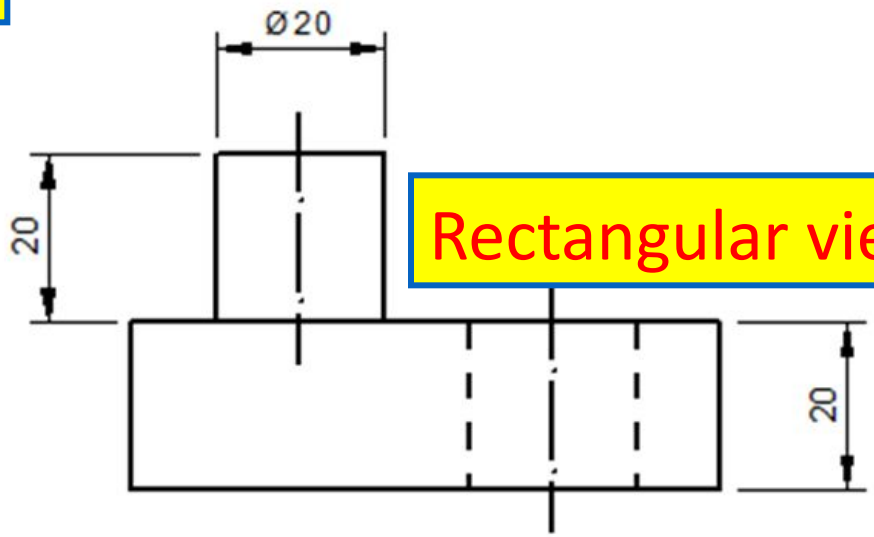


Find the hole and cylinder.

Which view is considered the circular view and which is considered the rectangular view?

Circular view

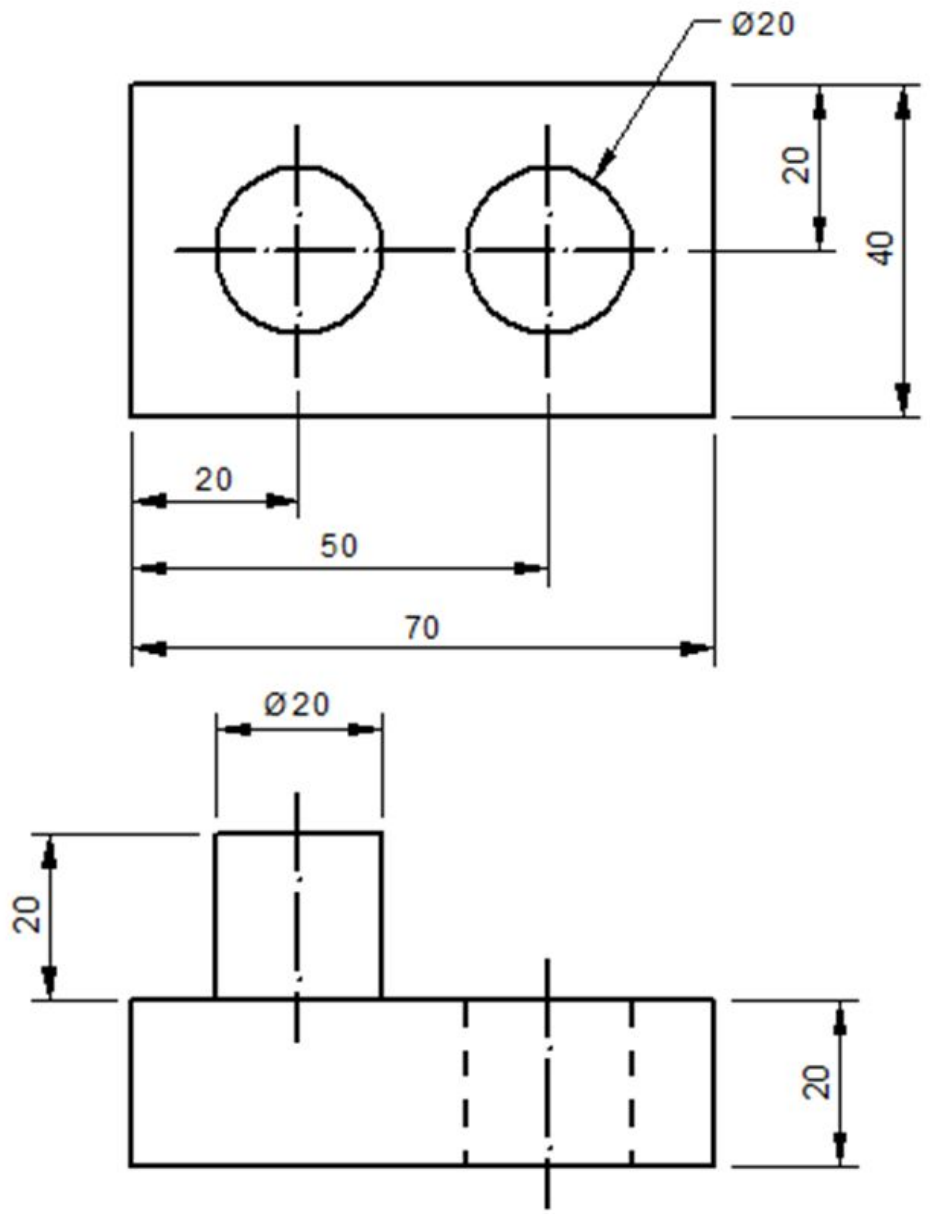




Rectangular view

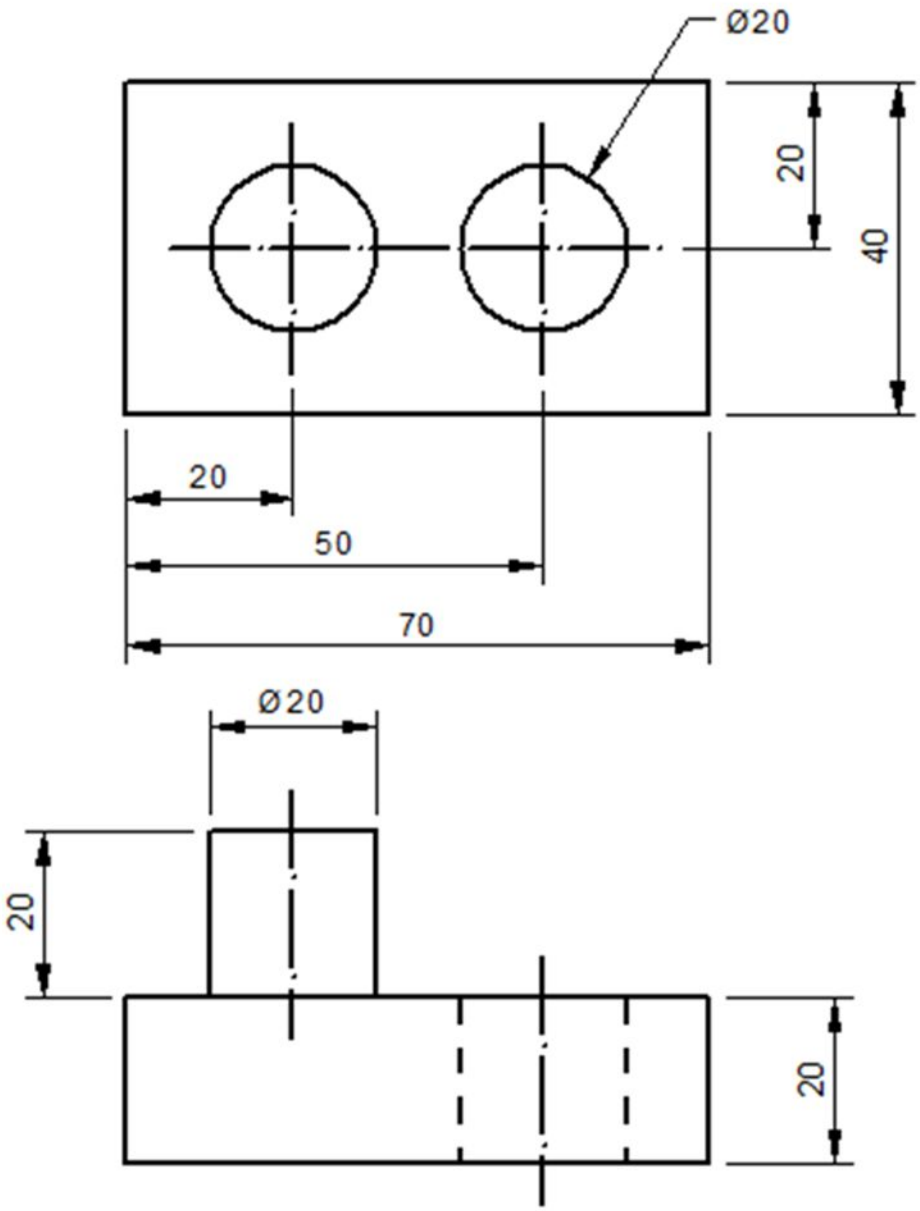
Looking at just the top view, can you tell the difference between the hole and the cylinder?

No



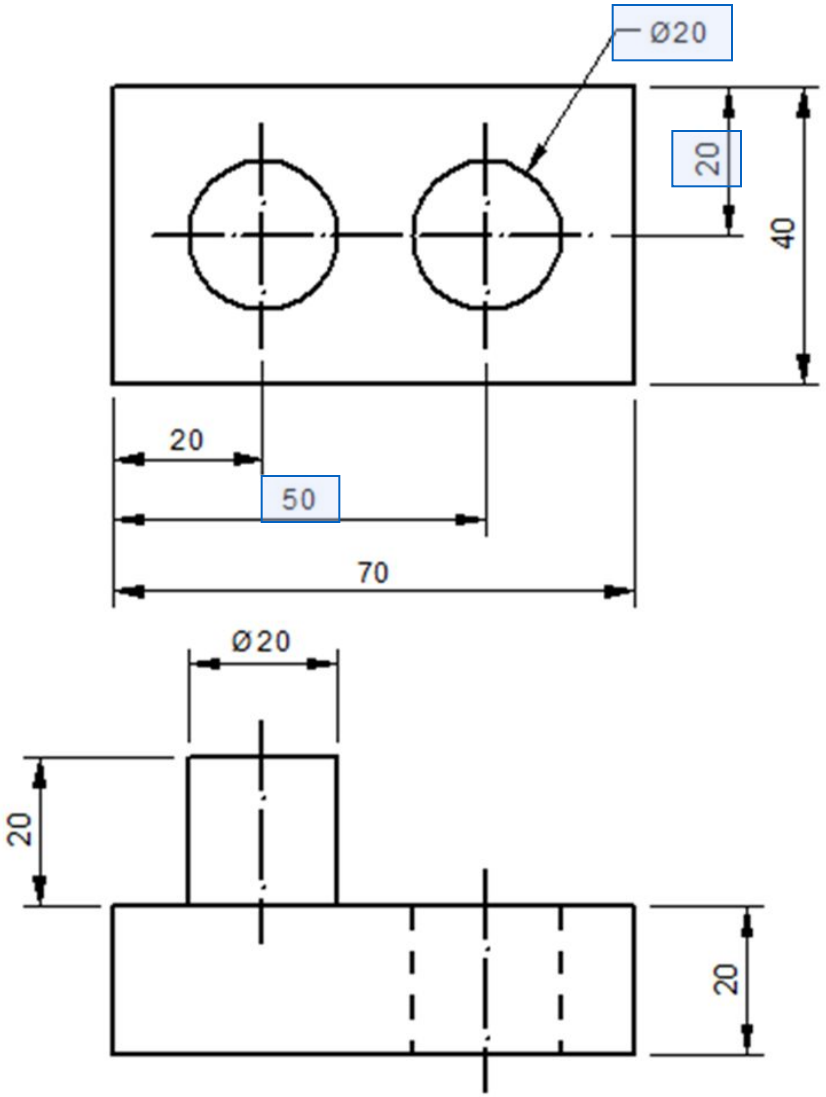
Why is the diameter of the hole given in the circular view and diameter of the cylinder given in the rectangular view?

So that the cylinder is not confused with a hole.



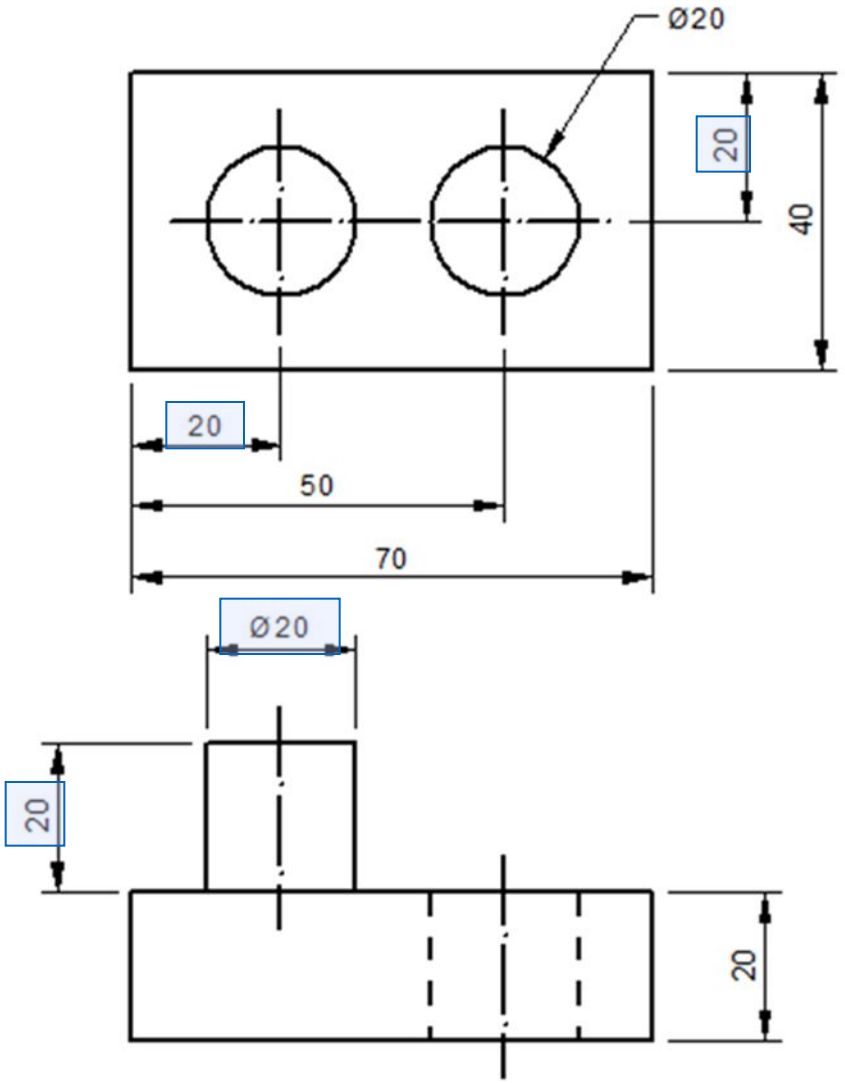
Dimensioning Features

b) Holes are dimensioned by giving their diameter and location in the circular view.



Dimensioning Features

c. A cylinder is dimensioned by giving its diameter and length in the rectangular view and is located in the circular view.



Dimensioning Features

d. Repetitive features or dimensions may be specified by using the symbol “X” along with the number of times the feature is repeated.

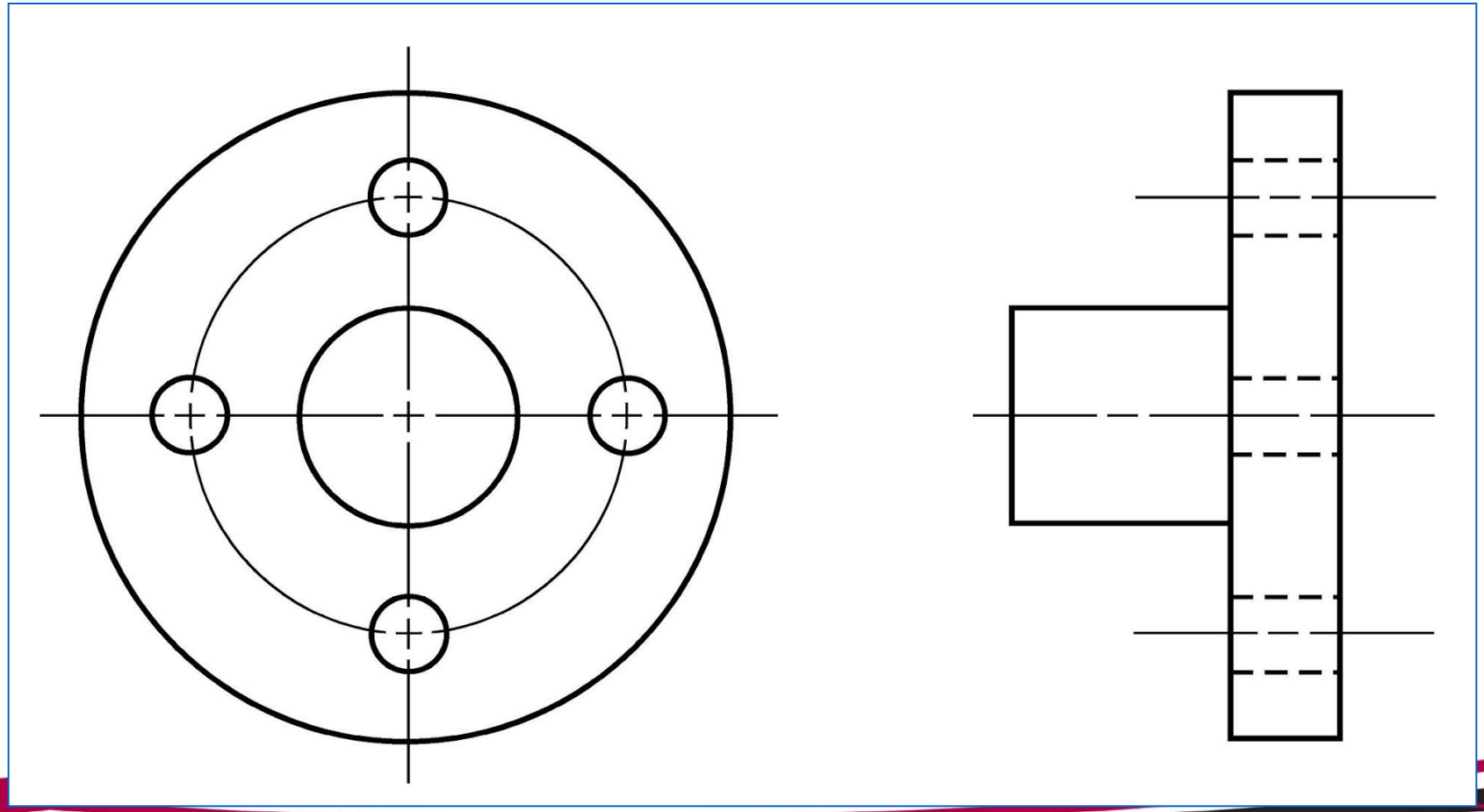
   - *There is no space between the number of times the feature is repeated and the “X” symbol, however, there is a space between the symbol “X” and the dimension.*

Exercise 2-7

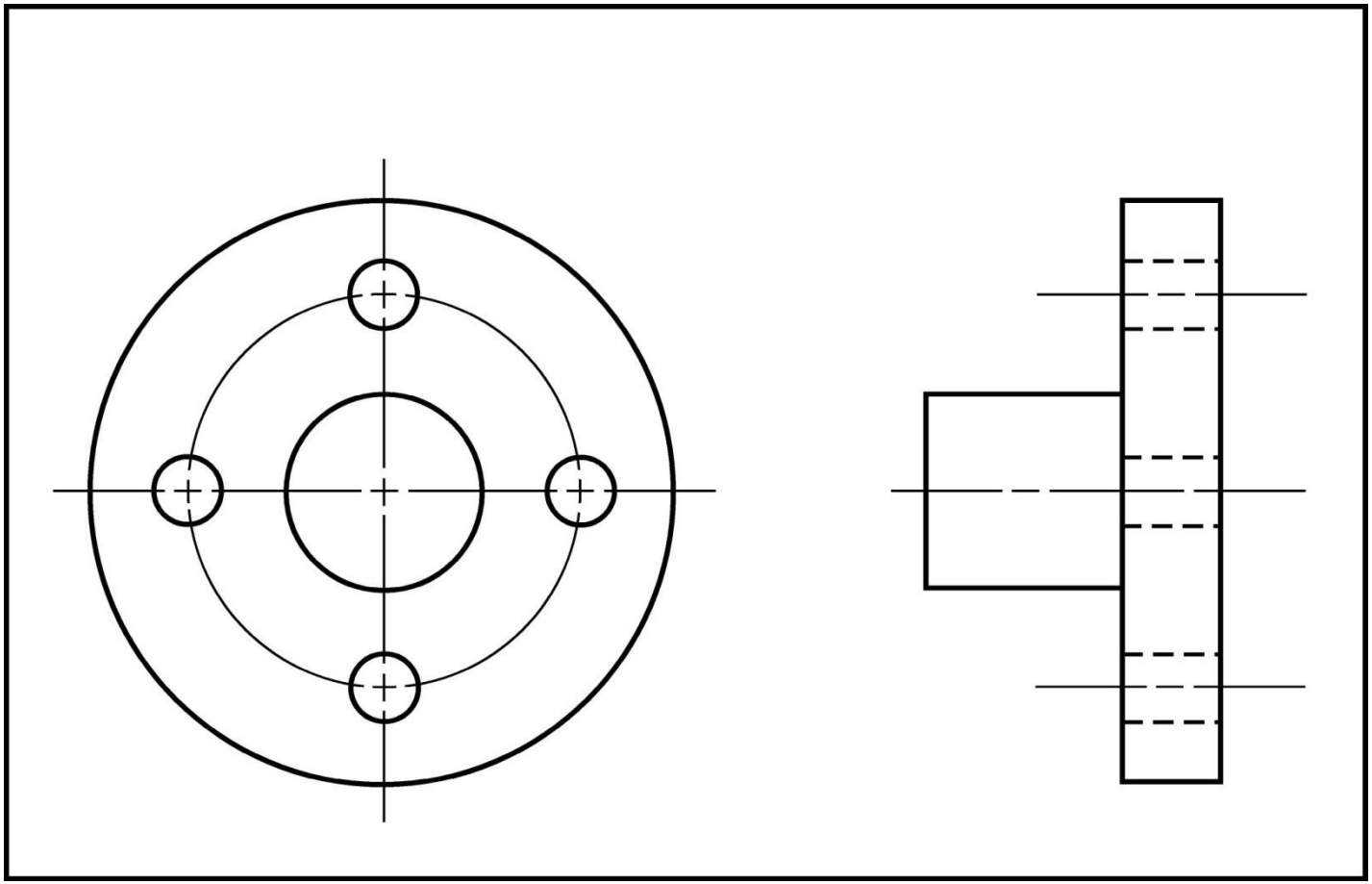
Dimensioning and locating features

Exercise 2-7

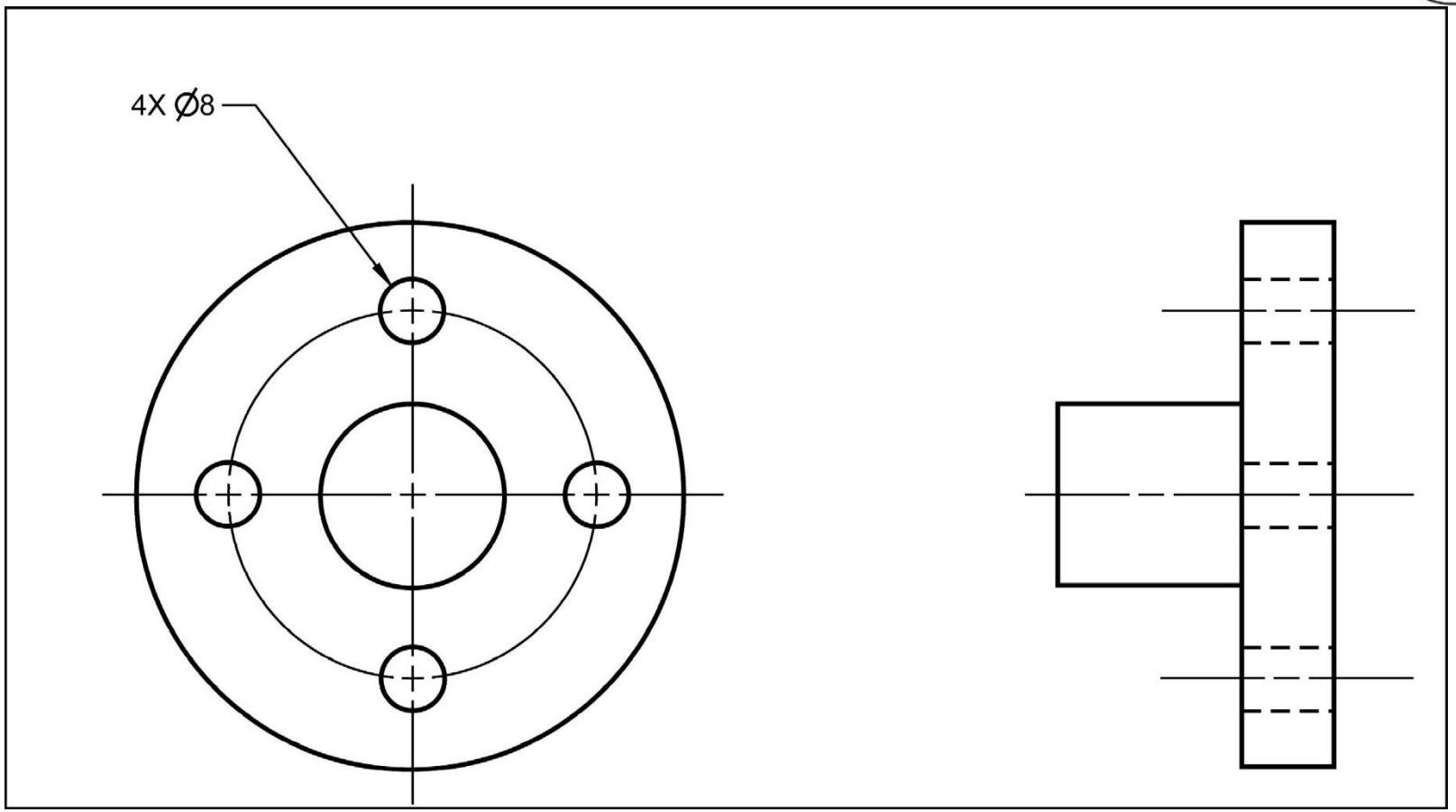
- Dimension the object shown.



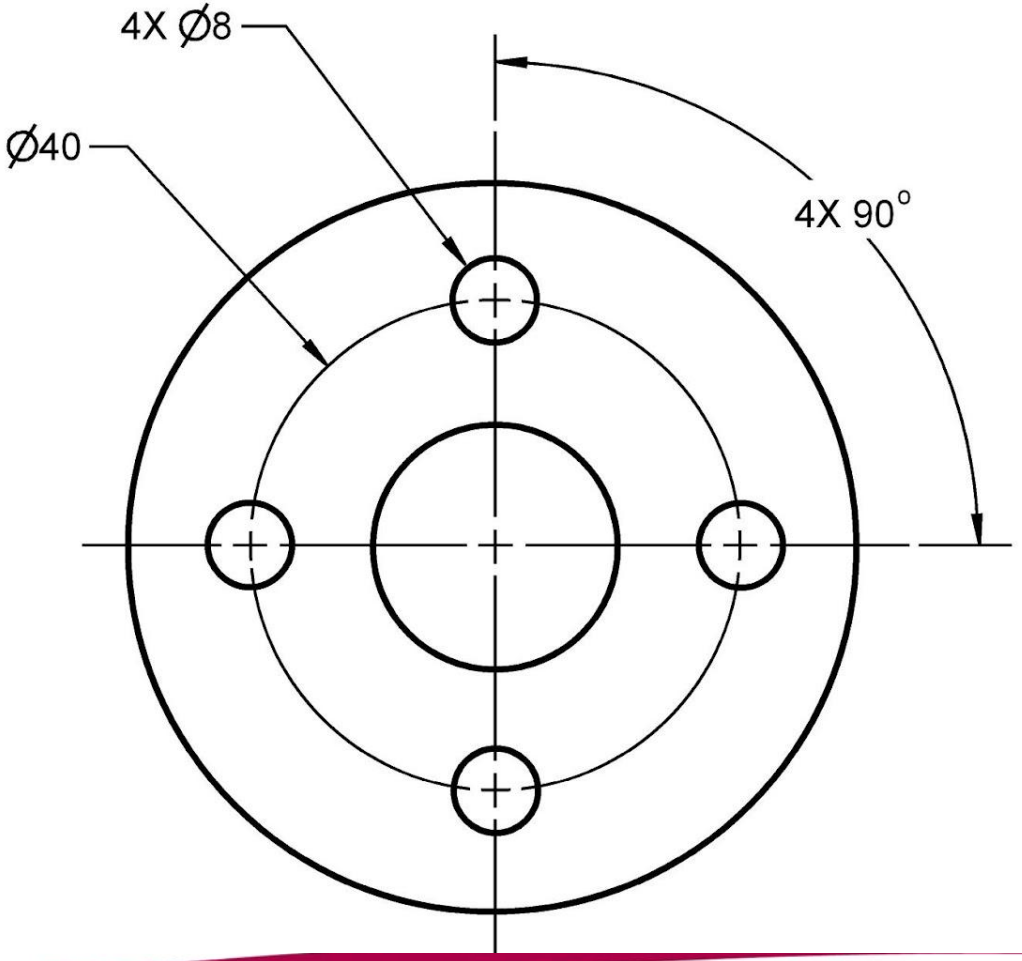
How do we dimension the hole diameters?

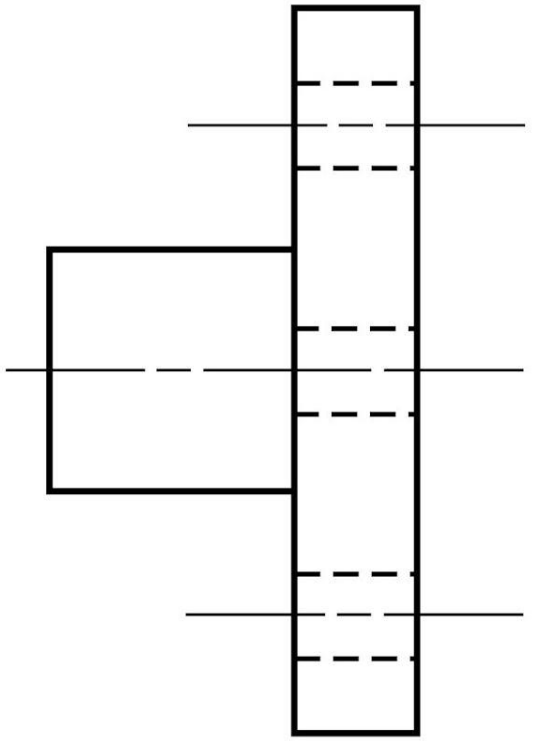


How do we locate the holes?

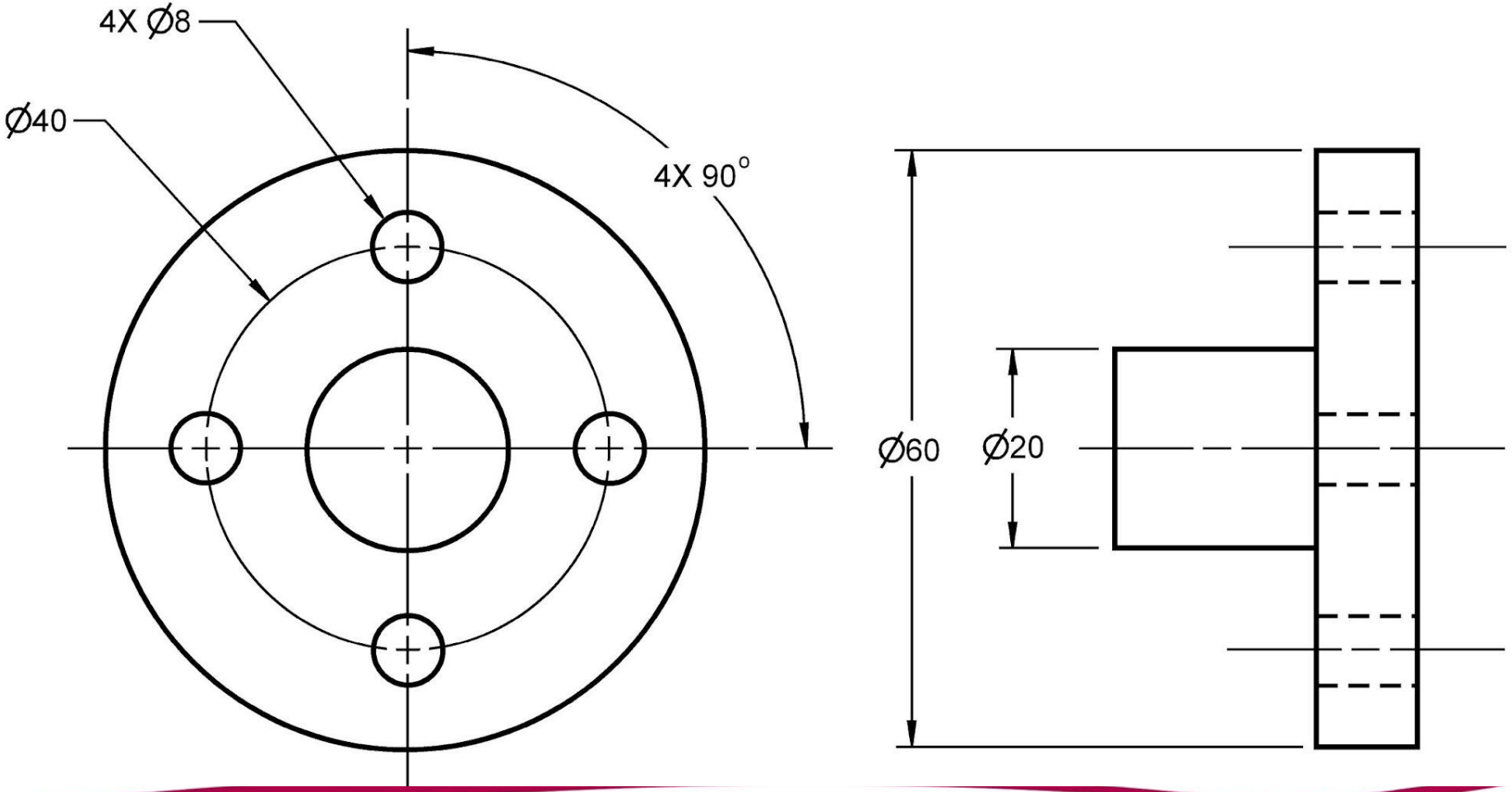


How do we dimension the cylinder diameters?



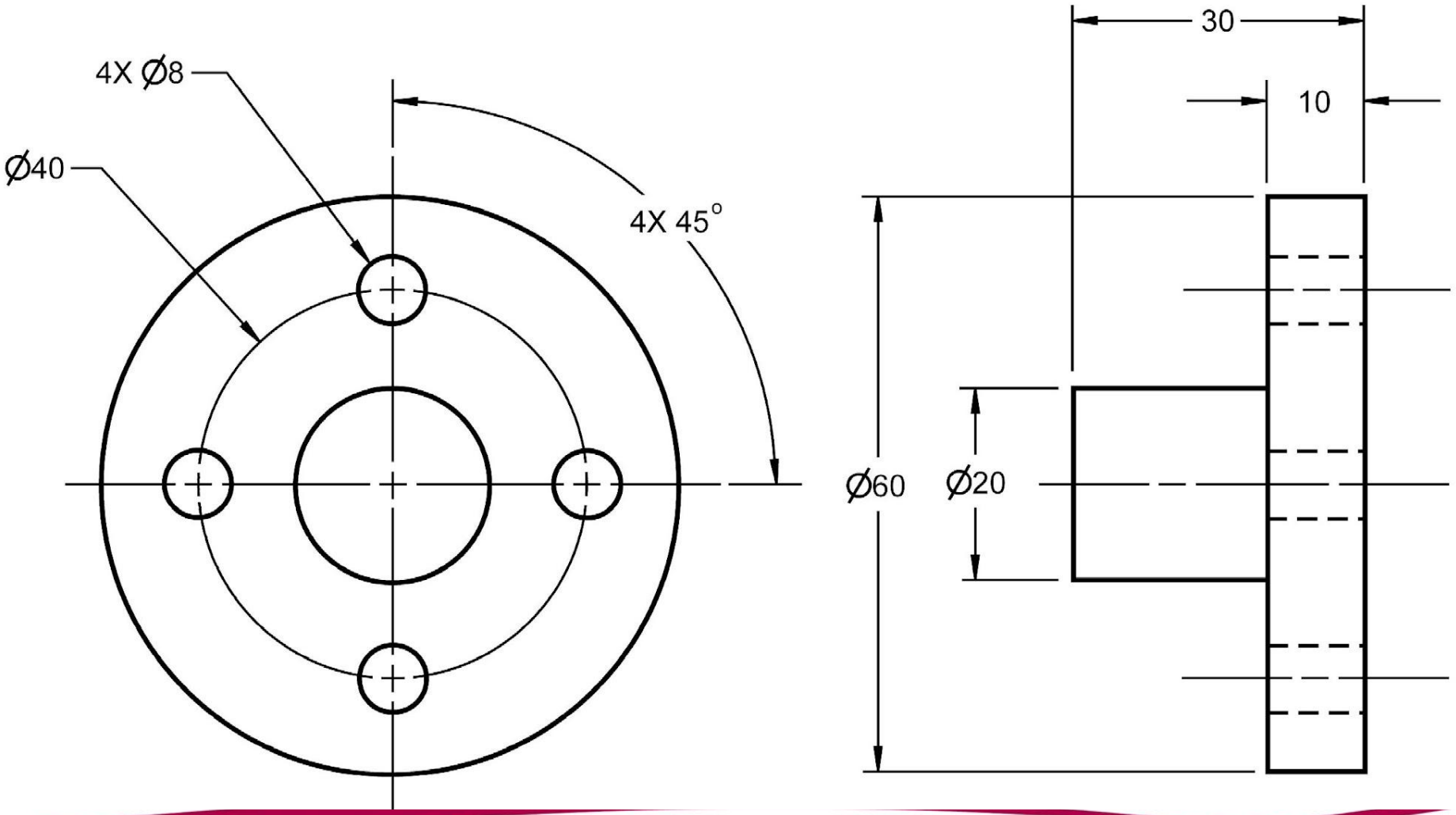


How do we dimension the cylinder heights?



Vertical Dimensions – ***Diameters*** – to ASME Y14.1 : 2004

Completely dimensioned.



Vertical Dimensions – ***Diameters*** – to ASME Y14.1 : 2004

Skip advanced topic

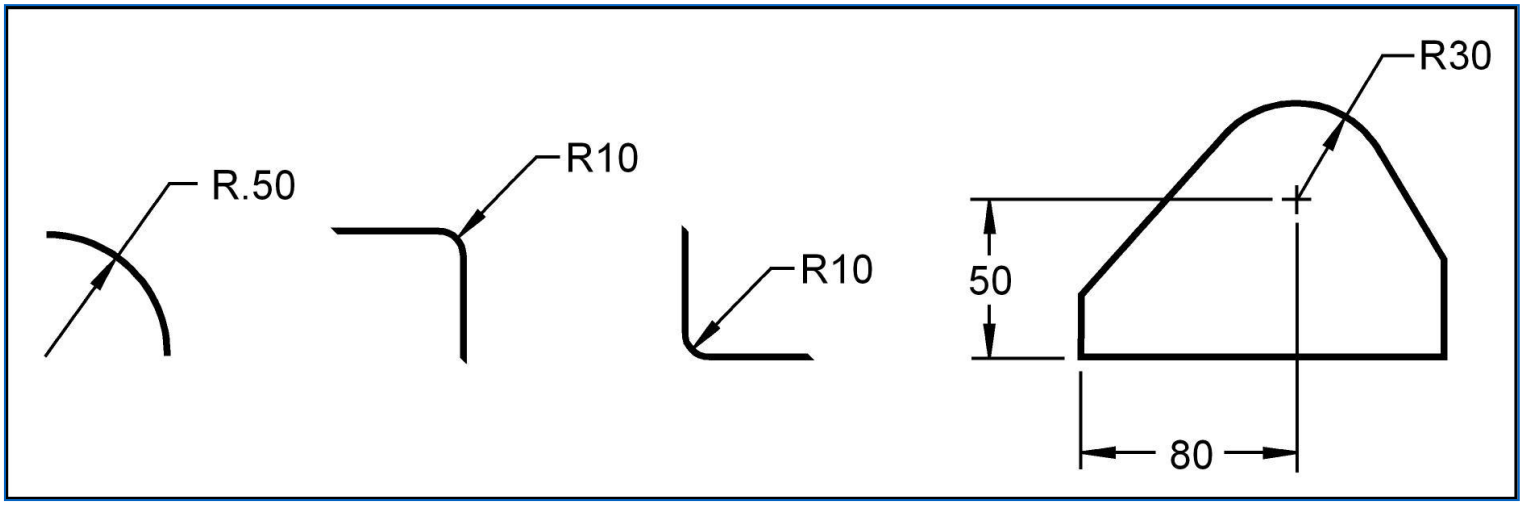


Dimensioning

2-5) Dimensioning and Locating Advanced Features

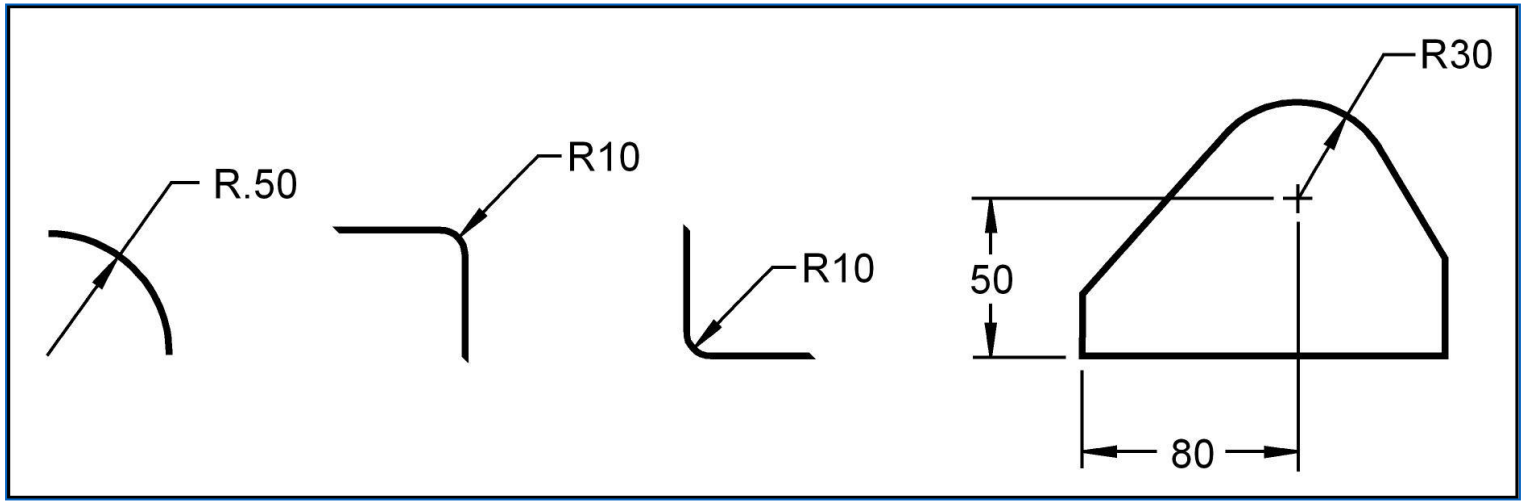
Dimensioning Features

a. If a dimension is given to the center of a radius, a small cross is drawn at the center.



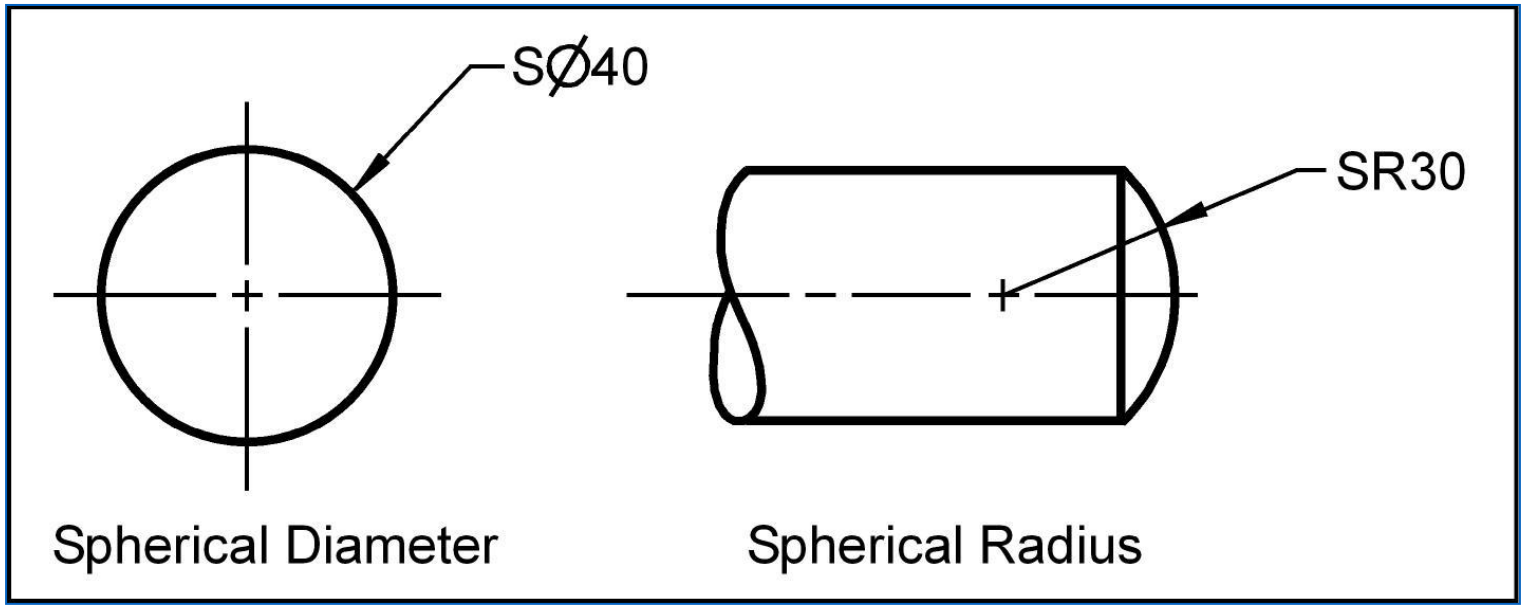
Dimensioning Features

b. Where the center location of the radius is unimportant, the drawing must clearly show that the arc location is controlled by other dimensioned features such as tangent surfaces.



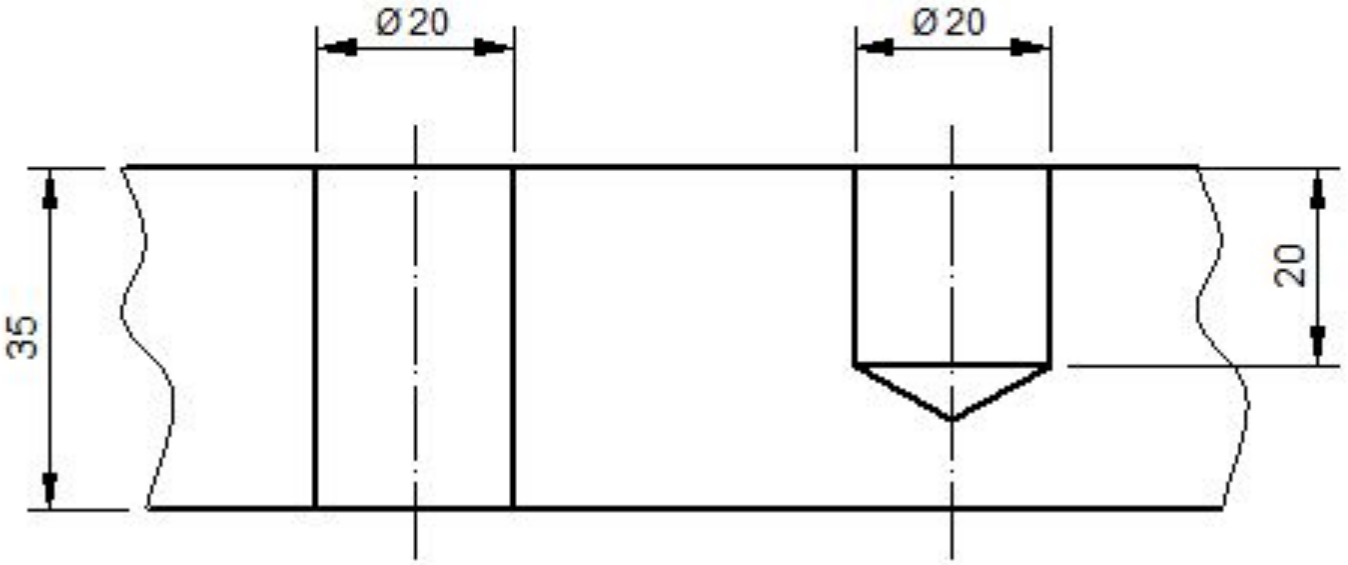
Dimensioning Features

c. A complete sphere is dimensioned by its diameter and an incomplete sphere by its radius.



Dimensioning Features

d. The depth of a blind hole may be specified in a note and is the depth of the full diameter from the surface of the object.

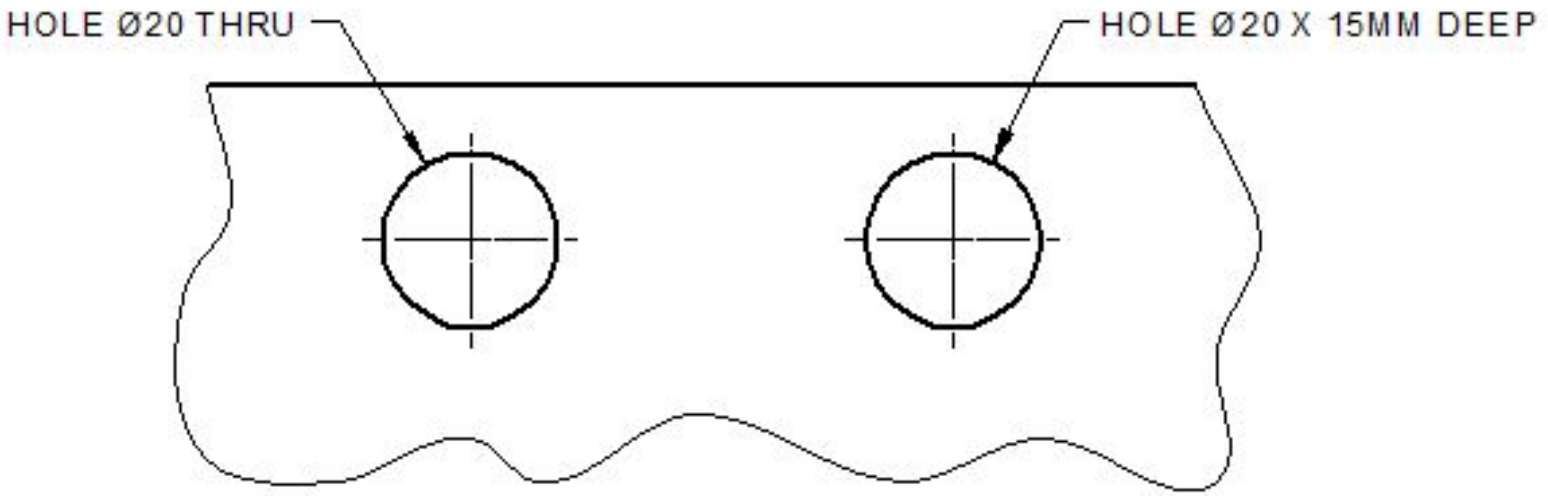


Hole drilled through the object

Blind hole drilled *part way* through the object

Dimensioning Features

e. If a hole goes completely through the feature and it is not clearly shown on the drawing, the abbreviation "THRU" follows the dimension.



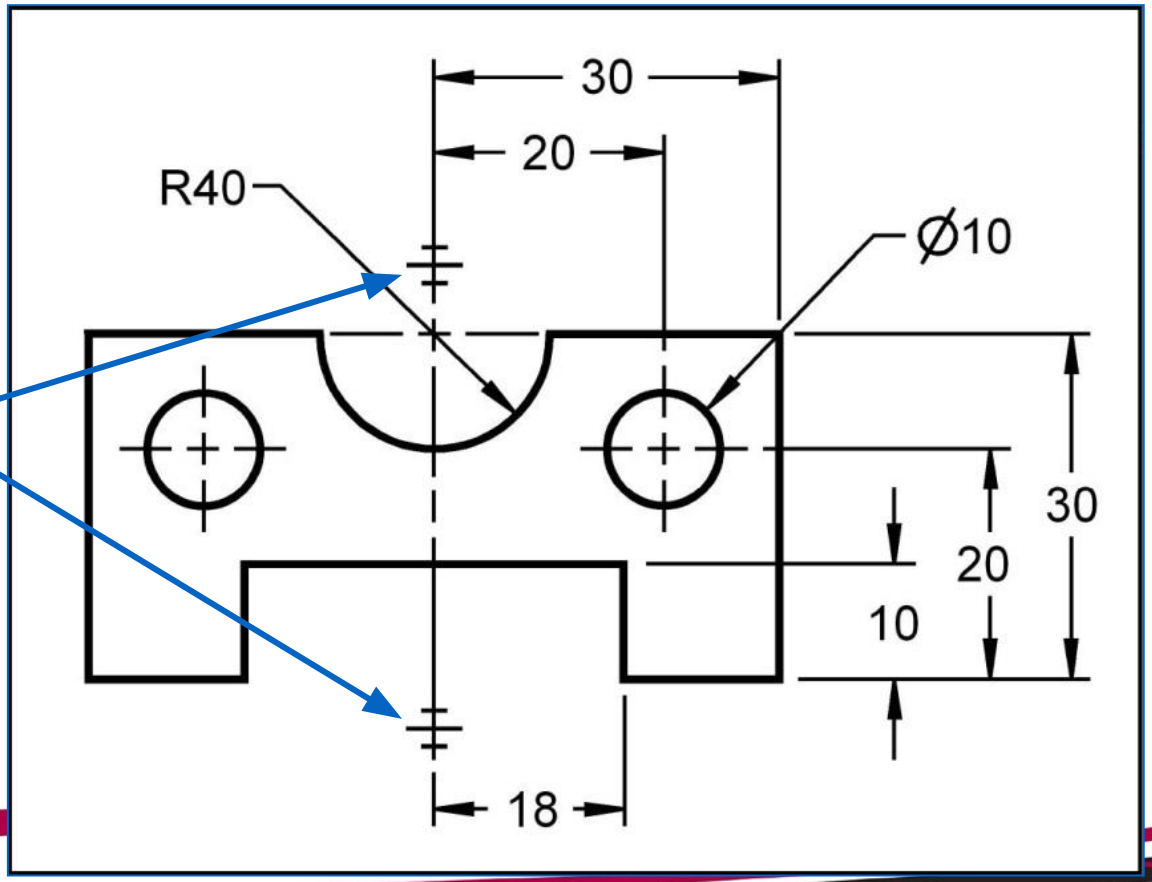
Hole drilled through the object

Blind hole drilled *part way* through the object

Dimensioning Features

f. If a part is **symmetric**, it is only necessary to dimension to one side of the center line of symmetry.

Symmetry symbol



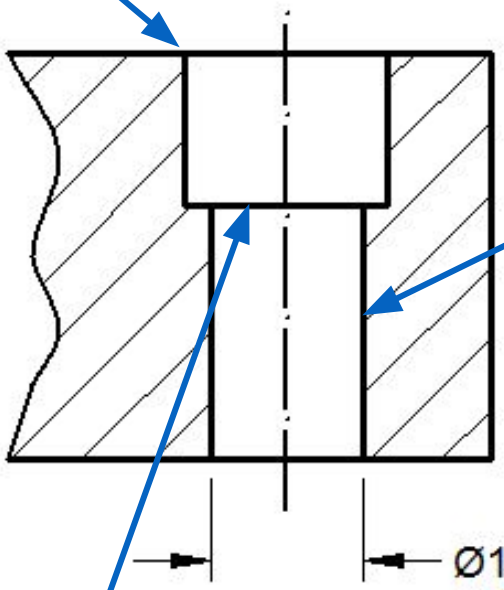
Dimensioning Features

g. Counterbored holes

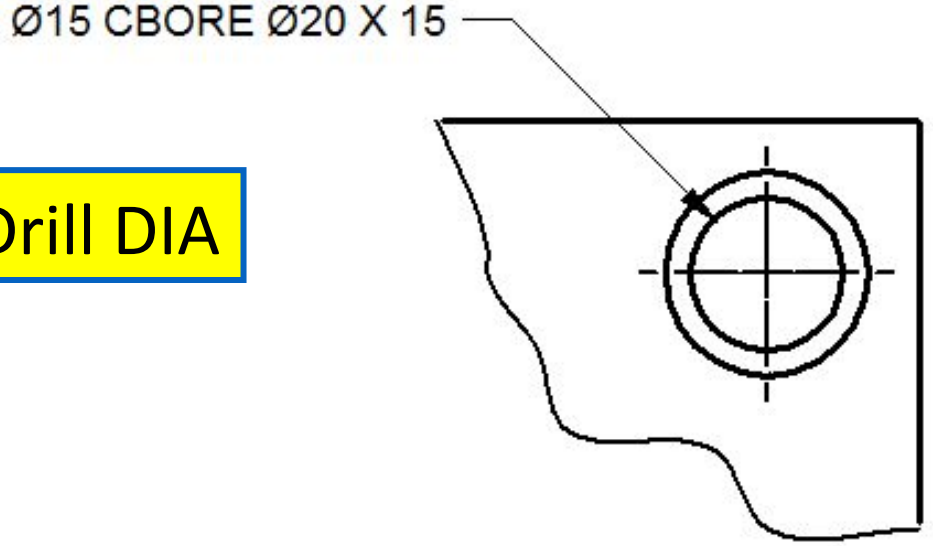
C'Bore DIA

Drill DIA

C'Bore depth

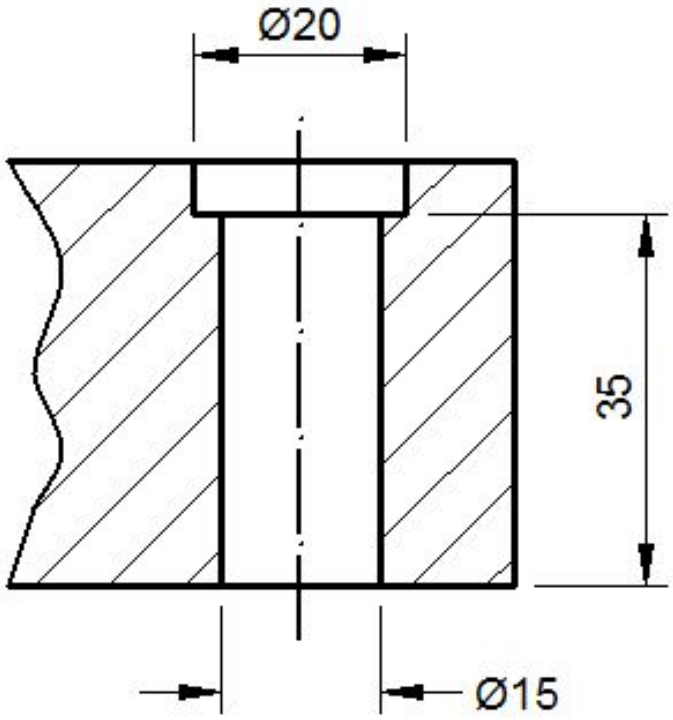


C'Bored hole detailed using a *Note and Leader*



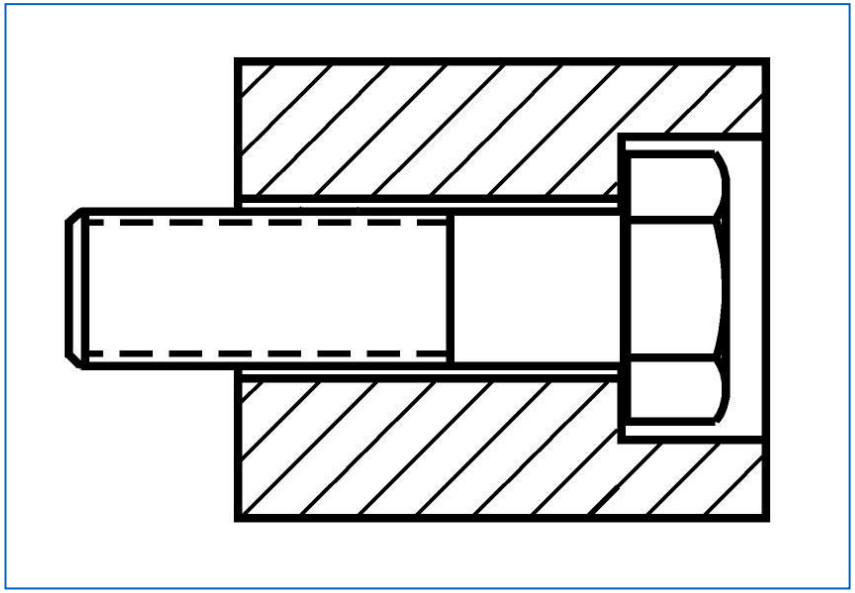
Dimensioning Features

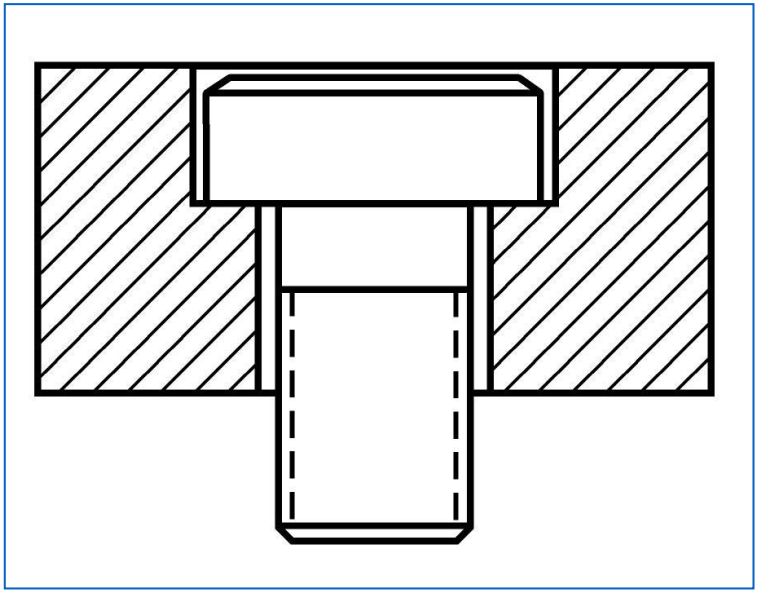
h. If the thickness of the material below the counterbore is significant, this thickness rather than the counterbore depth is given.



Application Question 2-1

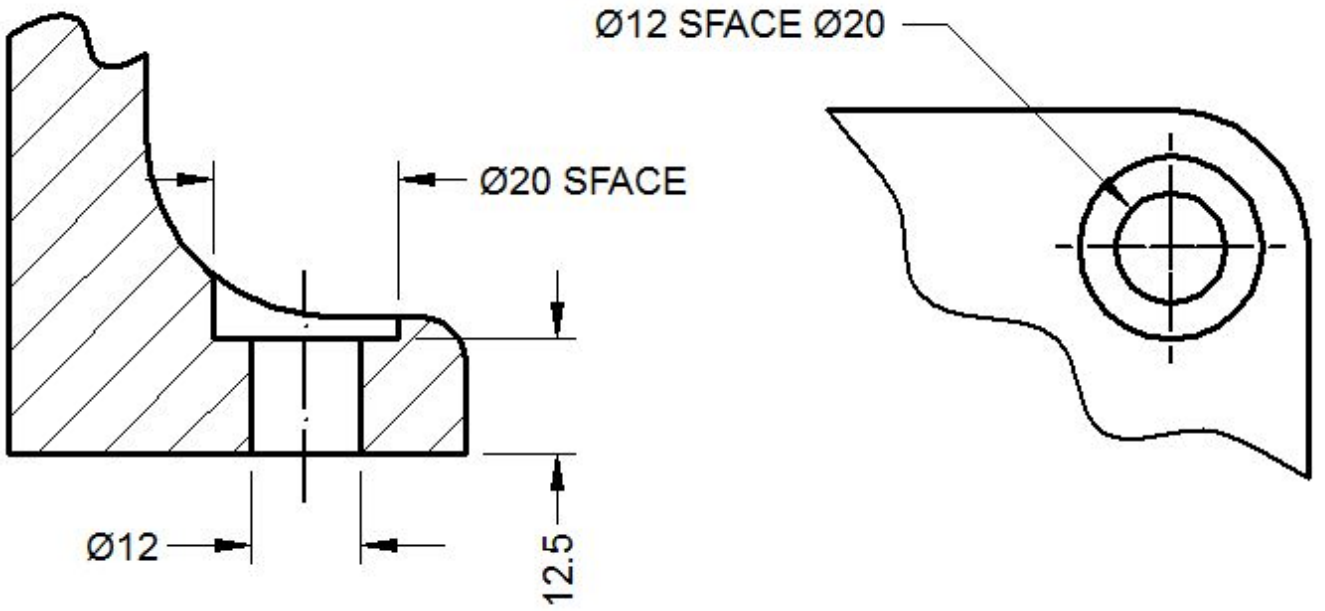
- *What do you think a counterbored hole is used for?*





Dimensioning Features

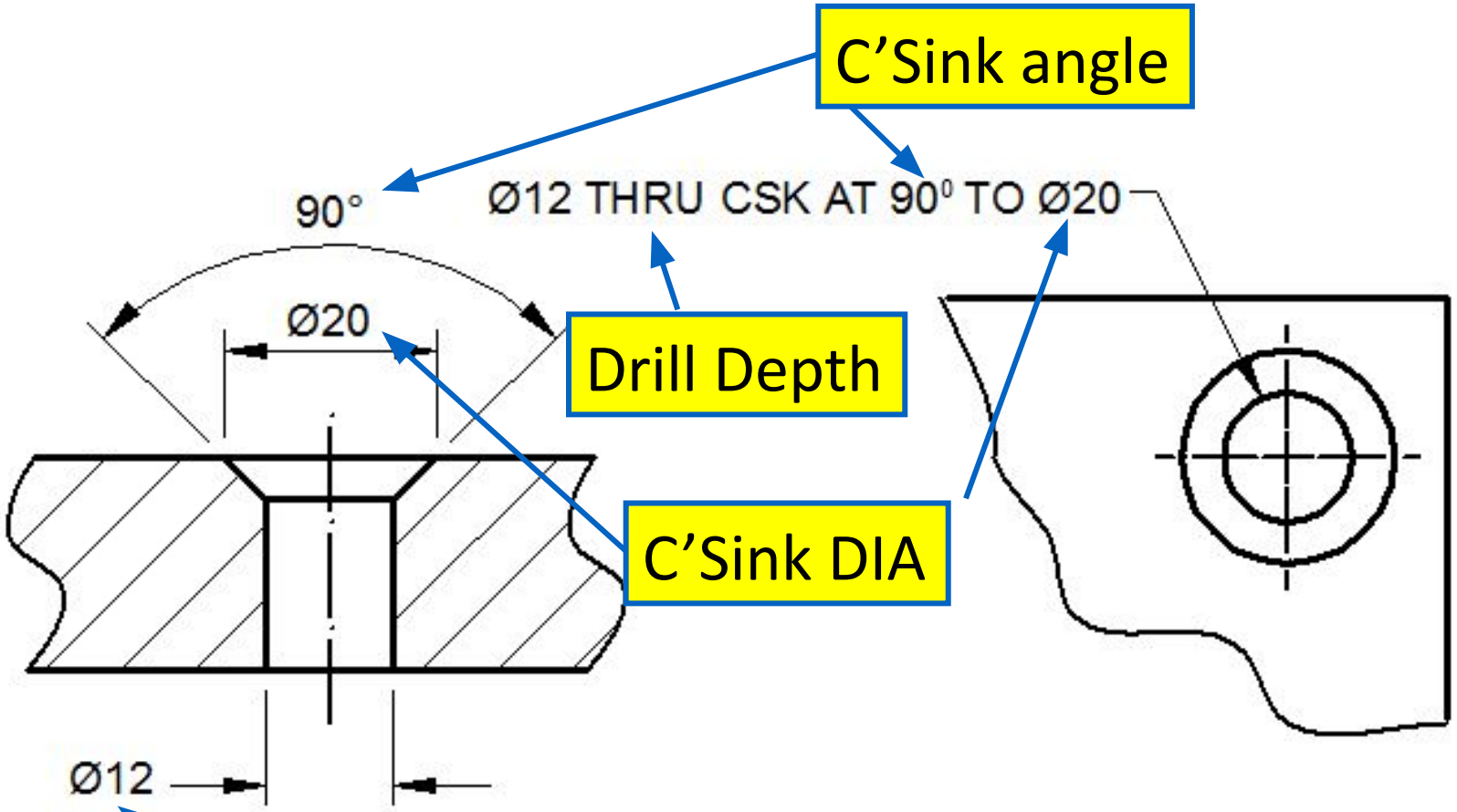
i. **<u>Spotfaced Holes:</u>** The difference between a C'BORE and a ***<u>Spotface</u>*** is that the machining operation occurs on a ***<u>curved surface</u>***.



Notice that the depth can not be specified in the note.

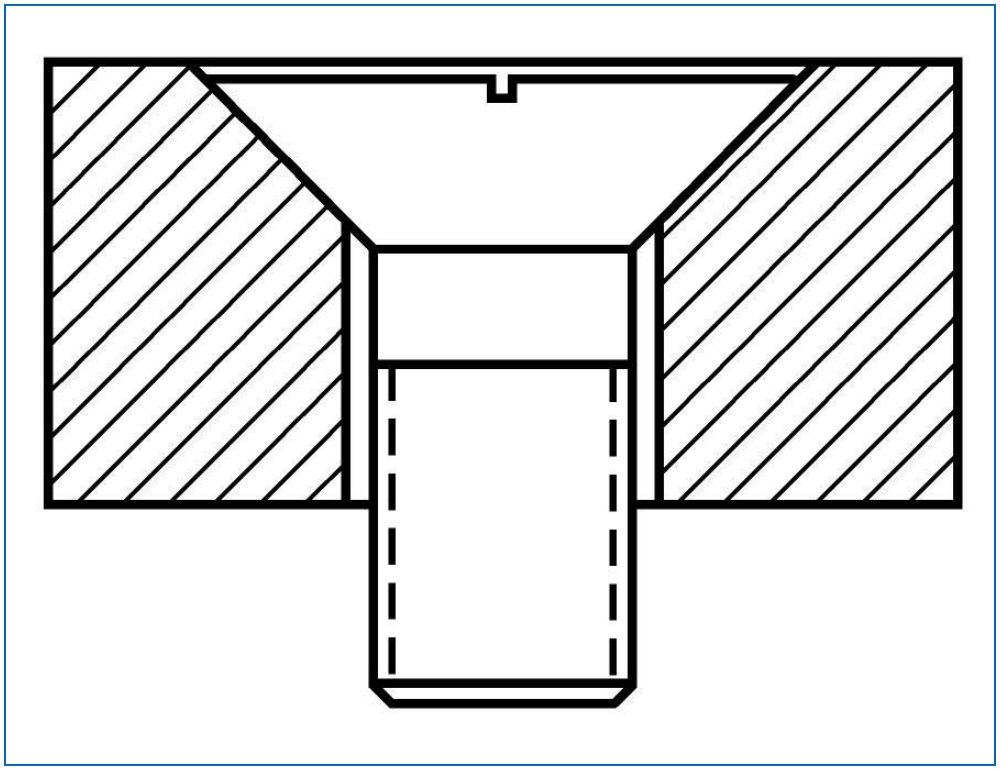
Dimensioning Features

j. Countersunk Holes



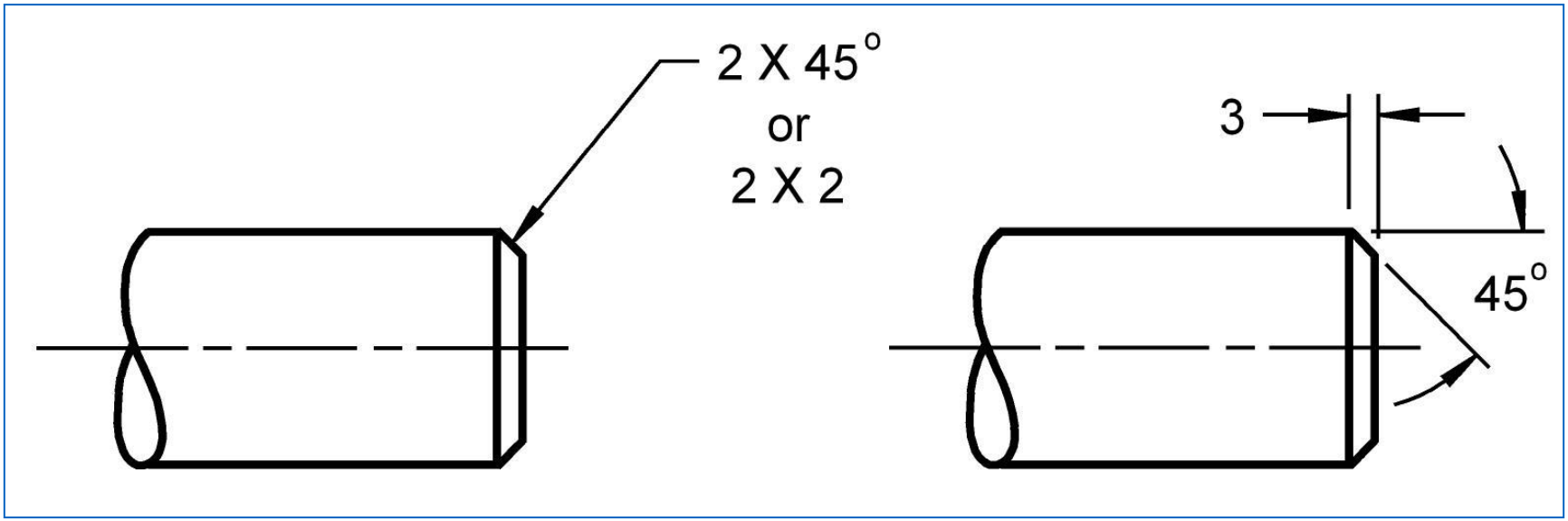
Application Question 2-2

- *What do you think a countersunk hole is used for?*



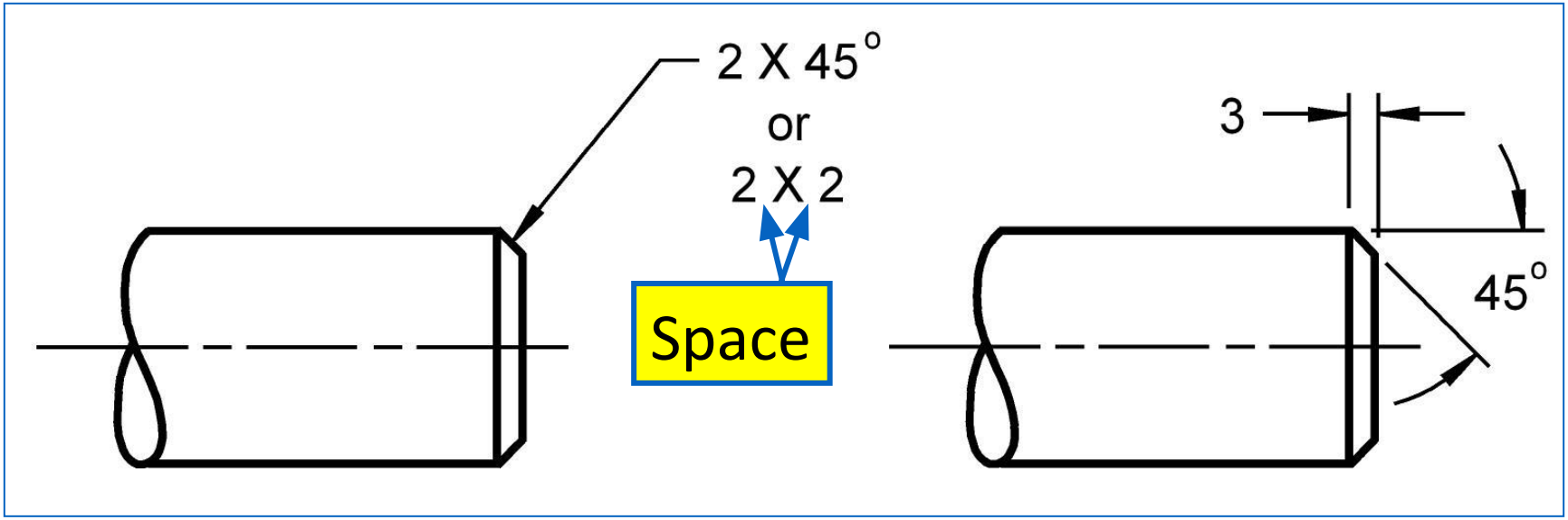
Dimensioning Features

k. **<u>Chamfers:</u>** Dimensioned by a linear dimension and an angle, or by two linear dimensions.



Dimensioning Features

I. **<u>Chamfers:</u>** Dimensioned by a linear dimension and an angle, or by two linear dimensions.



Application Question 2-3

- *What do you think a chamfer is used for?*

Safety.

Improve engagement of mating parts.

Drawing Notes

1. Drawing notes give additional information that is used to complement conventional dimension.
   - *manufacturing requirements*
   - *treatments and finishes*
   - *blanket dimensions (e.g. size of all rounds and fillets on a casting or a blanket tolerance).*
2. The note area is identified with the heading “NOTE:”.

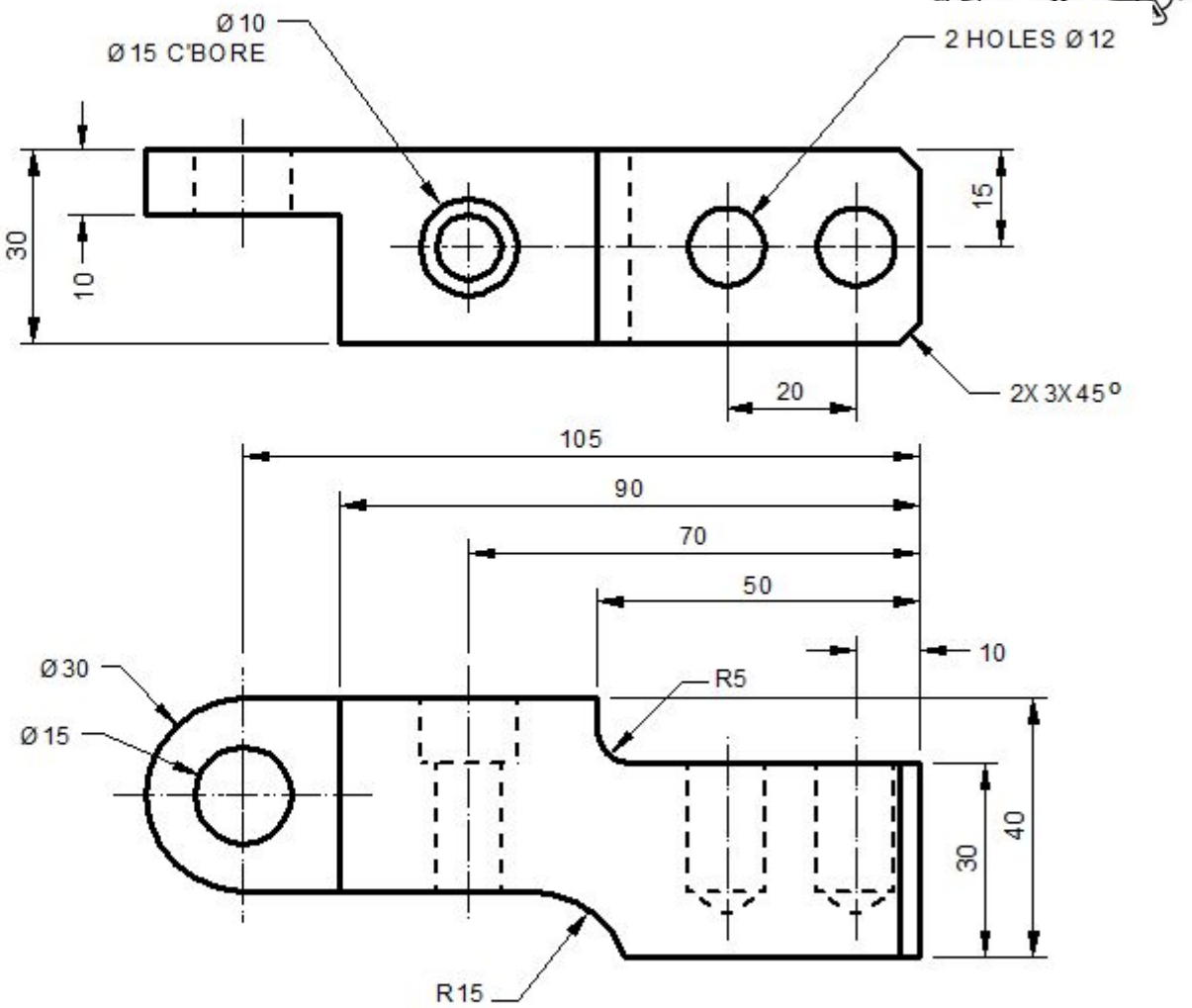


Exercise 2-8

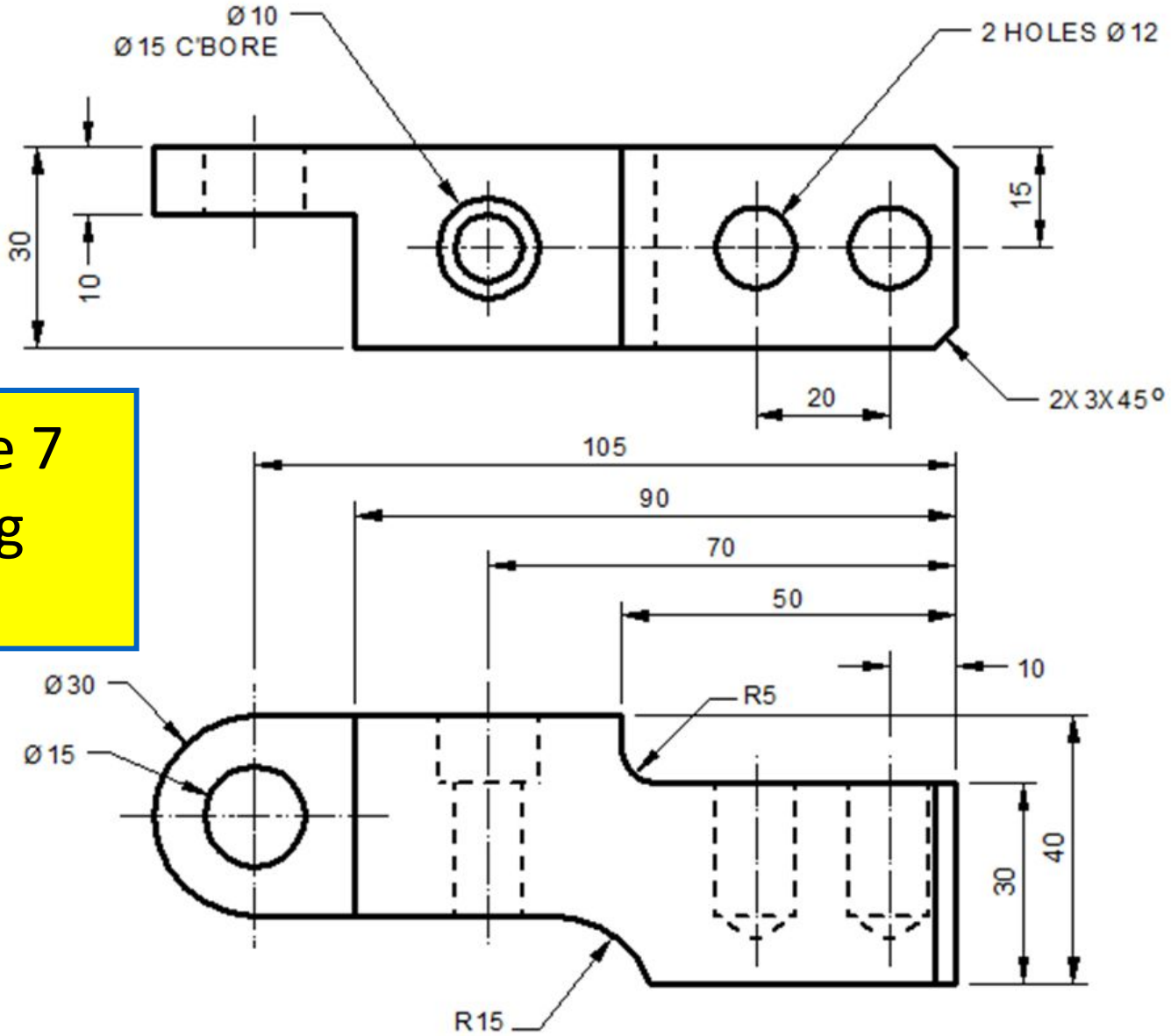
Advanced features

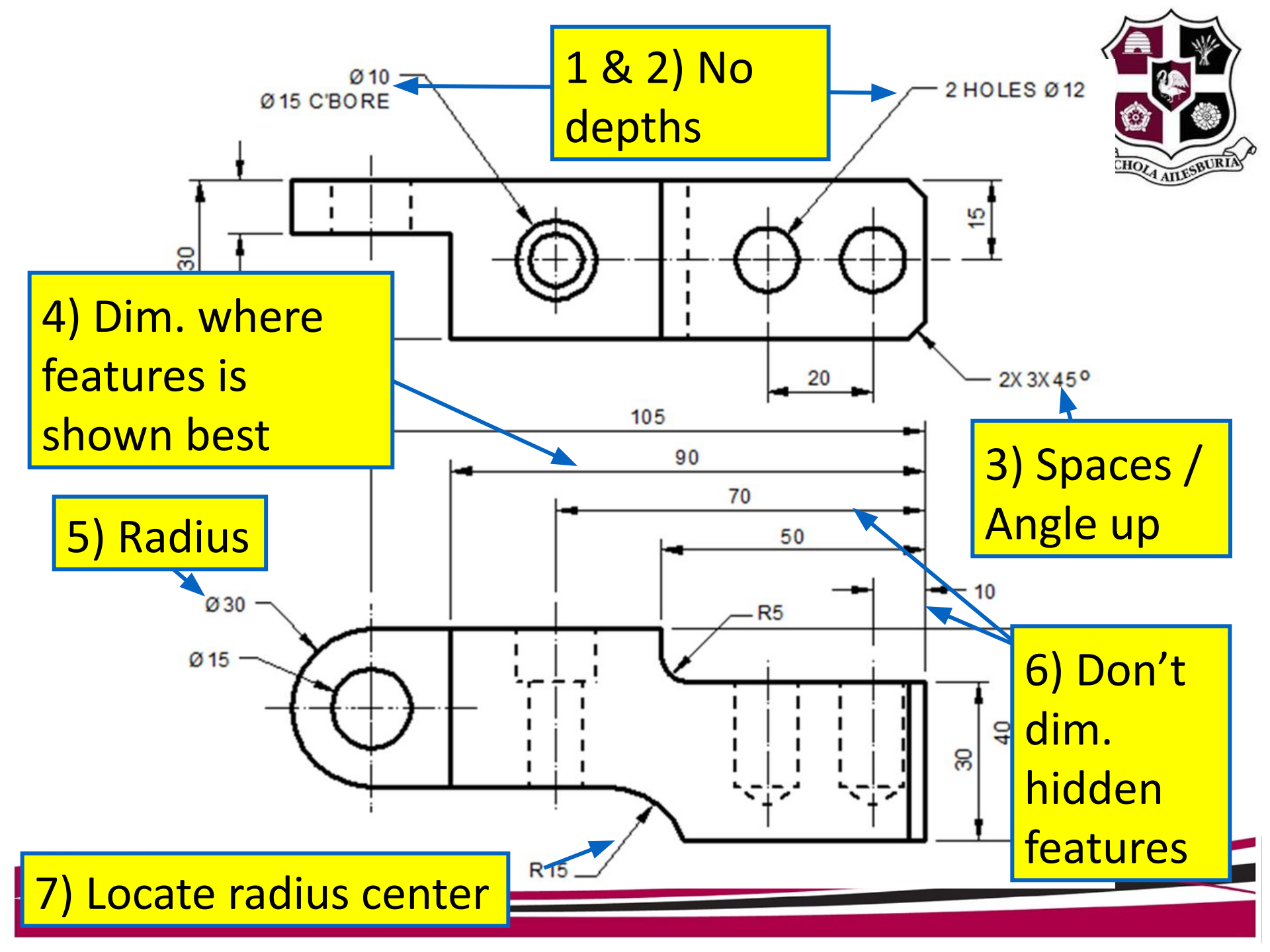
Exercise 2-8

- *List the dimensioning mistakes and then dimension the object correctly.*

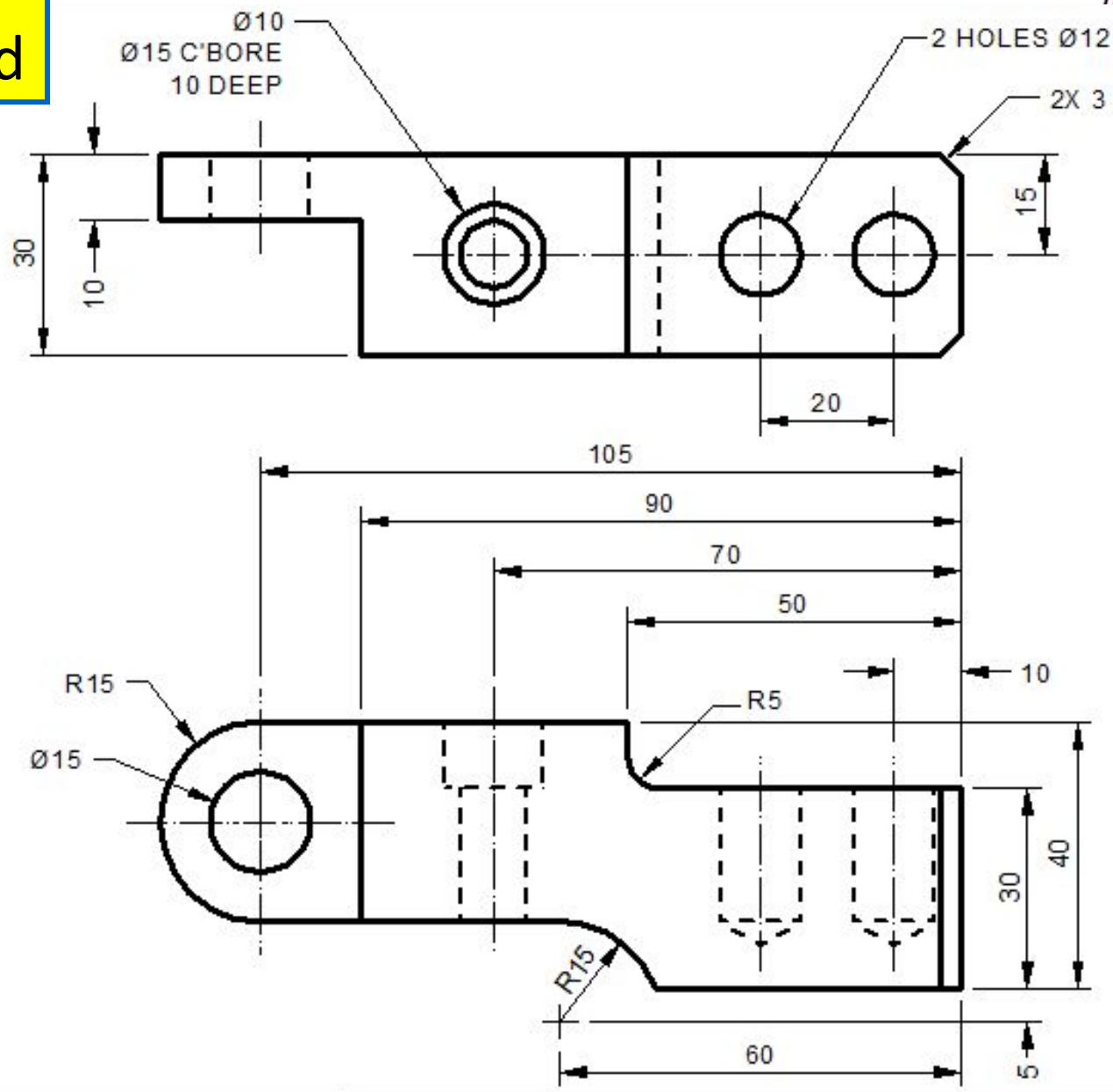


What are the 7 dimensioning mistakes?




1 & 2) No depths
Ø 10
Ø 15 C'BORE
2 HOLES Ø 12
30
15
20
2X 3X 45°
4) Dim. where features is shown best
105
90
70
50
3) Spaces / Angle up
5) Radius
Ø 30
Ø 15
R5
10
6) Don't dim. hidden features
30
40
R15
7) Locate radius center

Correctly Dimensioned



Dimensioning

2-6) Dimension Choice

Dimension Choice

1. Dimension placement and dimension text influences the manufacturing process used to make the part.
   - *Manufacturing process should not be specifically stated on the drawing.*

2. Choose dimensions based on function first then manufacturing.

Units and Decimal Places

a. Decimal dimensions should be used for all machining dimensions.
   - *You may encounter a drawing that specifies standard drills, broaches, and the like by size.*
   - *For drill sizes that are given by number or letter, a decimal size should also be given.*

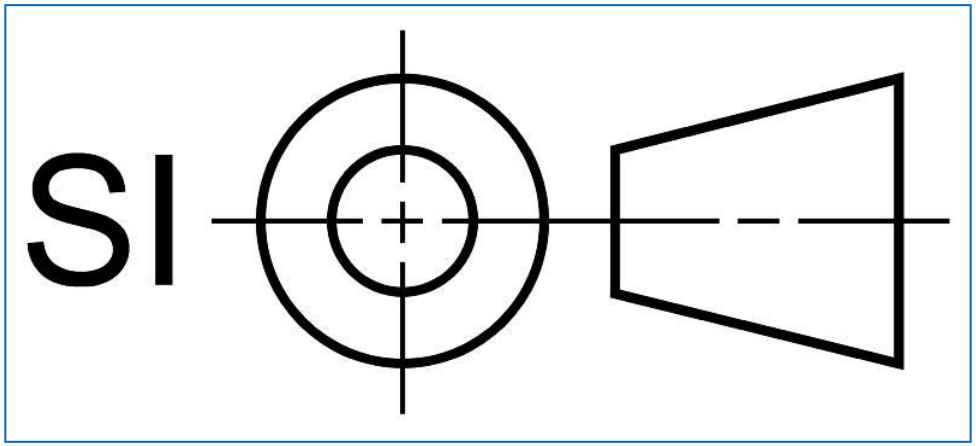
Units and Decimal Places

b. Metric dimensions are given in ‘mm’ and to 0 or 1 decimal place (e.g. 10, 10.2).

- *When the dimension is less than a millimeter, a zero should proceed the decimal point (e.g. 0.5).*

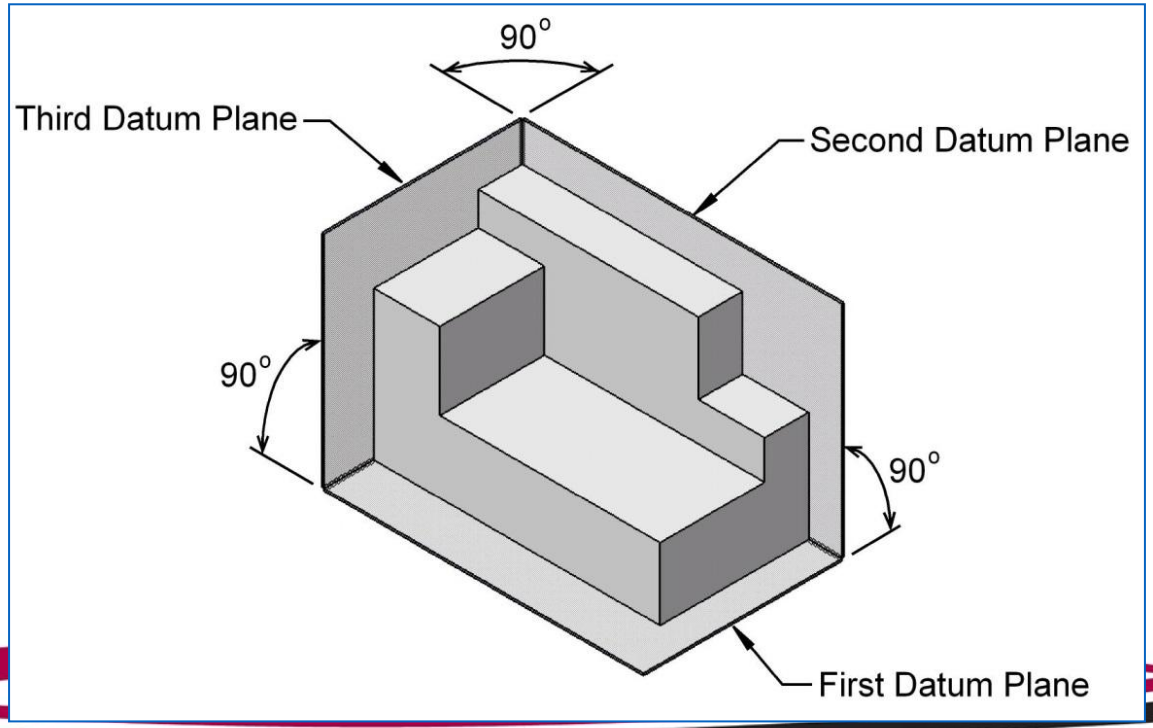
Units and Decimal Places

c. Metric 3rd angle drawings are designated by the SI symbol.



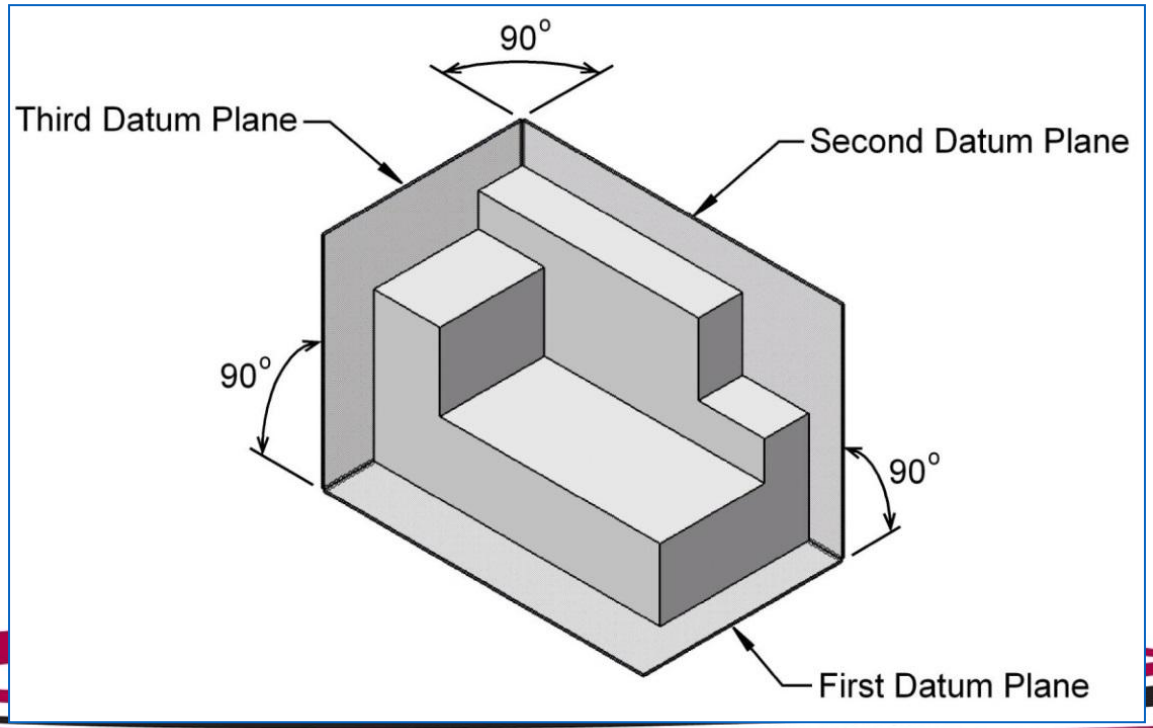
Locating Features Using Datums

- Consider three mutually perpendicular datum planes.
  - *These planes are imaginary and theoretically exact.*



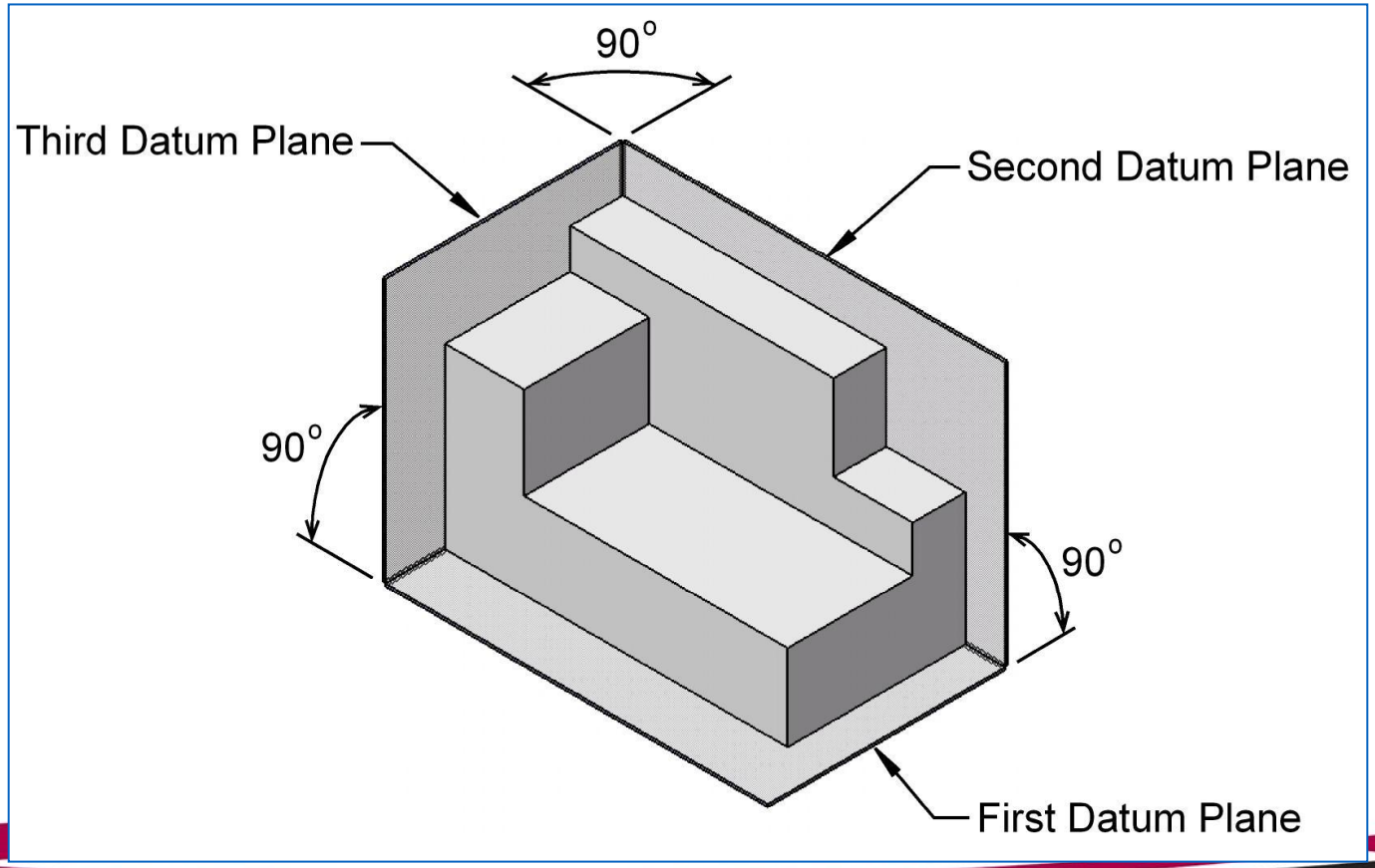
Locating Features Using Datums

- Now, consider a part that touches all three datum planes.
  - *The surfaces of the part that touch the datum planes are called **datum features**.*

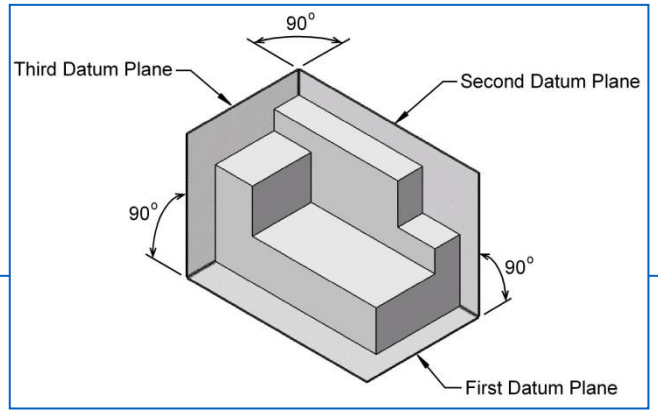


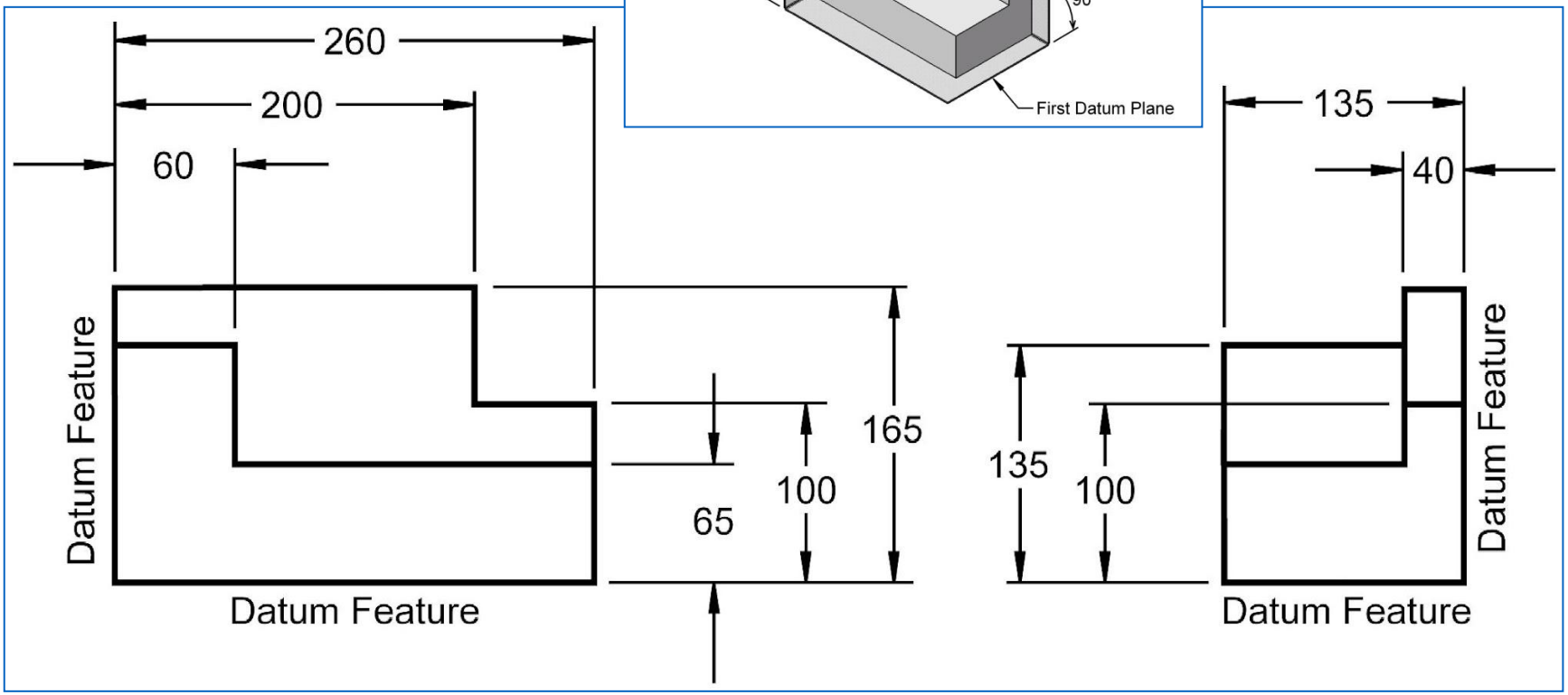
Locating Features Using Datums

- Most of the time, features on a part are located with respect to a datum feature.



Locating Features Using Datums





Locating Features Using Datums

- How do we choose which surface will be a datum feature?

- Good datum features are:
  - *functionally important surfaces*
  - *mating surfaces*
  - *big enough to permit its use in manufacturing the part*

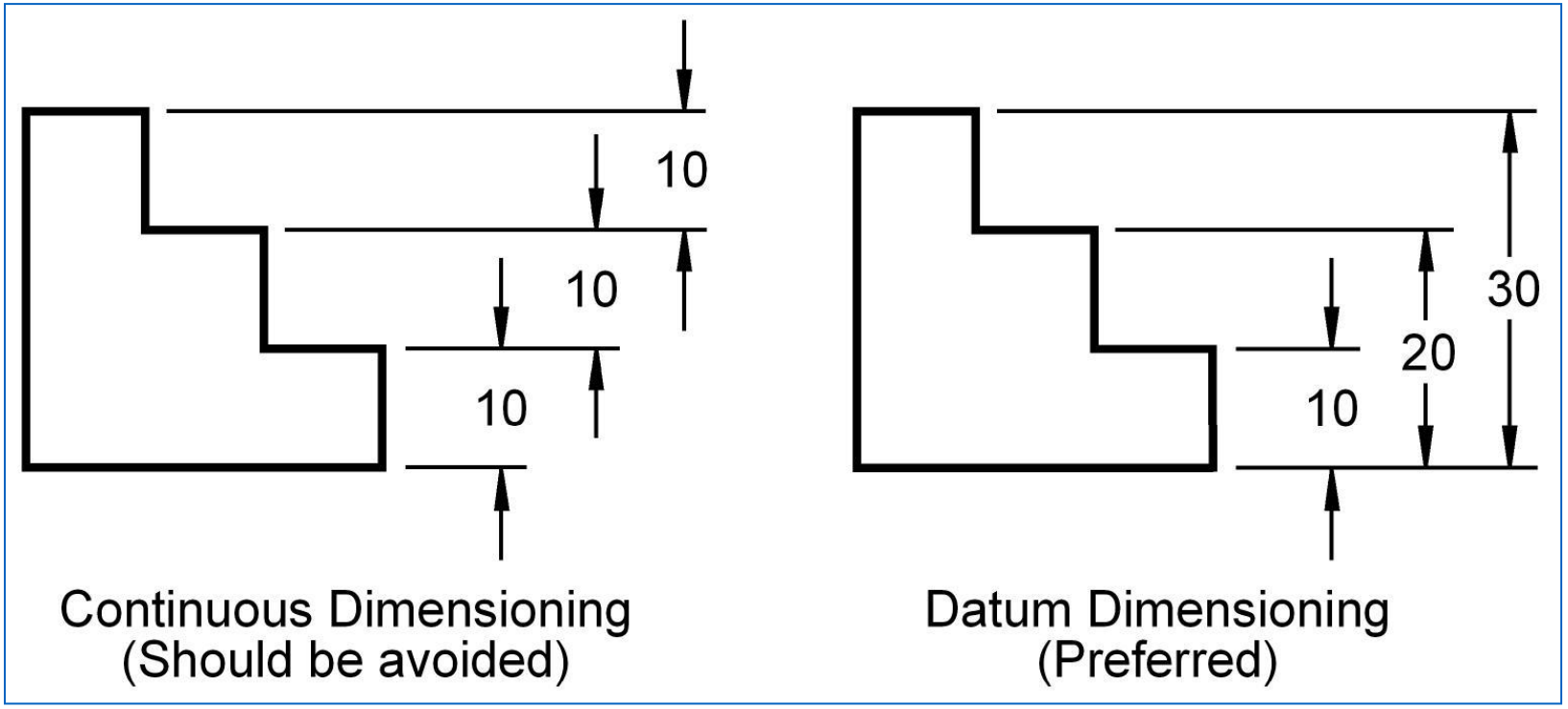
Locating Features Using Datums

- In a class setting, do we always know the function of the part?

- We need to make an educated guess as to the function of the part.

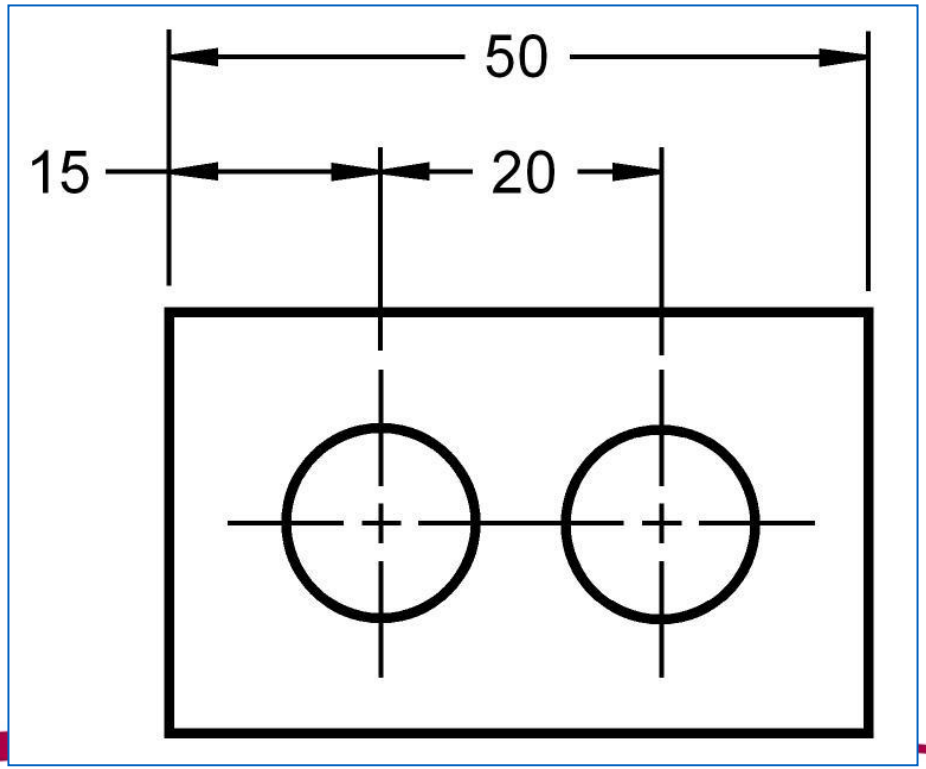
Locating Features Using Datums

a. Datum dimensioning is preferred over continuous dimensioning.



Locating Features Using Datums

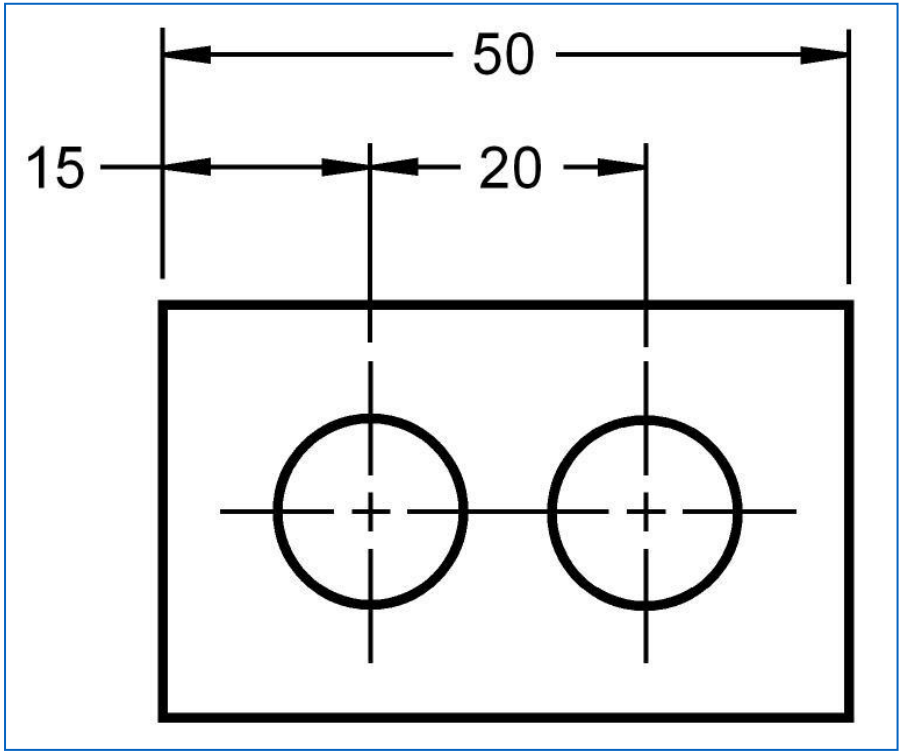
b. Dimensions should be given between points or surfaces that have a functional relation to each other
   - *Slots, mating hole patterns, etc...*



Application Question 2-4

- Why is the distance between the two holes functionally important?

If the hole pattern mates with 2 pins or bolts, the distance between the holes is more important than the distance from the edge to the second hole.



Dimension Accuracy

- There is no such thing as an "exact" measurement.
  - *Every dimension has an implied or stated tolerance associated with it.*
  - *A tolerance is the amount a dimension is allowed to vary.*

Dimensioning

Dimension Accuracy

Tolerances

When products are manufactured in industry, because of inaccuracy in manufacturing processes it is very difficult to achieve absolute accuracy in the size of the finished item. This creates problems when manufactured items have to fit accurately with other parts. When individual component parts are manufactured in batches of thousands or more, it is not economic to accept parts that do not fit and so cannot be assembled. To overcome this problem, items are manufactured with an acceptable margin of dimensional error called ***tolerance****.*

Quick Exercise

When products are manufactured in industry, because of inaccuracy in manufacturing processes it is very difficult to achieve absolute accuracy in the size of the finished item.

Working in pairs compile a <u>list</u> of things which could contribute to 'manufacturing inaccuracies'

For example – Tool Wear

Tolerances

*Tolerances that affect the size of an item or features contained within an item are known as **dimensional tolerances**. Tolerances can also be applied to the dimensions of location features on the item.*

For example – the position of holes and slots to ensure that one part can be assembled with the next part.

Functional and Non-Functional Dimensions

In practice, all dimensions in manufacturing drawings are subject to tolerances. There are however two types to consider:

- ***Functional Dimensions***
- ***Non-Functional Dimensions***

Functional Dimensions

Are dimensions that directly affect the function and fit of parts. An example is the diameter of a sink plug which must fit into a plug hole in a sink.

Non-Functional Dimensions

*Are dimensions that **do not** directly affect the function or fit of a part but may allow the product to meet other criteria such as final appearance or strength.*

For example – is the height of the sink plug, which could vary without affecting its function.

Non-Functional Dimensions

Non-functional dimensions *are generally represented in note form in the title block and are often referred to as general tolerances.*

For example:

TOLERANCES UNLESS OTHERWISE STATED LINEAR ±0.25, ANGULAR ±$0^0$30'

Functional Dimensions

*Tolerances on **functional dimensions** are represented by attaching information to individual dimensions on the drawing.*

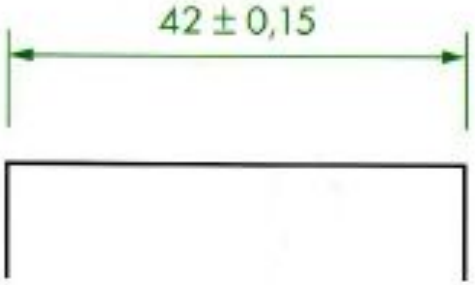
The methods used to indicate tolerances on individual dimensions are shown on the next two slides.

Functional Dimensions
Linear Dimensions

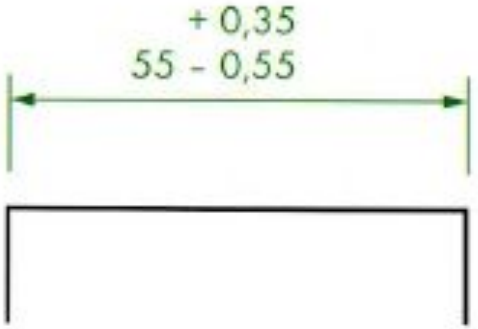
- The **common method** shows the upper limit of size placed above the lower limit.



- **Symmetrical tolerance** shows the nominal size and the symmetrical tolerance band.



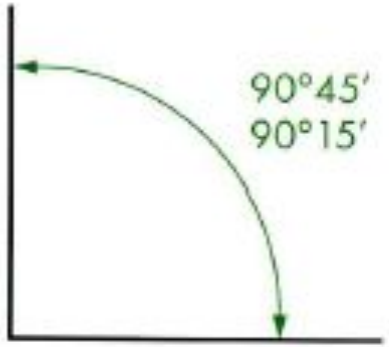
- **Asymmetrical tolerance** shows the nominal size plus the upper and lower limits of the tolerance band.



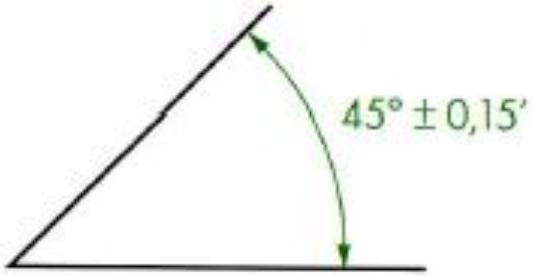
Functional Dimensions
Angular Dimensions

Tolerances of angular functional dimensions are indicated in much the same way as linear dimensions by giving information on individual dimensions, as shown.

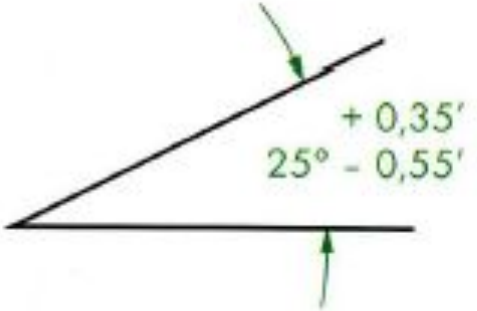
- **Common method**



- **Symmetrical tolerance**

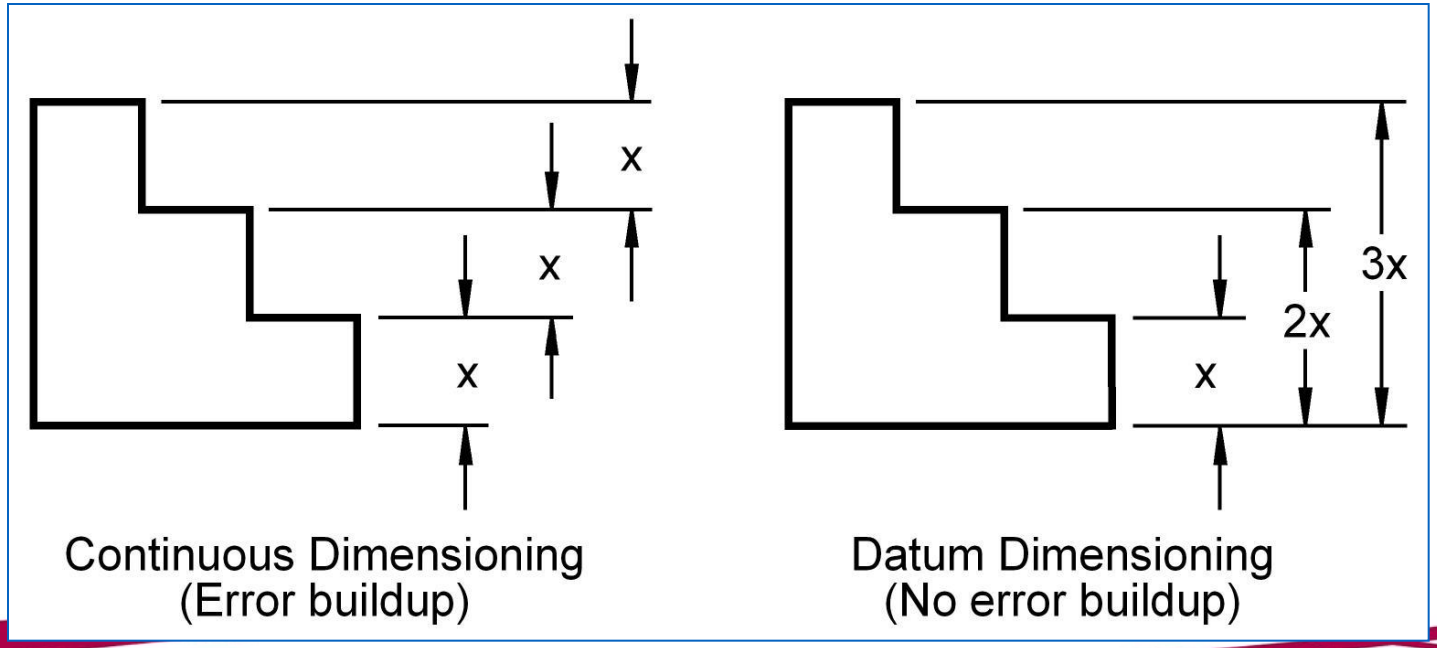


- **Asymmetrical tolerance**



Cumulative Tolerances

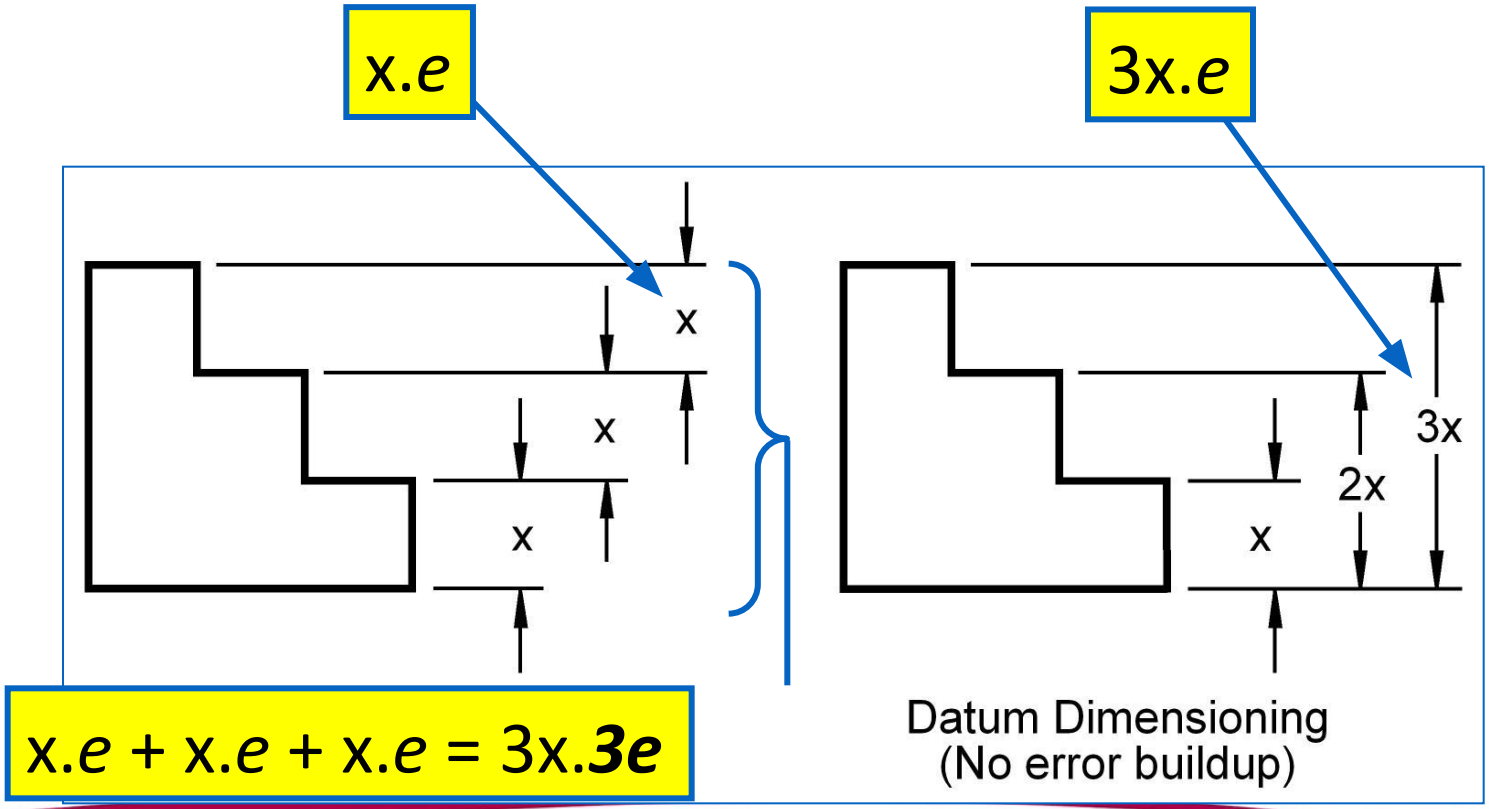
- Continuous dimensioning has the disadvantage of accumulating error.
  - *It is preferable to use datum dimensioning to reduce error buildup.*



Cumulative Tolerances

- What is error build up?

e = individual dimension error

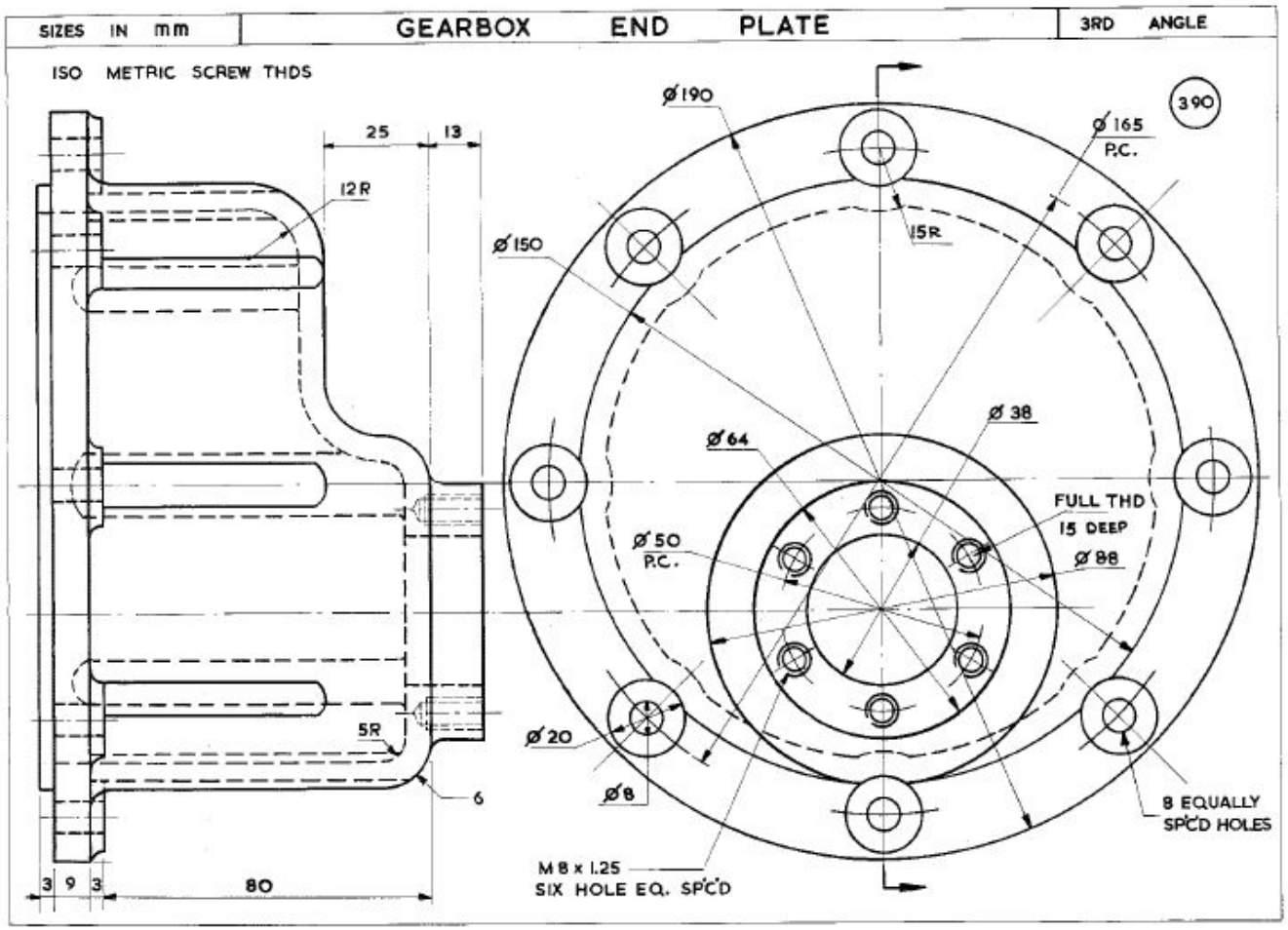


Exercise 2-9

Dimensional Accuracy

Exercise 2-9

- Using drawing equipment construct gear box end plate in third angle projection and fully dimension



Dimensioning

The End